

Undergraduate Lab Portal Requirements Specification



**Faculty of Engineering and Architecture
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CEN 302-Software Engineering**

Undergraduate Lab Portal Requirements Specification

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Table of Contents

1) Executive Summary	3
a) Project Overview	4
b) Purpose and Scope of this Specification	4
2) Product/Service Description	4
a) Product Context	4
b) User Characteristics	4
c) Assumptions	5
d) Constraints	6
e) Dependencies	6
3) Requirements	6
a) Functional Requirements	6
b) Non-Functional Requirements	9
i) Product Requirements	9
(1) User Interface Requirements	9
(2) Learnability	11
(3) Accessibility	11
(4) Efficiency	12
(5) Dependability/Memorability	12
(6) Errors	12
3.2.1.7 Satisfaction	12
(7) Security	13
ii) Organizational Requirements	13
(1) Availability	13
(2) Latency	13
Response time for critical operations (e.g., room booking, check-in, payment processing) should be under 2 seconds.	13
(3) Monitoring	14
(4) Maintenance	14
Regular software updates should be scheduled to ensure security and performance improvements.	14
(5) Operations	14
(6) Standards Compliance	14
(7) Portability	14
iii) External Requirements	14
(1) Protection	14
(2) Authorization and Authentication	15
(3) Legislative Requirements	15

Undergraduate Lab Portal Requirements Specification

c) Domain Requirements	15
4) User Scenarios/Use Cases	18
4.1 Requirements Analysis	18
4.1.1 User Scenarios List	18
4.1.2 User Scenarios Extended	19
4.1.3 User Cases	28
4.2 Behavioral Diagrams	48
4.2.1 Use Case Diagrams	48
4.2.2 Activity Diagrams	50
4.2.3 State Diagrams	69
4.2.4 Sequence Diagrams	77
4.4 Entity Relation	101
4.4.1 Database Schema Design	101
4.4.2 Entity Relation Diagram	102
4.5 Structural Diagrams	103
4.5.1 Class Diagram	103
4.5.3 Component Diagrams	109
4.5.4 Deployment Diagram	110
5. Implementation Technology	111
Implementation Technology	111

1. Executive Summary

a) Project Overview

The Undergraduate Lab Portal is a centralized, web-based platform designed to modernize and streamline the process by which undergraduate students discover, apply for, and secure research positions within their institution. Currently, the typical process for finding research opportunities relies

heavily on informal communication channels such as direct emails, word of mouth, and departmental bulletin boards. These fragmented methods create significant barriers for students seeking research experience, result in inconsistent application processes for faculty, and make it difficult for departments to monitor and promote research engagement at a systemic level.

The Undergraduate Lab Portal addresses these challenges by providing a unified digital environment that connects students and faculty through a structured, transparent, and intelligent recruitment system. By replacing ad-hoc email chains with a purpose-built portal, the platform enables students to build professional research profiles, browse open positions, and apply directly through the system. Faculty benefit from organized applicant management tools, GPA and skill-based filtering, and a digital roster for their lab teams. At the institutional level, department administrators gain oversight dashboards and reporting tools, while a dedicated External Programs department manages international mobility and scholarship opportunities such as Erasmus.

The portal leverages modern web technologies to deliver a user-friendly interface accessible from any device, facilitating seamless interaction among all stakeholders: students, professors, department administrators, and external program coordinators.

b) Purpose and Scope of this Specification

The purpose of this document is to define the software requirements for the Undergraduate Lab Portal, a web application that will digitalize and automate the research position recruitment pipeline within an academic institution. This application is intended to replace traditional, unstructured methods of student-faculty research matchmaking with a transparent, data-driven platform that promotes equal access to research opportunities.

The system will enable undergraduate students to create and maintain research profiles and submit applications; professors to post positions and manage applicants; administrators to monitor departmental research metrics and approve funding; and the External Programs department to coordinate international mobility and scholarship opportunities. This specification provides a comprehensive description of the platform's functional and non-functional requirements, intended for use by the development team, project stakeholders, and institutional reviewers.

2. Product/Service Description

This section describes general factors that affect the product and its requirements. It provides background context to explain the reasons behind specific requirements defined later in this document.

a) Product Context

Undergraduate Lab Portal Requirements Specification

The Undergraduate Lab Portal is designed for use within a university or academic institution, serving as a comprehensive and standalone system accessible to four primary user groups: students, professors, department administrators, and the External Programs and Partnerships department. The platform operates as an internal institutional tool, streamlining the full lifecycle of undergraduate research recruitment — from position posting and application, through to selection, onboarding, and institutional reporting.

The portal ensures seamless coordination between faculty labs, academic departments, and institutional offices, facilitating smooth operation of research programs and enabling data-driven decisions regarding research funding, student participation, and international engagement.

b) User Characteristics

1. Student

Undergraduate students are the primary applicants within the system. They interact with the portal to discover research opportunities and present their academic qualifications to faculty. Key actions include:

- Create and manage a professional research profile highlighting technical skills, academic history, and research interests.
 - Browse open lab and research positions posted by faculty across departments.
 - Apply directly through the platform without relying on external email communication.
 - Track the status of submitted applications and receive updates on their progress.
-

2. Professor

Professors use the portal to recruit undergraduate research assistants and manage their lab teams. Their interactions include:

- Post open research projects and lab assistant positions, including descriptions, requirements, and duration.
 - Specify required GPA thresholds and prerequisite qualifications for applicants.
 - Filter and search applicants by skills /GPA.
 - Maintain a digital roster of current lab members.
 - Review applicant history and manage the status of ongoing or past applications.
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3. Administrator (Department-Level)

Department administrators oversee institutional research activity, ensuring compliance, funding management, and accurate reporting. Their role includes:

Undergraduate Lab Portal Requirements Specification

- Monitor department-wide research engagement and student participation metrics.
 - Track research output and contribution data across labs and faculty.
 - Manage funding and stipend approvals for research positions.
 - Oversee lab safety compliance records and ensure adherence to institutional standards.
 - Generate detailed reports for institutional review boards, accreditation, and departmental planning.
-

4. External Programs & Partnerships Department (HR)

This user group manages opportunities beyond the institution's borders, connecting students with international research, exchange, and scholarship programs. Their responsibilities include:

- Manage applications related to Erasmus and other international student mobility programs.
- Inform students about government scholarships, funded research internships, and international projects.
- Coordinate with external partner institutions and funding bodies to keep opportunity listings current and accurate.

c) Assumptions

- Students will have an active institutional email account and will use it to register and authenticate on the portal without external assistance.
- Professors will take responsibility for keeping their posted positions up to date, including closing listings once a vacancy is filled or a project ends.
- Students are expected to maintain accurate and honest profiles, including correct GPA, skills, and academic standing.
- The university has the necessary IT infrastructure (computers, internet connectivity, institutional servers) to support the deployment and continuous operation of the portal.
- Department administrators are responsible for keeping funding records, stipend approvals, and compliance data current within the system.
- Professors will review and respond to applications within a reasonable timeframe, ensuring applicants are not left without a status update for extended periods.
- The External Programs & Partnerships (HR) department will regularly update scholarship, Erasmus, and international opportunity listings to reflect current availability and deadlines.
- Students are assumed to have basic digital literacy and will be able to navigate the portal, submit applications, and track their status without external assistance.
- The institution will provide proper onboarding or guidance to professors and administrators on how to use the portal's management and reporting features effectively.

a) Constraints

- The Undergraduate Lab Portal requires a stable internet connection to function properly, as it is a web-based solution.

Undergraduate Lab Portal Requirements Specification

- The system's effectiveness depends on students, professors, and administrators actively adopting and correctly using the platform.
- The performance and reliability of the portal are dependent on the university's IT infrastructure, including network stability, server availability, and device access across campus.
- The accuracy and completeness of student profile data — including GPA, skills, and experience — directly impact the quality of search results and the effectiveness of position filtering (**UC-S-05, UC-S-06, UC-S-11**).
- The matching and filtering system's ability to surface suitable candidates relies on professors providing detailed and accurate position requirements when posting opportunities (**UC-P-02**).
- The portal's user experience and functionality may be constrained by the limitations of the web technologies and frameworks used during development.
- The ability to generate institutional reports and track research participation metrics depends on accurate and timely data entry across all user roles (**UC-A-01**).
- Stipend and funding workflows are constrained by the completeness of funding request submissions and the availability of administrators to process them (**UC-P-11, UC-A-05**).

b) Dependencies

- The posting and visibility of research positions depends on professors actively creating and maintaining listings within the portal (**UC-P-02, UC-P-04**).
- The application process depends on students having a complete profile; incomplete profiles block submission entirely (**UC-S-06, UC-S-11**).
- Applicant status tracking depends on professors reviewing applications and updating decisions in a timely manner (**UC-P-07**).
- Lab roster management depends on professors promptly adding or removing members as their team changes (**UC-P-09, UC-P-10**).
- The generation of department-level research statistics and compliance reports depends on administrators maintaining accurate lab certification and participation records (**UC-A-01, UC-A-02, UC-A-03**).
- Stipend processing depends on professors submitting funding requests and administrators acting on pending stipend records (**UC-P-11, UC-A-04, UC-A-05**).
- The availability and visibility of international and scholarship opportunities depends on HR staff publishing and keeping program listings current before their deadlines (**UC-HR-01, UC-HR-02, UC-HR-03, UC-HR-06**).
- Program application decisions depend on HR staff reviewing applicants and updating statuses, which in turn triggers student notifications (**UC-HR-05**).

Req#	Requirement	Comments	Pr io	Date Rvwd	SME Reviewed / Approved
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Undergraduate Lab Portal Requirements Specification

			Priority		
FR-AUTH-01	The system shall allow users to register with a unique username, unique email address, password, and a role	Four distinct roles supported at registration.	1	27/05/2026	Sibora Kociu / Redi Cuka
FR-AUTH-02	The system shall reject registration if the password is shorter than 8 characters, missing an uppercase letter, missing a lowercase letter, or missing a digit.	Frontend and backend both validate password strength.	1	27/05/2026	Arlind Disha / Arli Asllani
FR-AUTH-03	The system shall authenticate users via username and password and issue a signed JWT access token upon success.	Token stored client-side in localStorage.	1	27/05/2026	Megi Sollaku / Sibora Kociu
FR-AUTH-04	The system shall enforce role-based access control; requests from unauthorized roles shall receive HTTP 403.	Each role sees only their permitted endpoints.	1	27/05/2026	Nikol Isaku / Megi Sollaku
FR-AUTH-05	The system shall support logout by removing the client-side token.	Token removed from localStorage on logout.	2	27/05/2026	Sibora Kociu / Redi Cuka
FR-STU-01	The system shall allow a student to create a profile with full name, GPA, major, and a list of skills.	GPA must fall within valid academic range.	1	27/05/2026	Arlind Disha / Arli Asllani
FR-STU-02	The system shall allow a student to update their profile at any time.	No restrictions on frequency of updates.	2	27/05/2026	Megi Sollaku / Sibora Kociu
FR-STU-03	The system shall display all lab positions with status OPEN to logged-in students.	Only OPEN positions visible; CLOSED and FILLED are hidden.	1	27/05/2026	Redi Cuka / Arlind Disha
FR-STU-04	The system shall prevent a student from submitting more than one application per position.	Duplicate check performed server-side; returns HTTP 409.	1	27/05/2026	Sibora Kociu / Redi Cuka
FR-STU-05	The system shall allow a student to view all their applications and their current statuses.	Statuses include SUBMITTED, UNDER_REVIEW, ACCEPTED,	2	27/05/2026	Arlind Disha / Arli Asllani

Undergraduate Lab Portal Requirements Specification

		REJECTED, WITHDRAWN.			
FR-STU-06	The system shall allow a student to upload documents	Multiple document types supported per student.	2	27/05/2026	Megi Sollaku / Sibora Kociu
FR-STU-07	The system shall display all published international programs to students.	DRAFT and CLOSED programs not visible to students.	1	27/05/2026	Arli Asllani / Arbiana Muka
FR-STU-08	The system shall allow a student to apply for a published program if their GPA meets the minimum requirement.	GPA check enforced server-side; returns HTTP 400 if unmet.	1	27/05/2026	Nikol Isaku / Megi Sollaku
FR-STU-09	The system shall display the student's lab memberships including lab name, assigned role, and join date.	Shown in profile or dedicated Lab tab.	2	27/05/2026	Arlind Disha / Arli Asllani
FR-PRO-01	The system shall allow a professor to update their department and title.	Basic profile management for faculty.	2	27/05/2026	Megi Sollaku / Sibora Kociu
FR-PRO-02	The system shall allow a professor to create a lab position with title, description, required GPA, associated lab, and optional stipend amount.	Stipend is optional; triggers FR-PRO-04 if provided.	1	27/05/2026	Redi Cuka / Arlind Disha
FR-PRO-03	The system shall allow a professor to update the status of their own positions (OPEN → CLOSED, OPEN → FILLED).	Professors cannot modify other professors' positions.	1	27/05/2026	Nikol Isaku / Megi Sollaku
FR-PRO-04	The system shall display all applications submitted to a professor's positions, with full student details.	Includes student profile and uploaded documents.	1	27/05/2026	Sibora Kociu / Redi Cuka
FR-PRO-05	The system shall allow a professor to transition an application to UNDER_REVIEW, ACCEPTED, or REJECTED.	Decision triggers notifications per FR-NOT-02.	1	27/05/2026	Arlind Disha / Arli Asllani
FR-PRO-06	The system shall allow a professor to view the complete roster of their lab(s), including student GPA, major, and assigned role.	Read-only view of current lab composition.	2	27/05/2026	Redi Cuka / Arlind Disha

Undergraduate Lab Portal Requirements Specification

FR-PRO-07	The system shall allow a professor to manually add a student to a lab by username or student ID.	Bypasses the application flow for direct additions.	2	27/05/2026	Arli Asllani / Arbiana Muka
FR-PRO-08	The system shall allow a professor to remove a student from a lab.	Removal does not affect application history.	2	27/05/2026	Nikol Isaku / Megi Sollaku
FR-PRO-09	The system shall allow a professor to submit a funding request with amount, purpose, and associated lab.	Request enters PENDING state awaiting admin review.	1	27/05/2026	Sibora Kociu / Redi Cuka
FR-PRO-10	The system shall allow a professor to view all their submitted funding requests and their statuses.	Statuses include PENDING, APPROVED, REJECTED.	2	27/05/2026	Arlind Disha / Arli Asllani
FR-HR-01	The system shall allow HR Staff to create an international program with title, description, application deadline, and minimum GPA.	Programs start in DRAFT and are not yet visible to students.	1	27/05/2026	Arbiana Muka / Nikol Isaku
FR-HR-02	The system shall allow HR Staff to close a PUBLISHED program, setting its status to CLOSED.	Prevents further student applications after closure.	1	27/05/2026	Nikol Isaku / Megi Sollaku
FR-HR-03	The system shall display all student applications for each program to HR Staff with full student profile details.	Includes GPA, major, skills, and uploaded documents.	1	27/05/2026	Sibora Kociu / Redi Cuka
FR-HR-04	The system shall allow HR Staff to accept or reject program applications.	Decision triggers notification per FR-NOT-03.	1	27/05/2026	Arlind Disha / Arli Asllani
FR-HR-05	The system shall notify the student when their program application status changes.	Notification delivered in-app per FR-NOT-03.	1	27/05/2026	Arbiana Muka / Nikol Isaku
FR-ADM-01	The system shall provide the admin with aggregate statistics: total labs, open positions, submitted applications, and total lab members.	Displayed on the Research Overview dashboard.	1	27/05/2026	Megi Sollaku / Sibora Kociu
FR-ADM-02	The system shall display all labs with their certification status, professor name, and current lab status (ACTIVE, INACTIVE, SUSPENDED).	Used for compliance monitoring.	1	27/05/2026	Redi Cuka / Arlind Disha

Undergraduate Lab Portal Requirements Specification

FR-ADM-03	The system shall display all pending funding requests to the admin.	Admin acts as the sole approver of funding requests.	1	27/05/2026	Nikol Isaku / Megi Sollaku
FR-ADM-04	The system shall allow the admin to view and process stipends (approve → process, or reject).	Stipend lifecycle: PENDING → APPROVED → PROCESSED or REJECTED.	1	27/05/2026	Sibora Kociu / Redi Cuka
FR-NOT-01	The system shall create an in-app notification for the professor when a student submits a position application.	Keeps professors informed of new applicants in real time.	2	27/05/2026	Arbiana Muka / Nikol Isaku
FR-NOT-02	The system shall create an in-app notification for the student when a professor changes their application status.	Covers UNDER_REVIEW, ACCEPTED, and REJECTED transitions.	1	27/05/2026	Megi Sollaku / Sibora Kociu
FR-NOT-03	The system shall create an in-app notification for the student when their program application status changes.	Triggered by HR Staff decisions per FR-HR-07.	1	27/05/2026	Redi Cuka / Arlind Disha
FR-NOT-04	The system shall display the count of unread notifications in the sidebar badge.	Badge updates dynamically without page refresh.	2	27/05/2026	Arli Asllani / Arbiana Muka

a) Non-Functional Requirements

i) Product Requirements

(1) User Interface Requirements

The user interface for the web application should be compatible with any browser in order for the user to access it from Desktop. The user interface is divided into 5 main groups:

Home/Login/Register Interface:

- A landing page with a navigation bar containing the application name and buttons for logging in and registering.
- The login interface contains fields for username, password and a login button.
- The register interface contains fields for username, email, password, and role selection (Student, Professor, HR Staff).
- All inputs will be thoroughly validated; if invalid, an error will display on screen instructing the user to enter valid details. Password strength requirements are shown inline with real-time feedback.

Student Interface:

- A navigation sidebar containing options such as "Browse Opportunities", "My Applications", "External Programs", "My Documents", "My Profile", "Notifications" and "Log Out".
- A dashboard overview showing available open positions, the student's active applications and their statuses.
- A Browse Opportunities section where students can view and filter all open lab positions by department and minimum GPA.
- An application form that appears when clicking Apply on a position, requiring no additional fields beyond confirmation.
- A My Applications section showing all submitted applications and their statuses (Submitted, Under Review, Accepted, Rejected, Withdrawn), with an option to withdraw eligible applications.
- An External Programs section where students can browse published international programs and submit or track program applications.
- A Documents section where students can upload files by selecting a file and document type.
- A Notifications panel accessible via a bell icon in the sidebar, showing all notifications with unread badge count.
- A My Profile section where students fill in their full name, GPA, major and skills.

Professor Interface:

- A navigation sidebar containing options such as "Manage Positions", "Lab Management", "Request Funding", "My Profile" and "Log Out".
- A dashboard overview showing the professor's active positions, total applicants and lab members.
- A Manage Positions section where the professor can view all their positions, create new ones via a form (title, description, required GPA, lab, stipend), and update position status (Open, Closed, Filled).
- An Applicants view per position, showing each applicant's profile and documents, with options to Accept or Reject.
- A Lab Management section showing the lab roster with member details (GPA, major, role), and options to add or remove members.
- A Request Funding section where the professor can submit funding requests and view their history.

HR Staff Interface:

- A navigation sidebar containing options such as "Manage Programs", "My Profile" and "Log Out".
- A dashboard overview showing total programs, published programs and pending applications.

Undergraduate Lab Portal Requirements Specification

- A Manage Programs section where HR staff can create new international programs via a form (title, description, deadline, minimum GPA), update program details, publish or close programs, and review and decide on student program applications.

Admin Interface:

- A navigation sidebar containing options such as "Research Overview", "Lab Compliance", "Funding & Stipends", "My Profile" and "Log Out".
 - A dashboard overview showing aggregate statistics: total labs, certified labs, pending funding requests and active stipends.
 - A Lab Compliance section listing all labs with their certification status, with options to certify or decertify each lab with an optional note.
 - A Funding & Stipends section listing all pending funding requests and stipends, with options to approve, process or reject each with optional notes.
-

(2) Learnability

The Lab Management System shall be designed to ensure that users can quickly and easily learn how to operate the system with minimal training. Key learnability features include:

- **Intuitive Interface:** The system will provide a clear, consistent and user-friendly interface tailored to each user role (Student, Professor, HR Staff, Admin).
 - **Guided Onboarding:** New users will receive tooltips or contextual help upon first login to familiarize them with core functionalities.
 - **Role-Based Dashboards:** Each user role will have a customized dashboard with only relevant functions, reducing complexity.
 - **Consistent Navigation:** Common actions (e.g., applying for positions, reviewing applicants) will follow standardized workflows across the system.
 - **Minimal Training Dependency:** The system will rely on familiar design patterns (e.g., form submissions, button placements) to reduce the need for extensive training.
-

(3) Accessibility

The system shall adhere to accessibility standards to ensure usability for all users, including those with disabilities:

- **Compliance:** The web interface will follow Web Content Accessibility Guidelines (WCAG) 2.1, including proper contrast, keyboard navigation and screen reader compatibility.
- **Responsive Design:** The system will be fully functional on desktops, tablets and mobile devices, adapting to screen sizes.
- **Text Alternatives:** Images and icons will include alt text for screen readers.
- **Adjustable Text:** Users can zoom in or adjust font sizes without breaking the interface.

(4) Efficiency

The system shall optimize task completion time and resource usage:

- **Quick Actions:** Frequent tasks (e.g., reviewing applications, updating position status) will be accessible within minimal clicks from the dashboard.
 - **Performance:** Pages will load within 2 seconds under normal network conditions.
 - **Search & Filters:** Users can quickly locate positions, programs or applicants using dynamic search and filtering by department, GPA and status.
 - **Automation:** Repetitive tasks (e.g., duplicate application checks, student notifications on status changes, stipend record creation) will be automated where possible.
-

(5) Dependability/Memorability

The system shall ensure users can easily recall how to perform tasks after periods of inactivity:

- **Consistent Layout:** Key features (e.g., application forms, position status updates) will maintain the same placement across sessions.
 - **Visual Cues:** Icons and labels will be standardized (e.g., a bell icon always represents notifications, a document icon always represents uploaded files).
 - **Saved Preferences:** User-specific settings and profile data will persist between logins.
 - **Task History:** Recently submitted applications and notifications will be highlighted for quick reference.
 - **Error Recovery:** Clear instructions will guide users to correct mistakes (e.g., "GPA does not meet the minimum requirement for this position").
-

(6) Errors

The system shall minimize user errors and provide clear recovery paths:

1. **Preventive Design:** Input validation will block invalid entries (e.g., GPA values outside the 0.0–4.0 range, unsupported file formats for document uploads).
 2. **Descriptive Messages:** Errors will explain the issue and suggest fixes (e.g., "You have already applied for this position", "Profile must be complete before applying").
 3. **Undo Actions:** Critical actions (e.g., withdrawing an application) will include confirmation dialogs before proceeding.
 4. **Logging:** All errors (user-facing and system) will be logged for troubleshooting, with no sensitive data exposed.
 5. **Graceful Degradation:** If a feature fails (e.g., file upload), the system will retain the user's progress and guide them to alternative steps.
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(7) Satisfaction

The Lab Management System is designed to be simple and easy to use for everyone, from students browsing positions to administrators managing lab compliance. The interface is intuitive, so users can quickly find what they need without spending too much time learning the system.

Everything is set up to make daily tasks smoother, whether it is applying for a lab position, reviewing applicants, or managing funding requests. Since the system works on both computers and mobile devices, users can access it anytime.

Overall, the goal is to create a hassle-free experience that helps university staff work efficiently and keeps students informed and engaged throughout the process.

(8) Capacity

The Lab Management System will be developed to meet all the operational needs of a university research environment. The system will function simultaneously for students, professors, HR staff and administrators. It will operate in real-time, ensuring that any updates or changes are immediately reflected for all authorized users.

All users will interact with the same database. If multiple requests are made to the server, they will be processed in a queue, which might cause slight delays. The application will be hosted on a web server. The system itself is lightweight and the database storage requirements are minimal, ensuring efficient performance.

ii) Organizational Requirements

(1) Availability

The system should be available 24/7 to support student applications and university operations. Regular backups should ensure data integrity and prevent data loss.

(2) Latency

Response time for critical operations (e.g., submitting applications, uploading documents, processing stipends) should be under 2 seconds.

(3) Monitoring

The system should include a dashboard for monitoring system performance, including uptime, user activity and application trends. Administrators should receive alerts for system failures, unauthorized access or unusual activity.

(4) Maintenance

Regular software updates should be scheduled to ensure security and performance improvements. System logs should be maintained to track user activity and troubleshoot issues.

(5) Operations

The system is designed to be intuitive, ensuring ease of use. Users returning after a long period can quickly re-establish proficiency. The interface includes icons, buttons and descriptions to enhance usability.

(6) Standards Compliance

The system is a newly developed platform designed to digitize university lab position and international program management. It must comply with data protection regulations and university administrative standards. Student and staff information will include all required fields according to academic management best practices. Every operation will be in accordance with data protection laws and standard university procedures.

(7) Portability

- The system will be a web-based application, ensuring it operates consistently across all operating systems.
- A computer or mobile phone with an internet connection is all that is required to use the system.
- The system will be accessible from the most commonly used browsers and will provide an optimal user experience across different devices.

iii) External Requirements

(1) Protection

To protect the system from unauthorized access, malicious modifications, disclosure, destruction or misuse, the following measures will be taken:

- Encrypting the most sensitive data, such as passwords, using hashing methods to maintain privacy.
- Logging and auditing user activities to ensure accountability in case of any issues or misuse.
- Validating all input data to prevent special characters and potential cyberattacks.
- Requiring user confirmation through pop-up windows for critical actions to avoid accidental mistakes.
- Each student will only be able to view their own personal information, applications and documents.

- Professors and staff will have limited access based on their roles to ensure the privacy and security of student data.

(2) Authorization and Authentication

- User authentication will use username, password and JWT tokens to enhance security.
- Authorization will be based on user roles, ensuring that each user can access only the information and functionalities assigned to their role.
- Sessions will be used to manage the currently logged-in user.
- If the user tries to log in with wrong credentials, an error message will be displayed.

(3) Legislative Requirements

The system must comply with applicable data protection legislation governing the storage and processing of personal data belonging to students and academic staff. All changes to application statuses, funding decisions and stipend records must be traceable through system logs, retaining before and after values for audit purposes. Uploaded documents and personal profile data must be accessible only to authorized roles as defined by the system's access control policy.

c) Domain Requirements

This web application operates in the field of university research and academic administration. The main purpose is to digitize and automate lab position management, student applications, international program coordination and research funding, making staff coordination more efficient.

Data security and access control are of utmost importance. Sensitive information such as student profiles, GPA records, uploaded documents and funding data must be protected and accessible only to authorized users. The application will ensure that each user has role-based access, allowing them to interact only with the relevant features.

This system is intended for use within a specific university and does not need to integrate with external systems. The web application will be responsive, ensuring accessibility across different devices, and it will implement modern authentication and authorization methods to guarantee data confidentiality and secure transactions.

4) User Scenarios/Use Cases

4.1 Requirements Analysis

4.1.1 User Scenarios List

Nr	Name	Description
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Undergraduate Lab Portal Requirements Specification

US-01	User Logs In	Student, Professor, HR Staff, and Admin log in using username and password.
US-02	Register an Account	Student registers by entering username, email, password, and role.
US-03	Manage Profile	Student creates and updates their research profile with GPA, major, and skills.
US-04	Browse Open Positions	Student browses all open lab and research positions available on the platform.
US-05	Filter Positions	Student filters available positions by department and minimum GPA.
US-06	Apply for a Position	Student with a complete profile applies for an open lab position.
US-07	View My Applications	Student views all submitted applications and their current statuses.
US-08	Withdraw Application	Student withdraws a pending or under-review application.
US-09	Upload Documents	Student uploads documents such as resume, transcript, or cover letter.
US-10	Browse External Programs	Student browses published international and scholarship programs.
US-11	Apply for a Program	Student applies for a published external program if GPA requirements are met.
US-12	View Program Applications	Student views their submitted program applications and statuses.
US-13	View Notifications	Student, Professor, HR Staff, Admin view in-app notifications.
US-14	Mark Notification as Read	User marks a specific notification as read.
US-15	View Lab Membership	Student views their current and past lab memberships.
US-16	Create a Lab Position	Professor creates a new open research or lab assistant position.
US-17	View My Positions	Professor views all positions they have posted.
US-18	Update Position Status	Professor closes or marks a position as filled.
US-19	View Applicants	Professor views all applicants for a specific position.
US-20	Review Applicant Detail	Professor views a specific applicant's profile and uploaded documents.
US-21	Accept or Reject Application	Professor accepts or rejects a student application.
US-22	View Lab Roster	Professor views the full roster of members in their lab.
US-23	Add Lab Member	Professor manually adds a student to their lab.
US-24	Remove Lab Member	Professor removes a student from their lab.

Undergraduate Lab Portal Requirements Specification

US-25	Submit Funding Request	Professor submits a funding request for their lab.
US-26	View Funding Requests	Professor views all their submitted funding requests and statuses.
US-27	Create International Program	HR Staff creates a new international or scholarship program.
US-28	Publish a Program	HR Staff publishes a draft program making it visible to students.
US-29	Update Program	HR Staff updates the details of an existing program.
US-30	View Program Applicants	HR Staff views all student applications for a program.
US-31	Decide Program Application	HR Staff accepts or rejects a student's program application.
US-32	Close a Program	HR Staff closes a published program, preventing further applications.
US-33	View Research Statistics	Admin views aggregate research statistics across the institution.
US-34	View All Labs	Admin views all labs with their certification and activity status.
US-35	Certify or Decertify Lab	Admin certifies or decertifies a lab with an optional note.
US-36	View Funding Requests	Admin views all pending funding requests submitted by professors.
US-37	Process Stipend	Admin approves, processes, or rejects a student stipend.

4.1.2 User Scenarios Extended

US-01: User Logs In

1. User navigates to the login page.
2. User enters username and password.
3. System validates credentials.
4. System issues a JWT token and grants access according to user role.
5. User is redirected to their role-specific dashboard.
6. If credentials are invalid: system displays an error message and prompts the user to try again (HTTP 401).

US-02: Register an Account

1. User navigates to the Register page.
2. User enters username, email, password, and selects role.
3. Frontend validates password strength (8+ characters, uppercase, lowercase, digit).
4. System creates the user account.

5. User is redirected to the Login page.
 6. If username or email already exists: system returns HTTP 409.
-

US-03: Manage Profile

1. Student navigates to My Profile.
 2. Student fills in full name, GPA, major, and skills.
 3. Student saves the profile.
 4. System marks profile as `is_complete = true` when all required fields are filled.
 5. Student can return and update profile information at any time.
-

US-04: Browse Open Positions

1. Student logs in and navigates to Browse Opportunities.
 2. System sends request to API to retrieve all positions with status OPEN.
 3. System displays the list of available positions with title, lab, department, and required GPA.
-

US-05: Filter Positions

1. Student is on the Browse Opportunities page.
 2. Student selects filter criteria: department and/or minimum GPA.
 3. System sends filtered request to API.
 4. System displays matching OPEN positions based on applied filters.
-

US-06: Apply for a Position

1. Student browses open positions and selects one.
 2. Student clicks Apply.
 3. System checks that student profile is complete (`is_complete = true`).
 4. System checks for duplicate application for the same position.
 5. System creates an Application record with status SUBMITTED.
 6. System sends an in-app notification to the professor.
 7. If profile incomplete: application blocked.
 8. If duplicate: system returns HTTP 409.
-

US-07: View My Applications

1. Student navigates to My Applications.
 2. System retrieves all applications submitted by the student.
 3. System displays each application with position title, lab, and current status (SUBMITTED, UNDER_REVIEW, ACCEPTED, REJECTED, WITHDRAWN).
-

US-08: Withdraw Application

1. Student navigates to My Applications.
 2. Student selects an application in SUBMITTED or UNDER_REVIEW status.
 3. Student clicks Withdraw.
 4. System updates application status to WITHDRAWN.
 5. If application is already decided: withdrawal is not permitted.
-

US-09: Upload Documents

1. Student navigates to their profile or documents section.
 2. Student selects a file and chooses document type (RESUME, TRANSCRIPT, COVER_LETTER, RECOMMENDATION, OTHER).
 3. Student submits the upload.
 4. System validates file format and size.
 5. System stores the file and creates a Document record linked to the student.
 6. If format or size is invalid: system returns HTTP 400.
-

US-10: Browse External Programs

1. Student navigates to the External Programs section.
2. System retrieves all programs with status PUBLISHED.
3. System displays programs with title, description, deadline, and minimum GPA requirement.

US-11: Apply for a Program

1. Student selects a published program and clicks Apply.
2. System checks that student profile is complete.
3. System validates that student GPA meets the program's minimum GPA requirement.
4. System checks for duplicate application for the same program.
5. System creates an Application record with status SUBMITTED.

6. If GPA too low: system returns HTTP 400.
 7. If duplicate: system returns HTTP 409.
-

US-12: View Program Applications

1. Student navigates to External Programs and selects My Applications.
 2. System retrieves all program applications submitted by the student.
 3. System displays each application with program name and current status.
-

US-13: View Notifications

1. User clicks the bell icon in the navigation sidebar.
 2. Notification panel opens to the right of the sidebar without overlapping it.
 3. System retrieves all notifications for the logged-in user.
 4. System displays notifications with unread badge count.
-

US-14: Mark Notification as Read

1. User opens the notification panel.
 2. User clicks on an unread notification.
 3. System marks the notification as read.
 4. System decrements the unread badge count.
-

US-15: View Lab Membership

1. Student navigates to My Profile or the Lab tab.
2. System retrieves all lab memberships associated with the student.
3. System displays lab name, assigned role, and join date for each membership.

US-16: Create a Lab Position

1. Professor navigates to Manage Positions.
2. Professor clicks Add Position.
3. Professor fills in the form: title, description, required GPA, associated lab, and optional stipend amount.
4. Professor submits the form.

Undergraduate Lab Portal Requirements Specification

5. System creates a Position record with status OPEN.
 6. If stipend amount is greater than zero, system automatically creates a Stipend record with status PENDING.
-

US-17: View My Positions

1. Professor navigates to Manage Positions.
 2. System retrieves all positions created by the professor.
 3. System displays each position with title, status, and associated lab.
-

US-18: Update Position Status

1. Professor navigates to Manage Positions and selects a position.
 2. Professor chooses a new status: CLOSED or FILLED.
 3. System updates the position status accordingly.
 4. Professors can only update positions they own.
-

US-19: View Applicants for a Position

1. Professor selects a position from Manage Positions.
 2. System retrieves all applications submitted for that position.
 3. System displays applicant list with student name, GPA, major, and application status.
-

US-20: Review Applicant Detail

1. Professor clicks on a specific applicant from the list.
2. System retrieves the full application detail.
3. System displays student profile information and all uploaded documents.

US-21: Accept or Reject Application

1. Professor reviews an applicant in SUBMITTED or UNDER_REVIEW status.
2. Professor selects ACCEPTED or REJECTED and optionally adds decision notes.
3. System updates the application status and records the decision timestamp.
4. If ACCEPTED: system automatically adds the student to the lab as a LabMember and notifies the student.

5. If REJECTED: system notifies the student.
-

US-22: View Lab Roster

1. Professor navigates to Lab Management.
 2. System retrieves the full roster of the professor's lab.
 3. System displays each member's name, GPA, major, assigned role, and join date.
-

US-23: Add Lab Member

1. Professor navigates to Lab Management.
 2. Professor enters a student's username or student ID and assigns a role.
 3. Professor submits the form.
 4. System creates a LabMember record for the student in the specified lab.
-

US-24: Remove Lab Member

1. Professor navigates to Lab Management and views the roster.
 2. Professor selects a lab member and clicks Remove.
 3. System deletes the LabMember record.
 4. Application history for the student remains unaffected.
-

US-25: Submit Funding Request

1. Professor navigates to Request Funding.
2. Professor fills in the form: amount, purpose, and associated lab.
3. Professor submits the request.
4. System creates a FundingRequest record with status PENDING awaiting admin review.

US-26: View Funding Requests

1. Professor navigates to Request Funding.
2. System retrieves all funding requests submitted by the professor.
3. System displays each request with amount, purpose, and current status (PENDING, APPROVED, REJECTED).

US-27: Create International Program

1. HR Staff navigates to Program Management.
 2. HR Staff clicks Add Program.
 3. HR Staff fills in the form: title, description, application deadline, and minimum GPA.
 4. HR Staff submits the form.
 5. System creates a Program record with status DRAFT, not yet visible to students.
-

US-28: Publish a Program

1. HR Staff navigates to Program Management and selects a DRAFT program.
 2. HR Staff clicks Publish.
 3. System updates the program status to PUBLISHED.
 4. Program becomes visible to all students on the Browse External Programs page.
-

US-29: Update Program

1. HR Staff navigates to Program Management and selects a program.
 2. HR Staff edits the relevant fields (title, description, deadline, minimum GPA).
 3. HR Staff saves the changes.
 4. System updates the program record.
 5. Updates are not permitted if the program is already CLOSED.
-

US-30: View Program Applicants

1. HR Staff navigates to Program Management and selects a program.
2. System retrieves all student applications for that program.
3. System displays applicants with full student profile details including GPA, major, skills, and documents.

US-31: Decide Program Application

1. HR Staff reviews a student application for a program.
2. HR Staff selects ACCEPTED or REJECTED.
3. System updates the application status.
4. System sends an in-app notification to the student informing them of the decision.

US-32: Close a Program

1. HR Staff navigates to Program Management and selects a PUBLISHED program.
 2. HR Staff clicks Close Program.
 3. System updates the program status to CLOSED.
 4. Students can no longer submit applications for this program.
-

US-33: View Research Statistics

1. Admin navigates to the Research Overview dashboard.
 2. System retrieves aggregate data from the database.
 3. System displays total number of labs, open positions, submitted applications, and total lab members.
-

US-34: View All Labs

1. Admin navigates to Lab Compliance.
 2. System retrieves all labs with their details.
 3. System displays each lab with certification status, professor name, and lab status (ACTIVE, INACTIVE, SUSPENDED).
-

US-35: Certify or Decertify Lab

1. Admin navigates to Lab Compliance and selects a lab.
 2. Admin toggles the certification status (certified or decertified).
 3. Admin optionally adds a certification note.
 4. System updates the lab's certification record.
-

US-36: View Funding Requests

1. Admin navigates to Funding & Stipends.
2. System retrieves all pending funding requests.
3. System displays each request with amount, purpose, associated lab, and professor name.

US-37: Process Stipend

1. Admin navigates to Funding & Stipends and selects a stipend in PENDING or APPROVED status.
2. Admin chooses a decision: APPROVED, PROCESSED, or REJECTED, and optionally adds notes.
3. System updates the stipend record.
4. System automatically sends an email notification to the student informing them of the outcome.

4.1.3 User Cases

Undergraduate Lab Portal Requirements Specification

UC-S-01: Register an Account

Field	Details
Use Case ID	UC-S-01
Use Case Name	Register an Account
Actors	• Student (Primary)
Dependencies	• None
Preconditions	• User does not already have an account • Registration page is accessible
Main Sequence	1. User navigates to and opens the Register page 2. User enters username, email, password, and selects role = STUDENT 3. Frontend validates password strength (min 8 characters, at least one uppercase, one lowercase, one digit) 4. User submits the registration form 5. System creates the new user account 6. User is redirected to the Login page
Alternative Sequences	1. Username or email already exists in the system → system returns HTTP 409 Conflict 2. Frontend validation fails (weak password, missing field) → inline error displayed, submission blocked 3. Backend is unavailable → generic error message shown to user
Non-Functional Requirements	• Password hashing must use <u>bcrypt</u> with a cost factor of at least 10 • Registration page must load within 2 seconds under normal load • System must handle concurrent registration attempts without race conditions • All form inputs must be sanitized to prevent SQL injection and XSS
<u>Postconditions</u>	• New user account created with role STUDENT • User is on the Login page

UC-S-02: Log In

Field	Details
Use Case ID	UC-S-02
Use Case Name	Log In
Actors	• Student (Primary)
Dependencies	• UC-S-01: Register an Account
Preconditions	• Student account exists in the system • Login page is accessible
Main Sequence	1. User opens the Login page 2. User enters their registered email/username and password 3. User submits the login form 4. Backend validates credentials against stored hashed password

Undergraduate Lab Portal Requirements Specification

Field	Details
	- Backend issues a signed JWT access token upon successful validation - Token is stored in the browser's <u>localStorage</u> - User is redirected to their Student Dashboard
Alternative Sequences	- Invalid credentials (wrong password or unknown email) → HTTP 401 Unauthorized; error message displayed - Account does not exist → HTTP 401 (generic message to avoid user enumeration) - Too many failed attempts → account temporarily locked for 15 minutes (brute-force protection)
Non-Functional Requirements	- JWT tokens must expire after 24 hours and must be signed with HS256 or RS256 - Login response time must be under 1 second under normal load - Passwords must never be logged or returned in any API response - HTTPS must be enforced on all authentication endpoints
<u>Postconditions</u>	- User is authenticated - JWT token stored in <u>localStorage</u> - User is on the Student Dashboard

UC-S-03: Manage Profile

Field	Details
Use Case ID	UC-S-03
Use Case Name	Manage Profile
Actors	Student (Primary)
Dependencies	UC-S-02: Log In
Preconditions	- Student is logged in - My Profile page is accessible
Main Sequence	- Student navigates to the My Profile page - System loads and displays existing profile data (if any) - Student fills in or updates full name, GPA, major, and skills - Student submits the profile form - System validates all required fields - System saves the profile; sets <u>is_complete</u> = true when all required fields are filled - Success confirmation is displayed to the student
Alternative Sequences	- Required fields are missing → validation error shown; profile saved as incomplete (<u>is_complete</u> = false) - GPA value is out of valid range (0.0–4.0) → inline validation error - Session has expired → user redirected to Login page
Non-Functional Requirements	- Profile save operation must complete within 2 seconds - GPA must be validated as a numeric value between 0.0 and 4.0 - Profile data must be persisted immediately upon save with no data loss on page refresh
<u>Postconditions</u>	- Profile updated and saved <u>is_complete</u> flag set to true if all required fields are filled

Undergraduate Lab Portal Requirements Specification

UC-S-04: Browse Open Positions

Field	Details
Use Case ID	UC-S-04
Use Case Name	Browse Open Positions
Actors	• Student (Primary)
Dependencies	• UC-S-02: Log In
Preconditions	• Student is logged in • At least one position with status OPEN exists
Main Sequence	1. Student navigates to the Browse Opportunities page 2. System sends GET /positions request with status filter = OPEN 3. Backend returns all positions with status OPEN 4. System renders the list of positions with title, lab name, required GPA, and stipend info 5. Student can scroll through and read position summaries
Alternative Sequences	1. No open positions exist → empty state message displayed ('No open positions available') 2. API call fails → error message shown with option to retry 3. Session has expired → user redirected to Login page
Non-Functional Requirements	• Position list must load within 2 seconds • Page must support pagination if results exceed 20 items • Position data must be up to date (max 30-second cache)
<u>Postconditions</u>	• List of all OPEN positions displayed to student

UC-S-05: Filter Positions

Field	Details
Use Case ID	UC-S-05
Use Case Name	Filter Positions
Actors	• Student (Primary)
Dependencies	• UC-S-04: Browse Open Positions
Preconditions	• Student is logged in • Position list is loaded on the Browse Opportunities page
Main Sequence	1. Student selects filter criteria from available options (department, minimum GPA) 2. Student applies the selected filters 3. Frontend sends GET /positions request with filter parameters 4. Backend returns only OPEN positions matching all specified filter criteria 5. System updates the position list display with filtered results
Alternative Sequences	1. No positions match the applied filters → empty state message displayed 2. Student clears all filters → full unfiltered list is restored 3. Invalid GPA filter value entered → validation message shown
Non-Functional Requirements	• Filter results must update within 1 second of submission • Multiple filters must be combinable (AND logic) • Filter state must persist if user navigates back to the list
<u>Postconditions</u>	• Position list filtered and updated to show only matching OPEN positions

Undergraduate Lab Portal Requirements Specification

UC-S-06: Apply for a Position

Field	Details
Use Case ID	UC-S-06
Use Case Name	Apply for a Position
Actors	• Student (Primary) • Professor (Notified)
Dependencies	• UC-S-02: Log In • UC-S-03: Manage Profile • UC-S-04: Browse Open Positions
Preconditions	• Student is logged in • Student profile <u>is_complete</u> = true • Target position has status OPEN
Main Sequence	1. Student browses to the desired position detail page 2. Student clicks the Apply button 3. System sends POST /applications with position ID and student ID 4. Service checks for duplicate application (same student + same position) 5. No duplicate found: system creates Application record with status SUBMITTED 6. System sends notification to the position's professor 7. Student is shown a confirmation message with application status
Alternative Sequences	1. Duplicate application exists → HTTP 409 Conflict; message: 'You have already applied for this position' 2. Student profile is incomplete → Apply button disabled with tooltip prompt to complete profile 3. Position is no longer OPEN → HTTP 400; message: 'This position is no longer accepting applications'
Non-Functional Requirements	• Application submission must be idempotent; duplicate check must be atomic • Professor notification must be delivered within 30 seconds • Application creation must complete within 2 seconds
<u>Postconditions</u>	• Application created with status SUBMITTED • Professor notified of new application

UC-S-07: View My Applications

Field	Details
Use Case ID	UC-S-07
Use Case Name	View My Applications
Actors	• Student (Primary)
Dependencies	• UC-S-06: Apply for a Position
Preconditions	• Student is logged in • Student has submitted at least one application
Main Sequence	1. Student navigates to the My Applications page 2. System sends GET /applications/mine 3. Backend returns all applications belonging to the authenticated student 4. System renders the application list showing position title, lab, date applied, and current status
Alternative Sequences	1. No applications exist → empty state message displayed 2. API error → error message with retry option
Non-Functional Requirements	• Application list must load within 2 seconds • Status indicators must be visually distinct (color-coded badges)
<u>Postconditions</u>	• All student applications displayed with current statuses

Undergraduate Lab Portal Requirements Specification

UC-S-08: Withdraw Application

Field	Details
Use Case ID	UC-S-08
Use Case Name	Withdraw Application
Actors	• Student (Primary)
Dependencies	• UC-S-06: Apply for a Position
Preconditions	• Student is logged in • Application exists with status SUBMITTED or UNDER_REVIEW
Main Sequence	1. Student opens My Applications page 2. Student clicks Withdraw on the target application 3. System displays a confirmation dialog 4. Student confirms withdrawal 5. System sends PATCH /applications/{id}/status with status = WITHDRAWN 6. Application status updated to WITHDRAWN 7. Application list refreshed to reflect the new status
Alternative Sequences	1. Application is already ACCEPTED or REJECTED → Withdraw option not shown 2. Student cancels the confirmation dialog → no change made 3. Concurrent status change by professor → HTTP 409; student shown updated status
Non-Functional Requirements	• Withdrawal must be immediate and reflected in the UI within 1 second • Withdrawn applications must remain visible in history (not deleted)
<u>Postconditions</u>	• Application status updated to WITHDRAWN

UC-S-09: Upload Documents

Field	Details
Use Case ID	UC-S-09
Use Case Name	Upload Documents
Actors	• Student (Primary)
Dependencies	• UC-S-02: Log In • UC-S-03: Manage Profile
Preconditions	• Student is logged in • Student profile exists
Main Sequence	1. Student navigates to the Documents section of their profile 2. Student clicks Upload and selects a file from their device 3. Student selects the document type (e.g., Resume, Transcript, Recommendation Letter) 4. Student submits the upload form 5. System sends POST /documents/upload as a multipart/form-data request 6. Service validates file format (allowed: PDF, DOCX) and file size (max 5 MB) 7. File is stored in secure storage; a Document record is created linked to the student 8. Success confirmation displayed; document appears in the Documents list
Alternative Sequences	1. Invalid file format → HTTP 400; message: 'Only PDF and DOCX files are accepted' 2. File exceeds size limit → HTTP 400; message: 'File must be under 5 MB' 3. Upload interrupted → error message with retry option; no partial record created
Non-Functional Requirements	• Uploaded files must be stored with a unique, non-guessable filename • File storage must be server-side; files must not be publicly accessible by direct URL • Upload endpoint must support files up to 5 MB without timeout • Virus/malware scanning should be performed on uploaded files
<u>Postconditions</u>	• File stored securely • Document record created in the database linked to student

Undergraduate Lab Portal Requirements Specification

UC-S-10: Browse External Programs

Field	Details
Use Case ID	UC-S-10
Use Case Name	Browse External Programs
Actors	• Student (Primary)
Dependencies	• UC-S-02: Log In
Preconditions	• Student is logged in • At least one program with status PUBLISHED exists
Main Sequence	1. Student navigates to the External Programs page 2. System sends GET /programs 3. Backend returns only programs with status PUBLISHED 4. System renders the list showing program title, deadline, and minimum GPA requirement
Alternative Sequences	1. No published programs → empty state message displayed 2. API failure → error with retry option
Non-Functional Requirements	• Program list must load within 2 seconds • Deadline dates must be displayed in the user's local <u>timezone</u>
<u>Postconditions</u>	• List of all PUBLISHED programs displayed to student

UC-S-11: Apply for a Program

Field	Details
Use Case ID	UC-S-11
Use Case Name	Apply for a Program
Actors	• Student (Primary) • HR Staff (Notified)
Dependencies	• UC-S-02: Log In • UC-S-03: Manage Profile • UC-S-10: Browse External Programs
Preconditions	• Program has status PUBLISHED • Student profile <u>is_complete</u> = true • Student GPA >= program <u>min_gpa</u>
Main Sequence	1. Student opens the desired program detail page 2. Student clicks Apply 3. System sends POST /programs/{id}/apply 4. Service validates student GPA against program <u>min_gpa</u> 5. Service checks for duplicate application 6. Application created with status SUBMITTED 7. HR Staff notified of new program application 8. Student shown confirmation message
Alternative Sequences	1. Student GPA below <u>min_gpa</u> → HTTP 400; message: 'You do not meet the minimum GPA requirement' 2. Duplicate application → HTTP 409 Conflict 3. Program has passed its deadline → Apply button disabled
Non-Functional Requirements	• GPA validation must be performed server-side regardless of client-side checks • Duplicate check must be atomic to prevent race conditions
<u>Postconditions</u>	• Program application created with status SUBMITTED • HR Staff notified

Undergraduate Lab Portal Requirements Specification

UC-S-12: View My Program Applications

Field	Details
Use Case ID	UC-S-12
Use Case Name	View My Program Applications
Actors	• Student (Primary)
Dependencies	• UC-S-11: Apply for a Program
Preconditions	• Student is logged in • Student has submitted at least one program application
Main Sequence	1. Student navigates to External Programs → My Applications tab 2. System sends GET /programs/mine/applications 3. Backend returns all program applications for the authenticated student 4. List rendered showing program name, date applied, and current status
Alternative Sequences	1. No program applications exist → empty state shown 2. API error → error message with retry
Non-Functional Requirements	• List must load within 2 seconds • Status must reflect the most recent decision by HR Staff
<u>Postconditions</u>	• All program applications displayed with current statuses

UC-S-13: View Notifications

Field	Details
Use Case ID	UC-S-13
Use Case Name	View Notifications
Actors	• Student (Primary)
Dependencies	• UC-S-02: Log In
Preconditions	• Student is logged in
Main Sequence	1. Student clicks the bell icon in the sidebar 2. Notification panel slides open to the right 3. System sends GET /notifications 4. Backend returns all notifications for the authenticated student 5. Notifications rendered in descending chronological order, with unread items visually highlighted 6. Unread badge count displayed on the bell icon
Alternative Sequences	1. No notifications exist → empty state message: 'You have no notifications' 2. API failure → error message shown inside the panel
Non-Functional Requirements	• Notification panel must open within 500ms • Unread badge count must be accurate and update in real time or on each panel open • Notifications must be retained for at least 30 days
<u>Postconditions</u>	• Notifications panel open with all notifications displayed

Undergraduate Lab Portal Requirements Specification

UC-S-14: Mark Notification as Read

Field	Details
Use Case ID	UC-S-14
Use Case Name	Mark Notification as Read
Actors	• Student (Primary)
Dependencies	• UC-S-13: View Notifications
Preconditions	• Student is logged in • At least one unread notification exists
Main Sequence	1. Student opens the notifications panel (UC-S-13) 2. Student clicks on an unread notification item 3. System sends PATCH /notifications/{id}/read 4. Backend marks the notification as read 5. Notification visual highlight removed 6. Unread badge count decremented by one
Alternative Sequences	1. Notification already marked as read → PATCH is idempotent; no error 2. Notification ID not found → HTTP 404
Non-Functional Requirements	• Read status update must be reflected in the UI immediately (optimistic update) • Badge count must never go below zero
<u>Postconditions</u>	• Notification marked as read • Unread badge count decremented

UC-S-15: View Lab Membership

Field	Details
Use Case ID	UC-S-15
Use Case Name	View Lab Membership
Actors	• Student (Primary)
Dependencies	• UC-S-02: Log In
Preconditions	• Student is logged in
Main Sequence	1. Student navigates to My Profile or the Lab tab 2. System sends GET /students/me/labs 3. Backend returns all lab memberships for the authenticated student 4. Memberships rendered showing lab name, role, and join date
Alternative Sequences	1. Student has no lab memberships → empty state message displayed 2. API error → error message with retry option
Non-Functional Requirements	• Lab membership list must load within 2 seconds • Data must be consistent with professor's lab roster view
<u>Postconditions</u>	• Lab memberships displayed with role and lab details

Professor Use Cases

UC-P-01: Update Professor Profile

Field	Details
Use Case ID	UC-P-01
Use Case Name	Update Professor Profile
Actors	• Professor (Primary)
Dependencies	• UC-S-02: Log In (Professor role)
Preconditions	• Professor is logged in with role PROFESSOR
Main Sequence	1. Professor navigates to their profile page 2. System checks if a Professor record exists; if not, system automatically creates one (get-or-create pattern) 3. Existing profile data loaded and displayed 4. Professor updates department and title fields 5. Professor submits the form via PATCH /professors/me 6. System saves the updated profile and confirms success
Alternative Sequences	1. Required fields left empty → validation error displayed 2. Session expired → redirected to Login page
Non-Functional Requirements	• Get-or-create operation must be atomic to prevent duplicate professor records • Profile update must complete within 2 seconds
<u>Postconditions</u>	• Professor profile created or updated with department and title

UC-P-02: Create a Lab Position

Field	Details
Use Case ID	UC-P-02
Use Case Name	Create a Lab Position
Actors	• Professor (Primary)
Dependencies	• UC-P-01: Update Professor Profile
Preconditions	• Professor is logged in • Professor profile exists • Lab exists and belongs to professor
Main Sequence	1. Professor navigates to Manage Positions and clicks Create Position 2. Professor fills in the position form: title, description, <u>required_gpa</u>, <u>lab_id</u>, <u>stipend_amount</u> 3. Professor submits via POST /positions 4. System creates a Position record with status OPEN 5. If <u>stipend_amount</u> > 0, system automatically creates a Stipend record with status PENDING 6. Success confirmation shown; professor redirected to position detail view
Alternative Sequences	1. Required fields missing → validation error; position not created 2. Invalid <u>lab_id</u> (lab not owned by professor) → HTTP 403 Forbidden 3. Negative stipend amount entered → validation error
Non-Functional Requirements	• Position and Stipend records must be created in a single atomic transaction • Required GPA must be validated as a value between 0.0 and 4.0
<u>Postconditions</u>	• Position created with status OPEN • Stipend record created (PENDING) if <u>stipend_amount</u> > 0

Undergraduate Lab Portal Requirements Specification

UC-P-03: View My Positions

Field	Details
Use Case ID	UC-P-03
Use Case Name	View My Positions
Actors	• Professor (Primary)
Dependencies	• UC-P-02: Create a Lab Position
Preconditions	• Professor is logged in
Main Sequence	1. Professor navigates to the Manage Positions page 2. System sends GET /positions/mine 3. Backend returns all positions created by the authenticated professor 4. List rendered showing title, lab, status, required GPA, and application count
Alternative Sequences	1. No positions created yet → empty state message displayed 2. API error → error message with retry option
Non-Functional Requirements	• Position list must load within 2 seconds • Application count per position must be accurate in real time
<u>Postconditions</u>	• All professor positions displayed with current statuses and counts

UC-P-04: Update Position Status

Field	Details
Use Case ID	UC-P-04
Use Case Name	Update Position Status
Actors	• Professor (Primary)
Dependencies	• UC-P-02: Create a Lab Position
Preconditions	• Professor is logged in • Position was created by this professor
Main Sequence	1. Professor navigates to a specific position in Manage Positions 2. Professor selects the new status from available options: CLOSED or FILLED 3. System sends PATCH /positions/{id}/status 4. Backend validates ownership of the position 5. Position status updated accordingly 6. If status set to FILLED or CLOSED, new applications can no longer be submitted
Alternative Sequences	1. Professor attempts to update a position they do not own → HTTP 403 Forbidden 2. Invalid status value → HTTP 400 3. Reverting from FILLED/CLOSED to OPEN is not permitted
Non-Functional Requirements	• Status change must propagate immediately; students must not be able to apply to CLOSED/FILLED positions • Status update must complete within 1 second
<u>Postconditions</u>	• Position status updated to CLOSED or FILLED as selected

Undergraduate Lab Portal Requirements Specification

UC-P-05: View Applicants for a Position

Field	Details
Use Case ID	UC-P-05
Use Case Name	View Applicants for a Position
Actors	• Professor (Primary)
Dependencies	• UC-P-02: Create a Lab Position • UC-S-06: Apply for a Position
Preconditions	• Professor is logged in • Position has at least one application
Main Sequence	1. Professor opens a position from their Manage Positions list 2. Professor clicks View Applicants 3. System sends GET /positions/{id}/applicants 4. Backend returns all applications for that position 5. List rendered showing student name, GPA, major, application date, and status
Alternative Sequences	1. No applicants exist → empty state message 2. Position not owned by professor → HTTP 403
Non-Functional Requirements	• Applicant list must load within 2 seconds • GPA and major data must be pulled from latest student profile
<u>Postconditions</u>	• Applicant list for the selected position displayed

UC-P-06: Review Applicant Detail

Field	Details
Use Case ID	UC-P-06
Use Case Name	Review Applicant Detail
Actors	• Professor (Primary)
Dependencies	• UC-P-05: View Applicants for a Position
Preconditions	• Professor is logged in • Applicant list is displayed • Application exists
Main Sequence	1. Professor clicks on an applicant in the applicant list 2. System sends GET /applications/{id} 3. Backend returns full application data including student profile and linked documents 4. System renders a detail view with: student name, GPA, major, skills, uploaded documents (with download links), and application date
Alternative Sequences	1. Application not found → HTTP 404 2. Professor does not own the associated position → HTTP 403 3. Student has no uploaded documents → documents section shows 'No documents uploaded'
Non-Functional Requirements	• Detail view must load within 2 seconds • Document download links must be authenticated (not public URLs) • Student personal data must only be accessible to the owning professor
<u>Postconditions</u>	• Full applicant detail view displayed including profile and documents

Undergraduate Lab Portal Requirements Specification

UC-P-07: Accept or Reject Application

Field	Details
Use Case ID	UC-P-07
Use Case Name	Accept or Reject Application
Actors	• Professor (Primary) • Student (Notified)
Dependencies	• UC-P-06: Review Applicant Detail
Preconditions	• Professor is logged in • Application is in status SUBMITTED or UNDER_REVIEW
Main Sequence	1. Professor reviews the applicant detail (UC-P-06) 2. Professor selects ACCEPTED or REJECTED and optionally enters review notes 3. Professor confirms the decision 4. System sends PATCH /applications/{id}/status with decision and notes 5. If ACCEPTED: system inserts a <u>LabMember</u> record linking the student to the lab; student is notified 6. If REJECTED: student is notified with the decision and any notes provided 7. Application status updated accordingly
Alternative Sequences	1. Application already ACCEPTED or REJECTED → cannot be changed; error shown 2. Professor does not own the position → HTTP 403 3. Notification delivery fails → decision still saved; retry notification queued
Non-Functional Requirements	• Decision and <u>LabMember</u> creation (if ACCEPTED) must be atomic • Student notification must be delivered within 30 seconds • Decision cannot be reversed once confirmed
<u>Postconditions</u>	• Application status updated • <u>LabMember</u> record created (if ACCEPTED) • Student notified

UC-P-08: View Lab Roster

Field	Details
Use Case ID	UC-P-08
Use Case Name	View Lab Roster
Actors	• Professor (Primary)
Dependencies	• UC-P-01: Update Professor Profile
Preconditions	• Professor is logged in • Lab exists and belongs to professor
Main Sequence	1. Professor navigates to Lab Management 2. Professor selects a lab 3. System sends GET /labs/{id}/roster 4. Backend returns all <u>LabMember</u> records for the selected lab 5. Roster rendered showing student name, GPA, major, and lab role
Alternative Sequences	1. Lab has no members → empty roster with option to add members 2. Lab not owned by professor → HTTP 403
Non-Functional Requirements	• Roster must load within 2 seconds • GPA and major data must reflect latest student profile
<u>Postconditions</u>	• Lab roster displayed with student details and roles

Undergraduate Lab Portal Requirements Specification

UC-P-09: Add Lab Member

Field	Details
Use Case ID	UC-P-09
Use Case Name	Add Lab Member
Actors	• Professor (Primary)
Dependencies	• UC-P-08: View Lab Roster
Preconditions	• Professor is logged in • Lab belongs to professor • Target student has an account
Main Sequence	1. Professor navigates to Lab Management → Roster 2. Professor enters the student's username or student ID and assigns a role 3. Professor submits via POST /labs/{id}/members 4. Backend validates the student exists and is not already a member 5. <u>LabMember</u> record created 6. Roster updated to show the new member
Alternative Sequences	1. Student username/ID not found → HTTP 404; message: 'Student not found' 2. Student already a member of this lab → HTTP 409 Conflict 3. Professor does not own the lab → HTTP 403
Non-Functional Requirements	• Duplicate membership check must be atomic • Roster update must be reflected immediately after addition
<u>Postconditions</u>	• <u>LabMember</u> record created • Student appears in the lab roster

UC-P-10: Remove Lab Member

Field	Details
Use Case ID	UC-P-10
Use Case Name	Remove Lab Member
Actors	• Professor (Primary)
Dependencies	• UC-P-08: View Lab Roster
Preconditions	• Professor is logged in • Lab belongs to professor • Target member exists in the roster
Main Sequence	1. Professor opens the Lab Roster page 2. Professor selects a member and clicks Remove 3. System displays a confirmation dialog 4. Professor confirms removal 5. System sends DELETE /labs/{id}/members/{ <u>member_id</u> } 6. <u>LabMember</u> record deleted 7. Roster refreshed to reflect the removal
Alternative Sequences	1. Professor cancels the confirmation dialog → no change 2. Member not found → HTTP 404 3. Professor does not own the lab → HTTP 403
Non-Functional Requirements	• Removal must be immediate and reflected in the roster within 1 second • Removed member's historical application records must not be deleted
<u>Postconditions</u>	• <u>LabMember</u> record deleted • Student no longer appears in the lab roster

Undergraduate Lab Portal Requirements Specification

UC-P-11: Submit Funding Request

Field	Details
Use Case ID	UC-P-11
Use Case Name	Submit Funding Request
Actors	• Professor (Primary) • Admin (Notified)
Dependencies	• UC-P-01: Update Professor Profile
Preconditions	• Professor is logged in • Professor profile exists • Associated lab exists
Main Sequence	1. Professor navigates to Request Funding 2. Professor fills in the funding form: amount, purpose description, and <u>lab_id</u> 3. Professor submits via POST /funding 4. System creates a <u>FundingRequest</u> record with status PENDING 5. Admin is notified of the new funding request 6. Professor shown confirmation with request ID
Alternative Sequences	1. Required fields missing → validation error 2. Invalid or unowned <u>lab_id</u> → HTTP 403 3. Amount is zero or negative → validation error
Non-Functional Requirements	• Funding request must be created within 2 seconds • Admin notification must be delivered within 30 seconds • Request amount must support values up to 2 decimal places
<u>Postconditions</u>	• <u>FundingRequest</u> created with status PENDING • Admin notified

UC-P-12: View My Funding Requests

Field	Details
Use Case ID	UC-P-12
Use Case Name	View My Funding Requests
Actors	• Professor (Primary)
Dependencies	• UC-P-11: Submit Funding Request
Preconditions	• Professor is logged in • Professor has submitted at least one funding request
Main Sequence	1. Professor navigates to the Request Funding page 2. System sends GET /funding/mine 3. Backend returns all funding requests submitted by the authenticated professor 4. List rendered showing purpose, amount, lab, date submitted, and current status
Alternative Sequences	1. No requests exist → empty state displayed 2. API error → error message with retry
Non-Functional Requirements	• List must load within 2 seconds • Status must reflect the latest Admin decision in real time
<u>Postconditions</u>	• All professor funding requests displayed with current statuses

HR Staff Use Cases

UC-HR-01: Create International Program

Field	Details
Use Case ID	UC-HR-01
Use Case Name	Create International Program
Actors	• HR Staff (Primary)
Dependencies	• UC-S-02: Log In (HR role)
Preconditions	• HR Staff is logged in with role HR_STAFF
Main Sequence	1. HR Staff navigates to the Programs management page and clicks Create Program 2. HR Staff fills in the program form: title, description, application deadline, <u>min_gpa</u> 3. HR Staff submits via POST /programs 4. System creates a Program record with status DRAFT 5. Success confirmation shown; HR Staff redirected to program detail view
Alternative Sequences	1. Required fields missing → validation error; program not created 2. Deadline date is in the past → validation error 3. <u>min_gpa</u> out of range → validation error
Non-Functional Requirements	• Program creation must complete within 2 seconds • DRAFT programs must not be visible to students
<u>Postconditions</u>	• Program created with status DRAFT; not yet visible to students

UC-HR-02: Publish a Program

Field	Details
Use Case ID	UC-HR-02
Use Case Name	Publish a Program
Actors	• HR Staff (Primary)
Dependencies	• UC-HR-01: Create International Program
Preconditions	• HR Staff is logged in • Program exists with status DRAFT
Main Sequence	1. HR Staff opens the program in the management page 2. HR Staff clicks Publish 3. System sends PATCH /programs/{id} with body {status: PUBLISHED} 4. System updates program status to PUBLISHED 5. Program becomes visible to all logged-in students on the Browse External Programs page
Alternative Sequences	1. Program is not in DRAFT status → HTTP 400; cannot publish a CLOSED program 2. HR Staff does not own the program → HTTP 403
Non-Functional Requirements	• Status change must propagate immediately; program must be <u>browseable</u> by students within 5 seconds • Publish action must be logged with timestamp and HR Staff user ID for audit purposes
<u>Postconditions</u>	• Program status = PUBLISHED • Program visible to students on Browse External Programs

Undergraduate Lab Portal Requirements Specification

UC-HR-03: Update Program

Field	Details
Use Case ID	UC-HR-03
Use Case Name	Update Program
Actors	• HR Staff (Primary)
Dependencies	• UC-HR-01: Create International Program
Preconditions	• HR Staff is logged in • Program is not CLOSED
Main Sequence	1. HR Staff navigates to the program detail view 2. HR Staff edits desired fields (title, description, deadline, <u>min_gpa</u>) 3. HR Staff submits via PATCH /programs/{id} 4. System validates updated fields 5. Program record updated and success confirmed
Alternative Sequences	1. Program is CLOSED → update not allowed; HTTP 400 2. Deadline set to a past date → validation error 3. Concurrent update conflict → HTTP 409; HR Staff asked to reload
Non-Functional Requirements	• Updates to PUBLISHED programs must be applied immediately and visible to students • All field updates must be validated server-side
<u>Postconditions</u>	• Program fields updated as specified

UC-HR-04: View Program Applicants

Field	Details
Use Case ID	UC-HR-04
Use Case Name	View Program Applicants
Actors	• HR Staff (Primary)
Dependencies	• UC-HR-01: Create International Program • UC-S-11: Apply for a Program
Preconditions	• HR Staff is logged in • Program has at least one application
Main Sequence	1. HR Staff opens the program detail view 2. HR Staff navigates to the Applicants tab 3. System sends GET /programs/{id}/applicants 4. Backend returns all applications with student details 5. List rendered showing student name, GPA, major, date applied, and current status
Alternative Sequences	1. No applicants yet → empty state message 2. API error → error message with retry
Non-Functional Requirements	• Applicant list must load within 2 seconds • GPA and profile data must reflect latest student-submitted info
<u>Postconditions</u>	• Full applicant list for the program displayed with student details

Undergraduate Lab Portal Requirements Specification

UC-HR-05: Decide Program Application

Field	Details
Use Case ID	UC-HR-05
Use Case Name	Decide Program Application
Actors	• HR Staff (Primary) • Student (Notified)
Dependencies	• UC-HR-04: View Program Applicants
Preconditions	• HR Staff is logged in • Application is in SUBMITTED or UNDER_REVIEW status
Main Sequence	1. HR Staff views the applicant list (UC-HR-04) and opens an application 2. HR Staff selects ACCEPTED or REJECTED 3. HR Staff optionally adds decision notes 4. HR Staff confirms decision via PATCH /programs/applications/{id}/status 5. System updates the application status 6. System notifies the student of the decision
Alternative Sequences	1. Application already decided → cannot be changed; HTTP 400 2. Notification delivery fails → decision saved; retry queued 3. Program is CLOSED → applications may still be decided
Non-Functional Requirements	• Decision must be irreversible once confirmed • Student notification must be delivered within 30 seconds • All decisions must be audit-logged
<u>Postconditions</u>	• Application status updated • Student notified of decision

UC-HR-06: Close a Program

Field	Details
Use Case ID	UC-HR-06
Use Case Name	Close a Program
Actors	• HR Staff (Primary)
Dependencies	• UC-HR-02: Publish a Program
Preconditions	• HR Staff is logged in • Program has status PUBLISHED
Main Sequence	1. HR Staff navigates to the program detail view 2. HR Staff clicks Close Program 3. System displays a confirmation dialog 4. HR Staff confirms 5. System sends PATCH /programs/{id} with body {status: CLOSED} 6. Program status updated to CLOSED 7. Students can no longer submit new applications to this program
Alternative Sequences	1. Program is already CLOSED → idempotent; no error 2. Program is in DRAFT → cannot close a draft; HTTP 400 3. HR Staff cancels confirmation → no change made
Non-Functional Requirements	• Closure must take effect immediately; POST /programs/{id}/apply must return HTTP 400 instantly after closure • Close action must be audit-logged
<u>Postconditions</u>	• Program status = CLOSED • No new applications can be submitted

Admin Use Cases

UC-A-01: View Research Statistics

Field	Details
Use Case ID	UC-A-01
Use Case Name	View Research Statistics
Actors	• Admin (Primary)
Dependencies	• UC-S-02: Log In (Admin role)
Preconditions	• Admin is logged in with role ADMIN
Main Sequence	1. Admin navigates to the Research Overview page 2. System sends GET /admin/stats 3. Backend aggregates and returns statistics: total labs, certified labs, open positions, total applications, active students, pending funding requests 4. Statistics displayed as aggregate counts and summary cards on the dashboard
Alternative Sequences	1. No data exists yet → all counts display as 0 2. API error → error message with retry option
Non-Functional Requirements	• Statistics must be computed from live data; max acceptable staleness is 60 seconds • Dashboard must load within 3 seconds even with large datasets • Only users with role ADMIN may access this endpoint
<u>Postconditions</u>	• Aggregate research statistics displayed on the Admin dashboard

UC-A-02: View All Labs

Field	Details
Use Case ID	UC-A-02
Use Case Name	View All Labs
Actors	• Admin (Primary)
Dependencies	• UC-S-02: Log In (Admin role)
Preconditions	• Admin is logged in
Main Sequence	1. Admin navigates to the Lab Compliance page 2. System sends GET /admin/labs 3. Backend returns all lab records across all professors 4. Labs rendered in a table showing lab name, owning professor, certification status, and last certification date
Alternative Sequences	1. No labs exist → empty state message 2. API error → error message with retry
Non-Functional Requirements	• Lab list must load within 3 seconds regardless of total lab count • List must support search and sorting by lab name, professor, and certification status
<u>Postconditions</u>	• All labs displayed with certification status

Undergraduate Lab Portal Requirements Specification

UC-A-03: Certify or Decertify Lab

Field	Details
Use Case ID	UC-A-03
Use Case Name	Certify or Decertify Lab
Actors	• Admin (Primary)
Dependencies	• UC-A-02: View All Labs
Preconditions	• Admin is logged in • Lab exists in the system
Main Sequence	1. Admin opens the Lab Compliance page 2. Admin finds the target lab and clicks the certification toggle 3. Admin enters an optional certification note 4. Admin confirms the action 5. System sends PATCH /admin/labs/{id}/certification with certified status and note 6. Lab record updated with new certification status and timestamp 7. Lab list refreshed to reflect the change
Alternative Sequences	1. Admin cancels the confirmation → no change 2. Lab not found → HTTP 404 3. Note exceeds max length → validation error
Non-Functional Requirements	• Certification change must be logged with timestamp and Admin user ID for audit purposes • Change must propagate immediately
<u>Postconditions</u>	• Lab certification status updated • Audit log entry created

UC-A-04: View Funding Requests

Field	Details
Use Case ID	UC-A-04
Use Case Name	View Funding Requests
Actors	• Admin (Primary)
Dependencies	• UC-P-11: Submit Funding Request
Preconditions	• Admin is logged in • At least one PENDING funding request exists
Main Sequence	1. Admin navigates to Funding & Stipends 2. System sends GET /funding/pending 3. Backend returns all <u>FundingRequest</u> records with status PENDING 4. Requests listed showing professor name, lab, purpose, amount requested, and submission date
Alternative Sequences	1. No pending requests → empty state displayed 2. API error → error message with retry
Non-Functional Requirements	• List must load within 2 seconds • Amounts must be displayed with two decimal places and currency symbol
<u>Postconditions</u>	• All pending funding requests displayed for Admin review

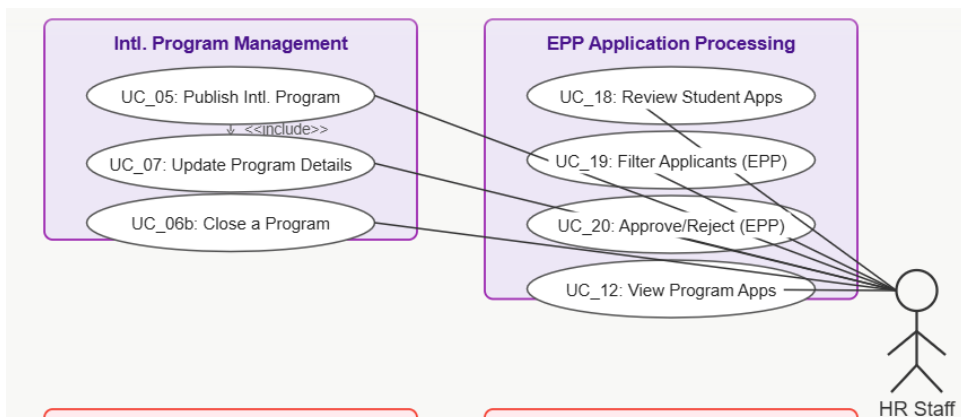
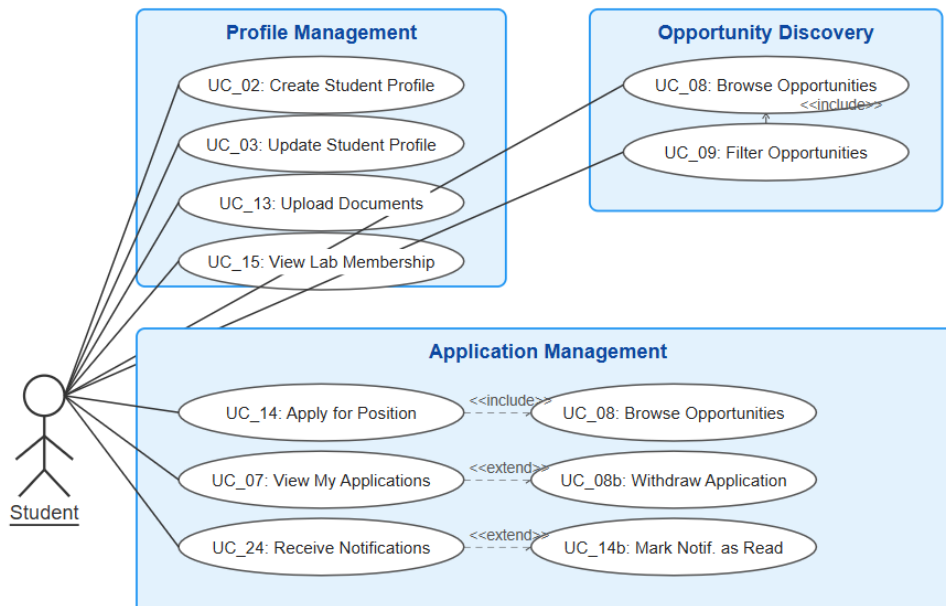
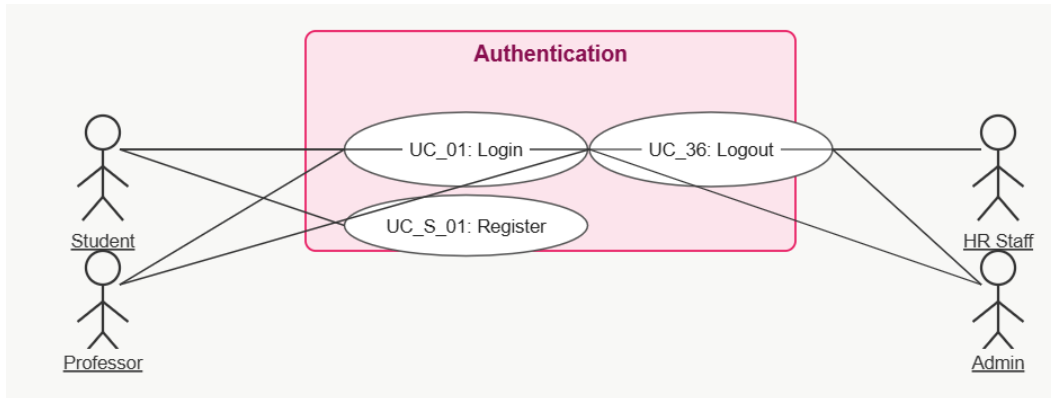
Undergraduate Lab Portal Requirements Specification

UC-A-05: Process Stipend

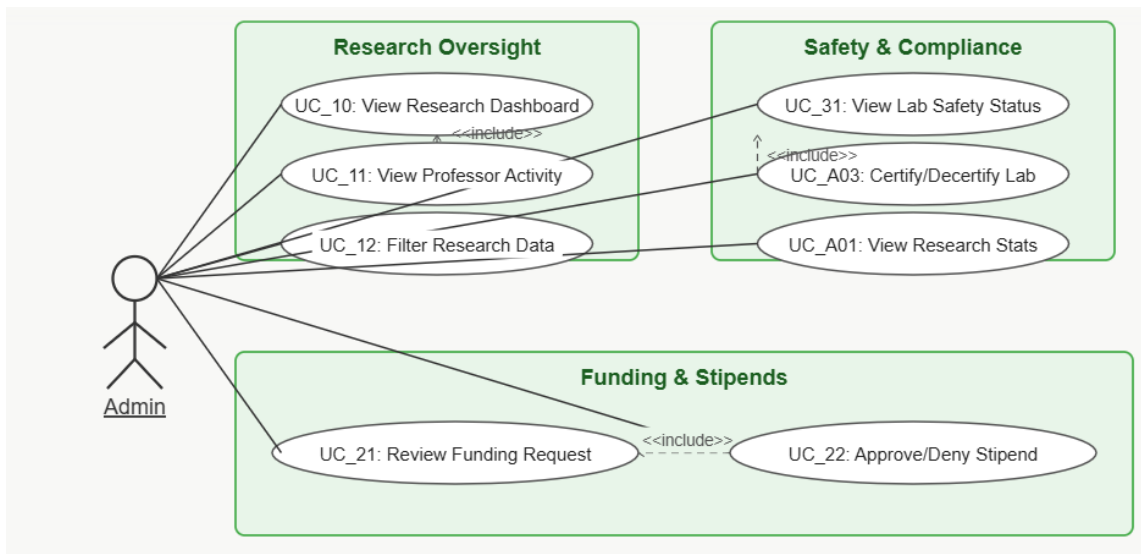
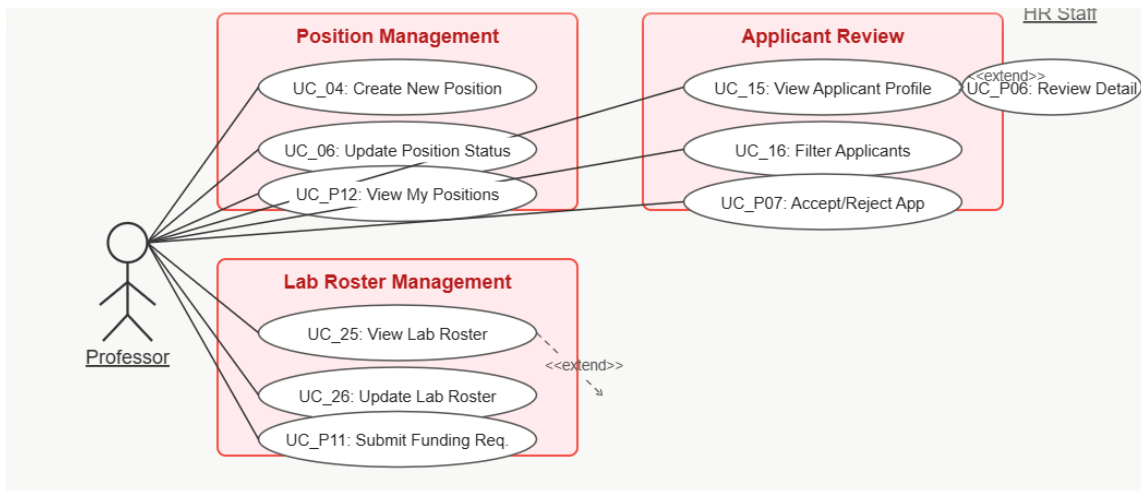
Field	Details
Use Case ID	UC-A-05
Use Case Name	Process Stipend
Actors	• Admin (Primary) • Student (Notified)
Dependencies	• UC-P-02: Create a Lab Position (with stipend) • UC-A-04: View Funding Requests
Preconditions	• Admin is logged in • Stipend exists with status PENDING or APPROVED
Main Sequence	1. Admin navigates to Funding & Stipends and selects a stipend 2. Admin reviews the stipend details (student, lab, amount) 3. Admin selects a decision: APPROVED, PROCESSED, or REJECTED 4. Admin optionally enters decision notes 5. Admin confirms via PATCH /stipends/{id} 6. Stipend status updated 7. Email service sends notification to the associated student
Alternative Sequences	1. Stipend already PROCESSED or REJECTED → cannot be changed; HTTP 400 2. Email service unavailable → decision saved; email retry queued 3. Admin enters notes exceeding max length → validation error
Non-Functional Requirements	• Decision and status update must be atomic • Email notification to student must be delivered within 5 minutes • All stipend decisions must be audit-logged with Admin user ID and timestamp
<u>Postconditions</u>	• Stipend status updated • Student notified via email

4.2 Behavioral Diagrams

4.2.1 Use Case Diagrams

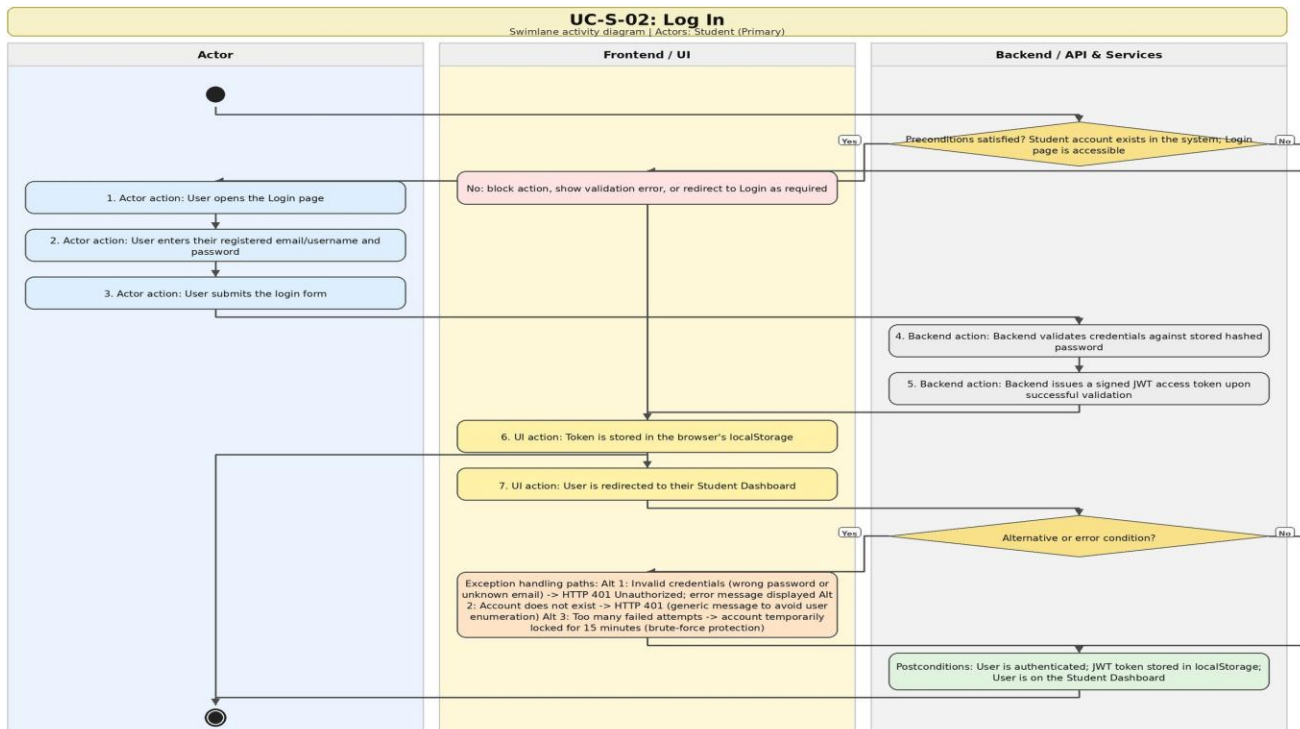
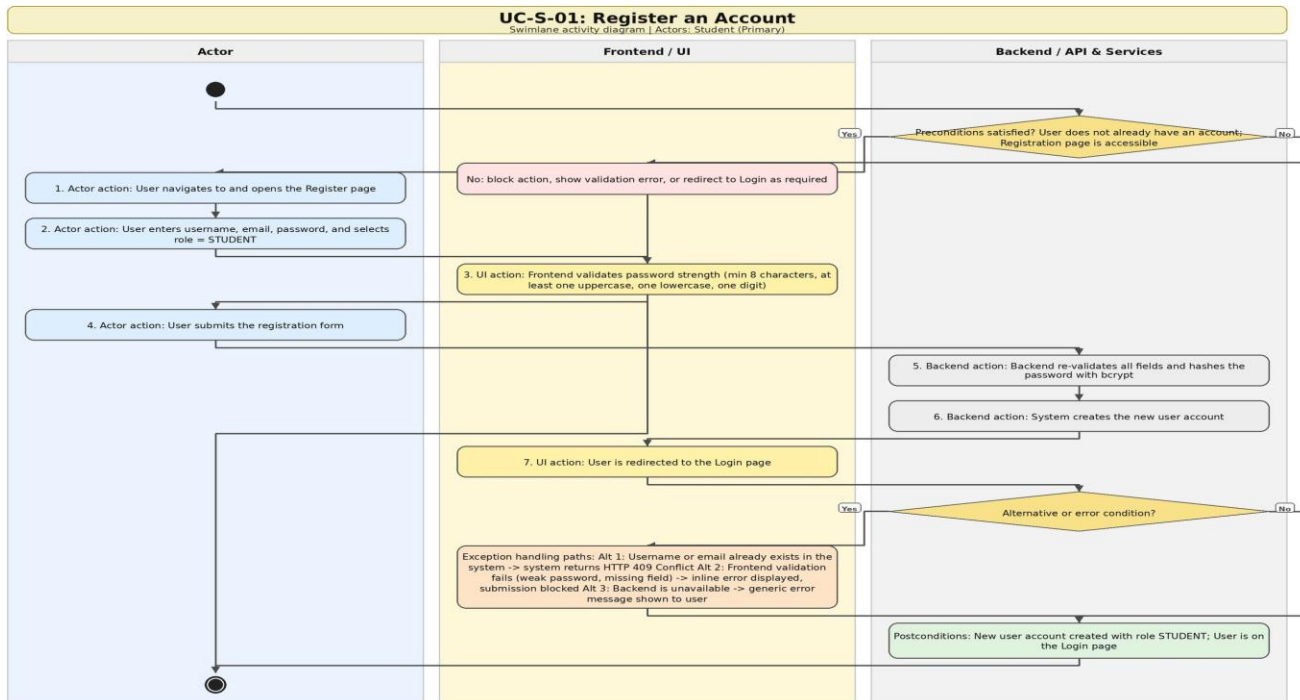


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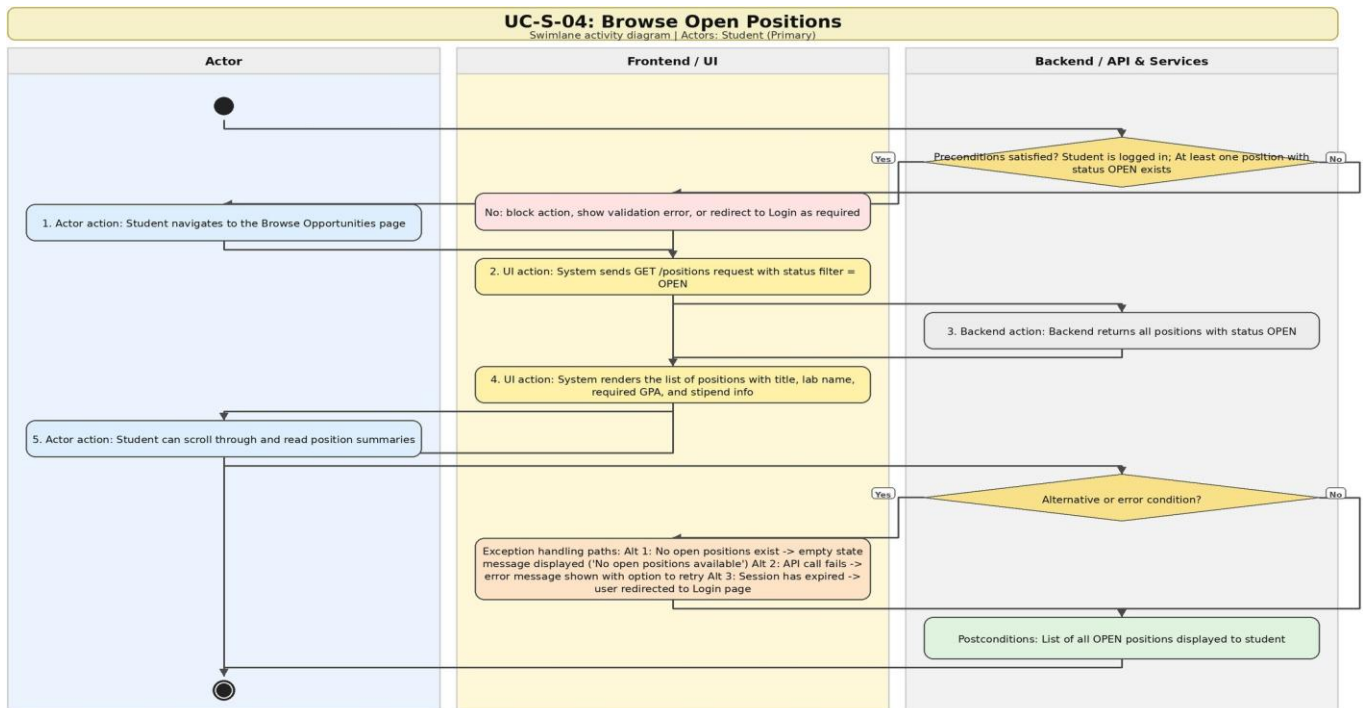
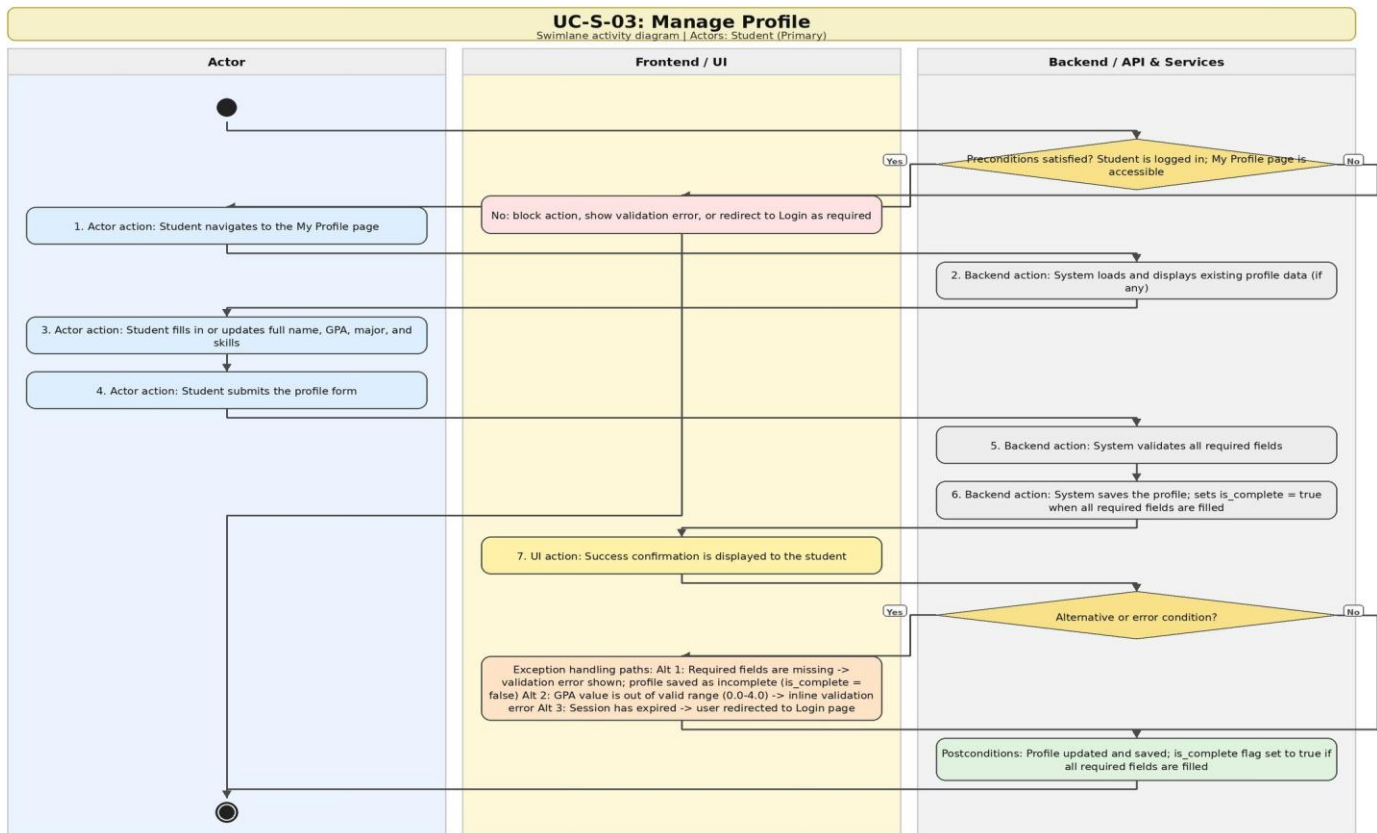


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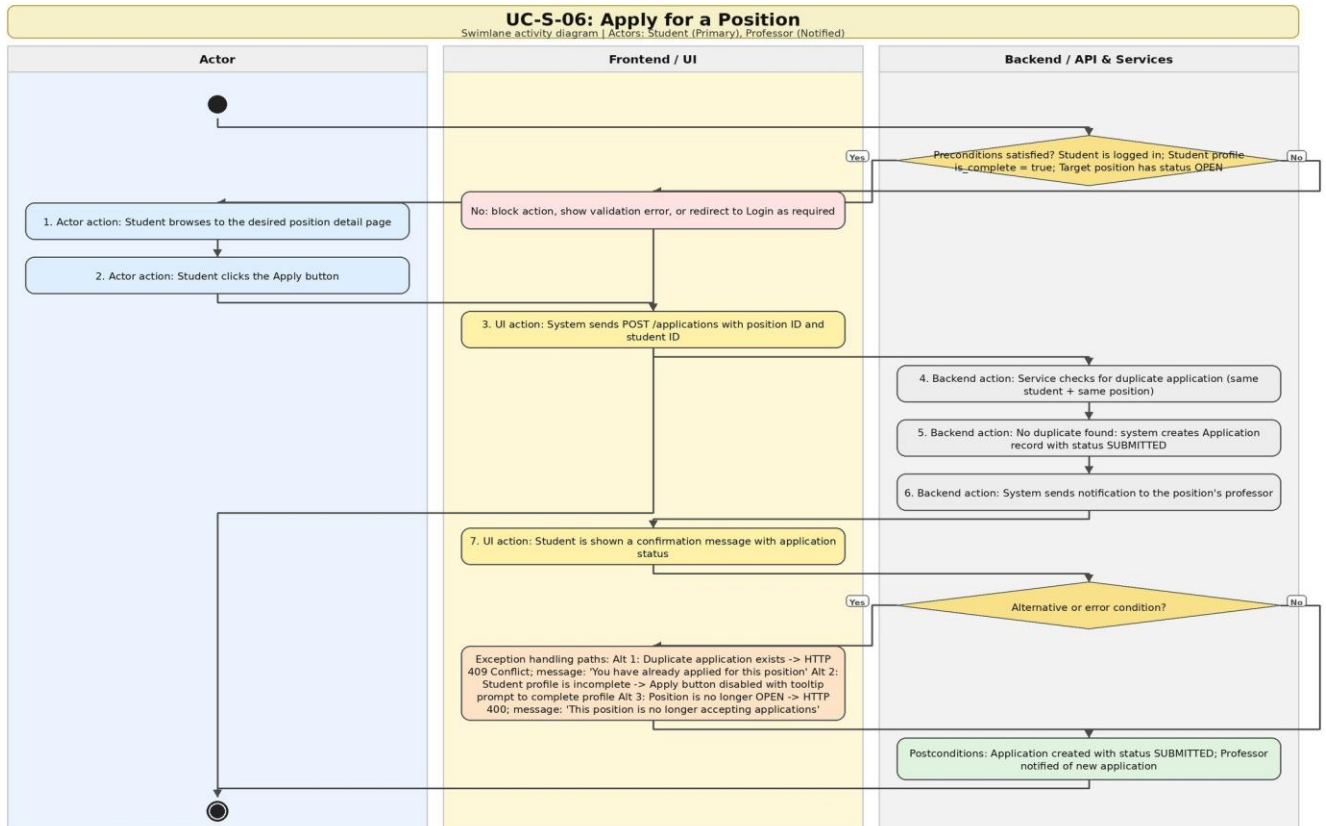
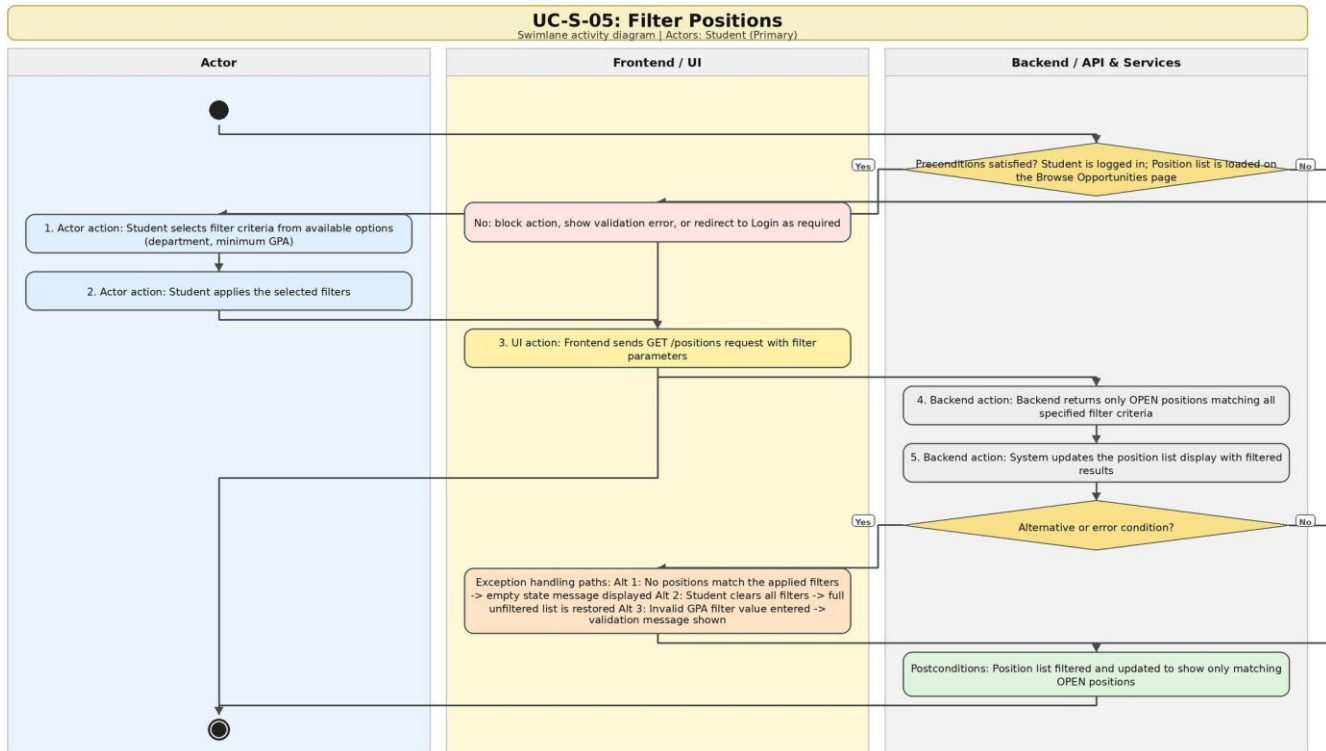
4.2.2 Activity Diagrams



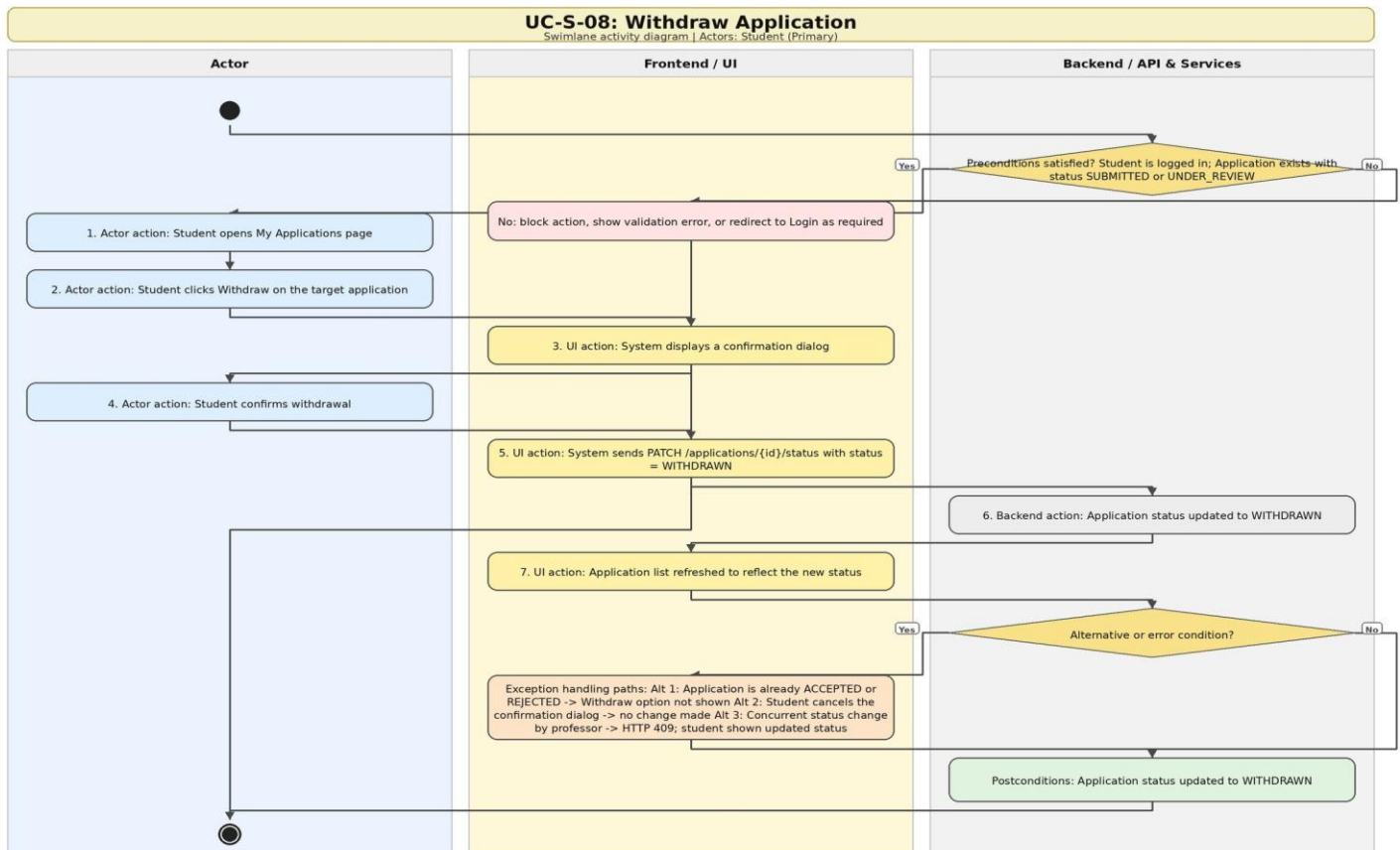
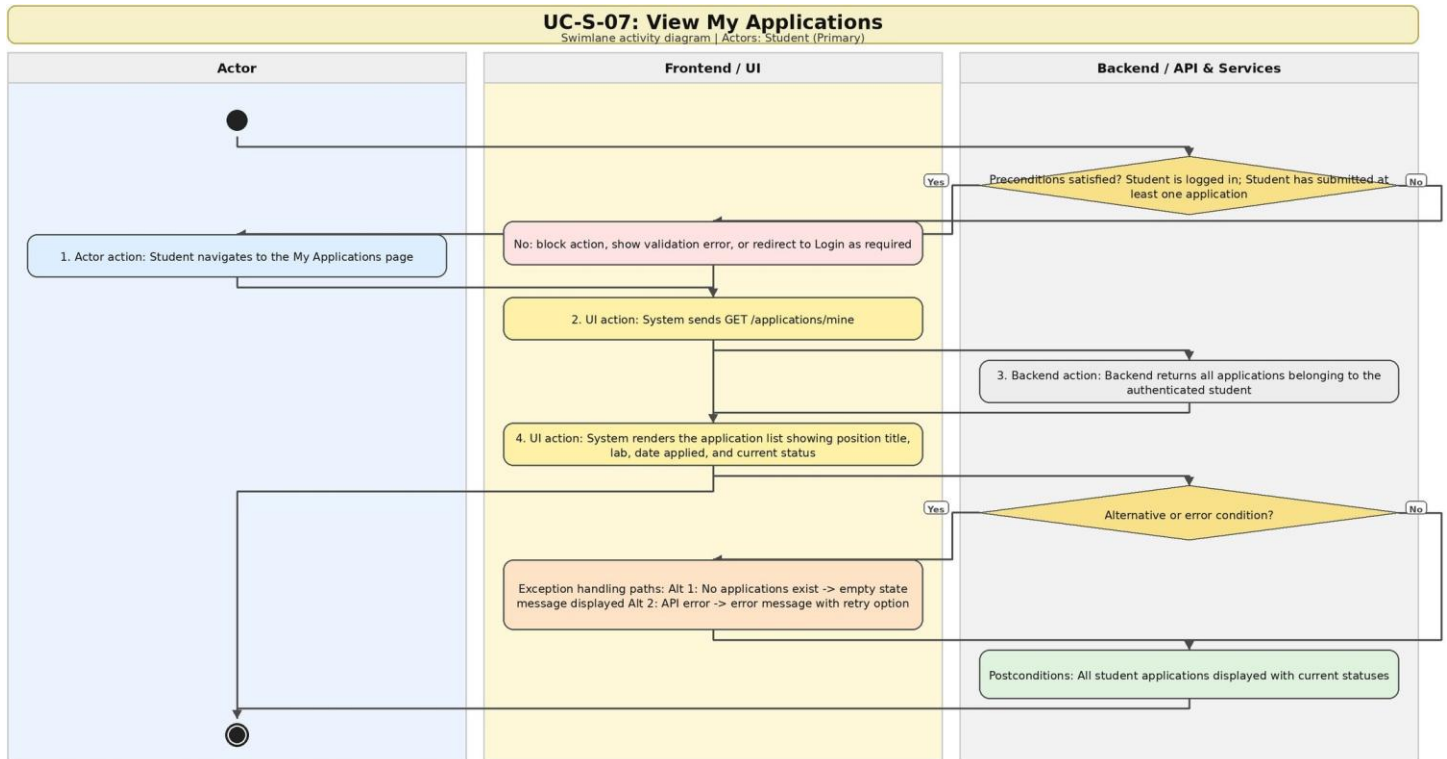
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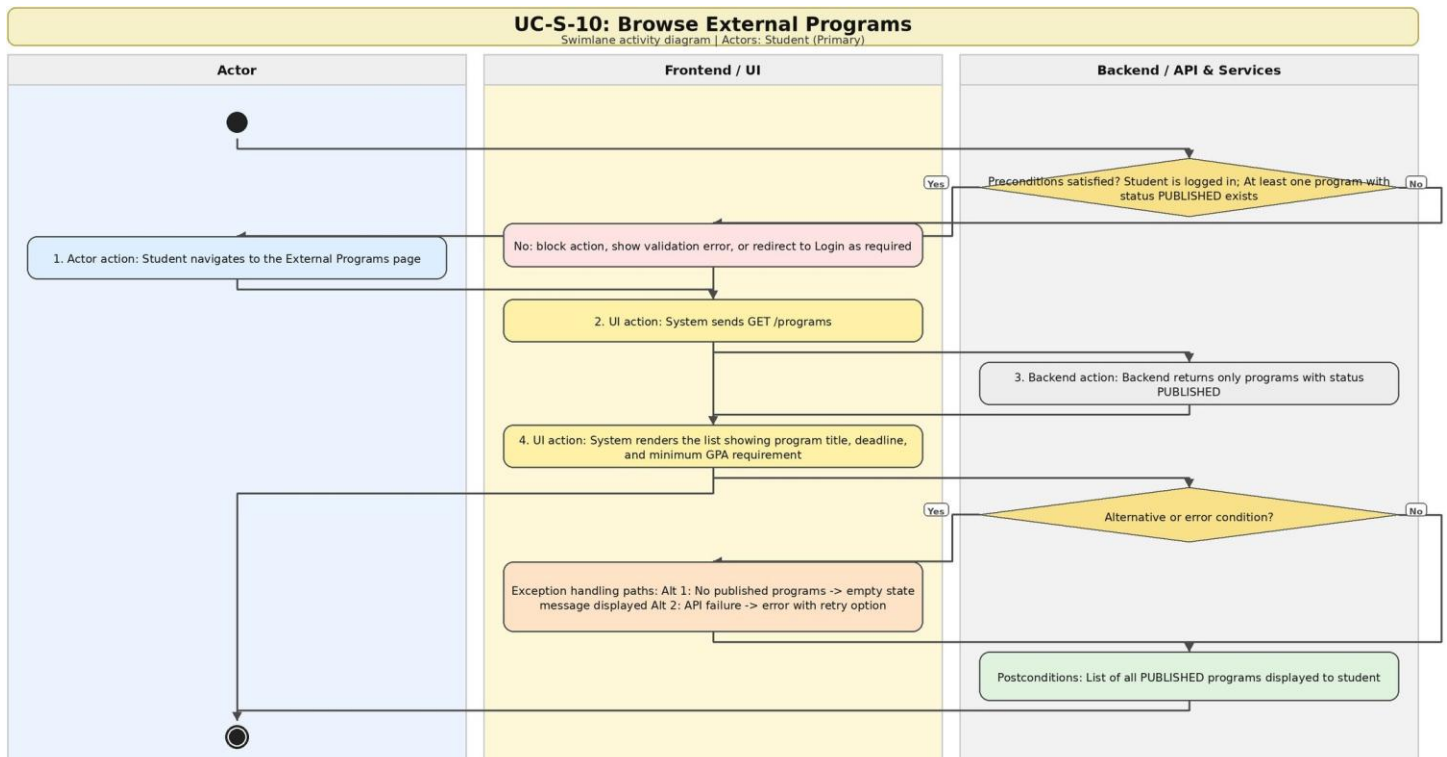
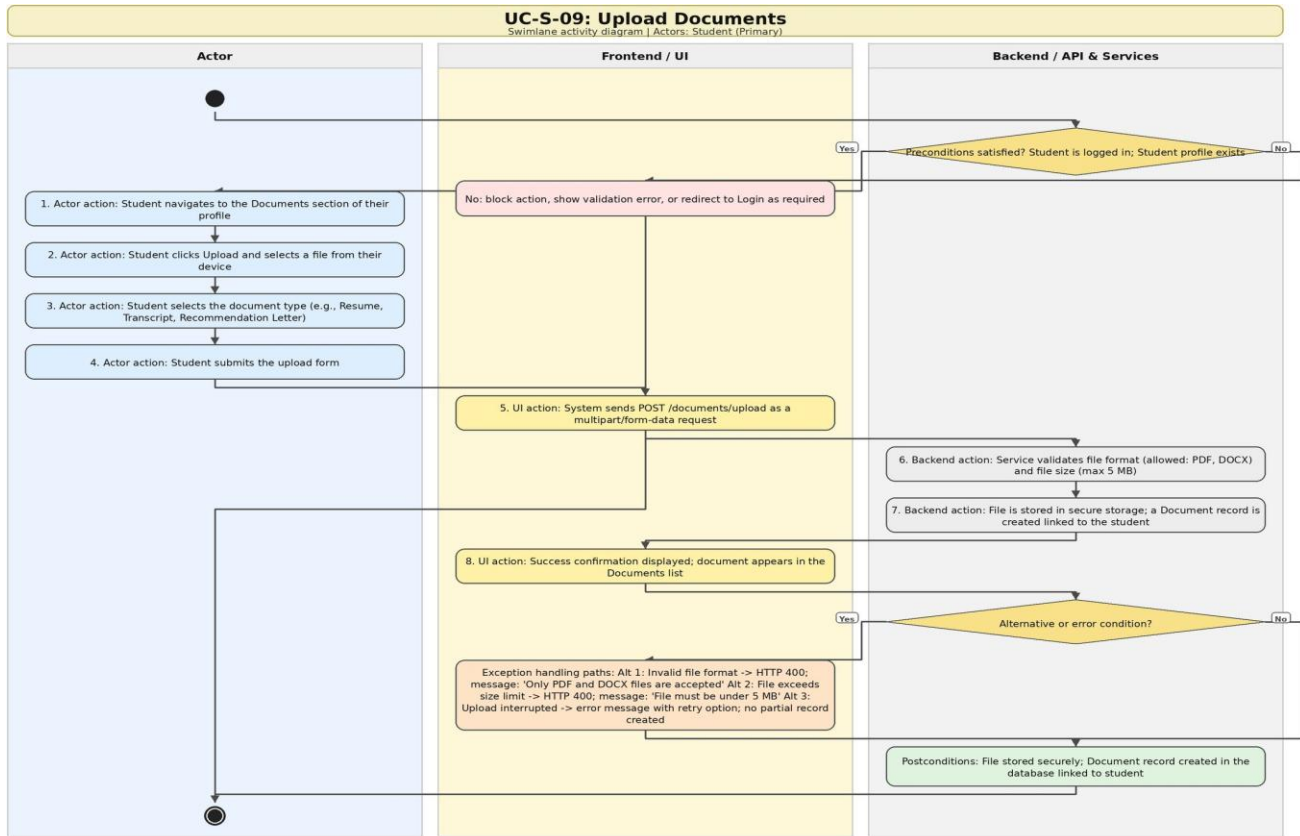
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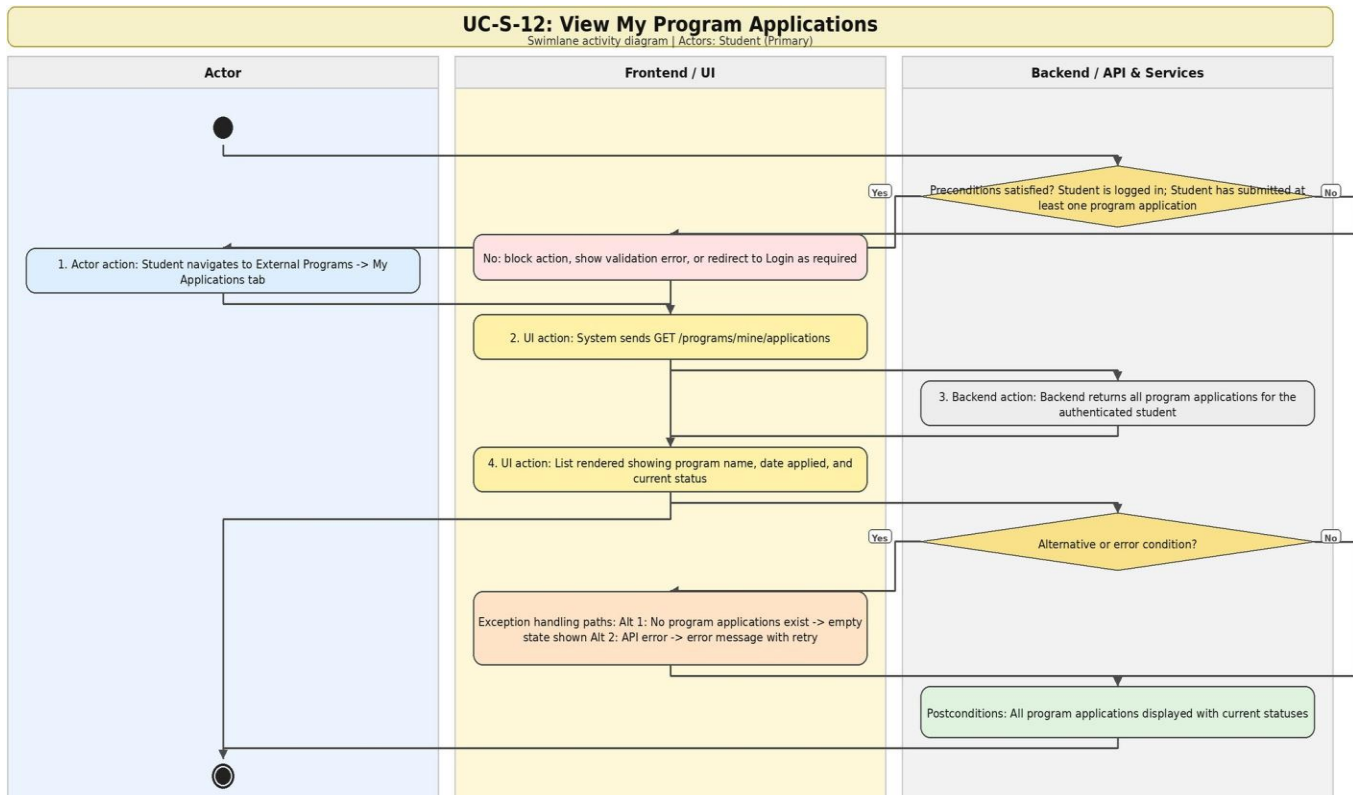
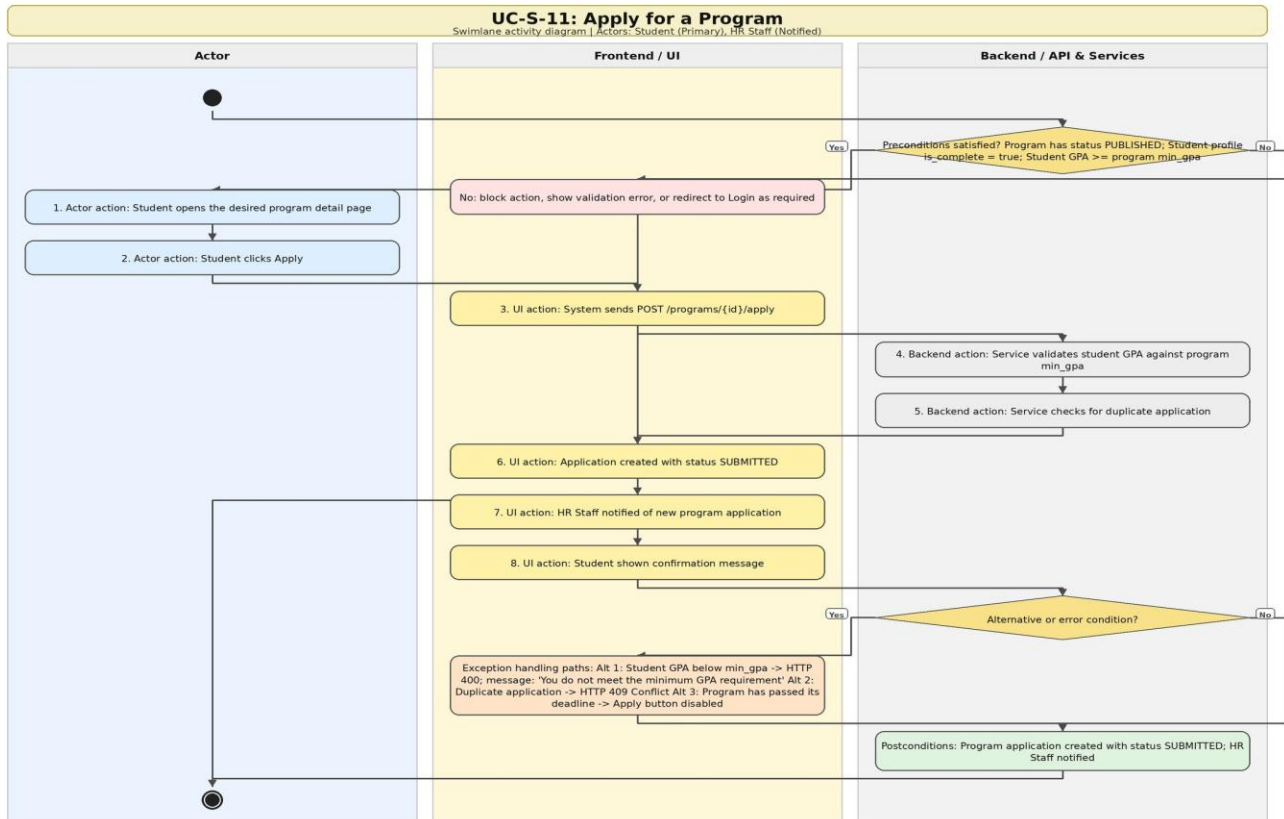
Undergraduate Lab Portal Requirements Specification



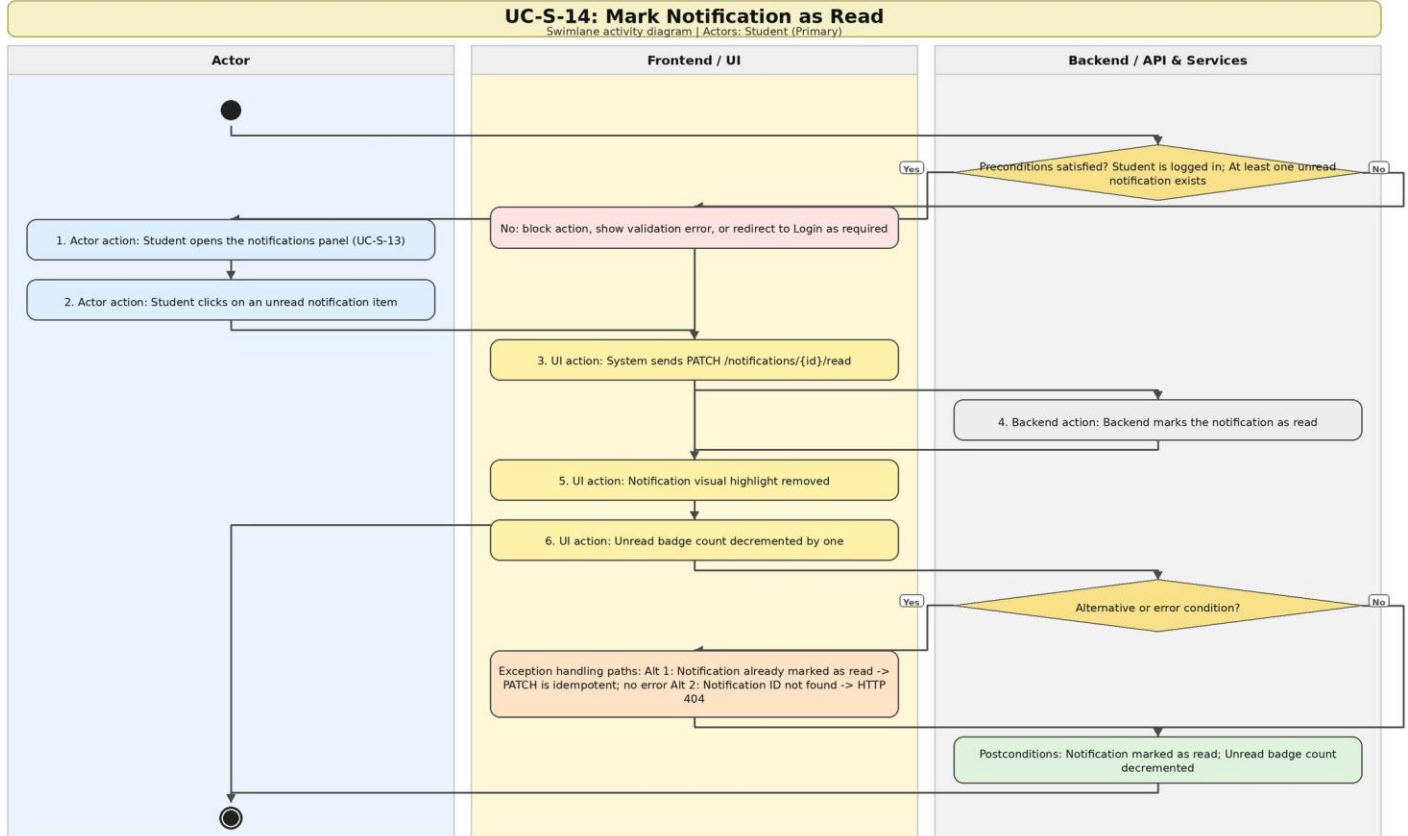
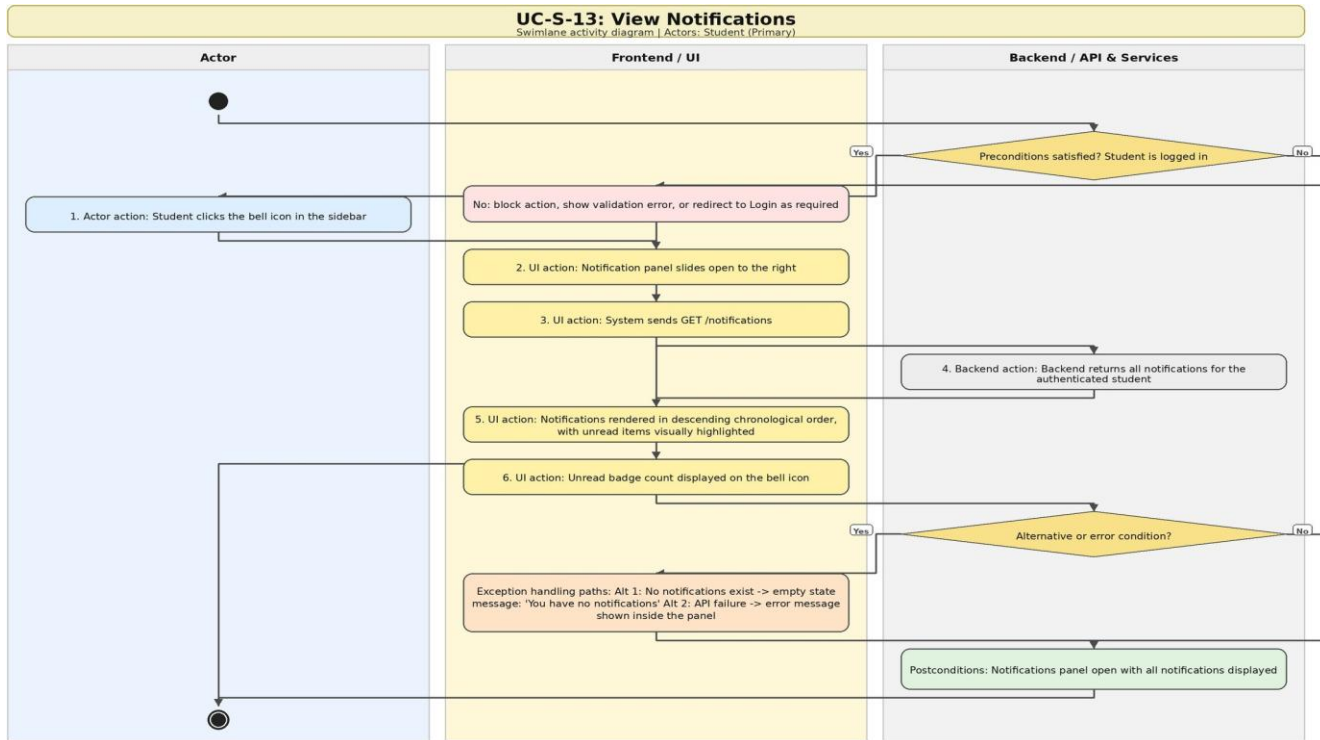
Undergraduate Lab Portal Requirements Specification



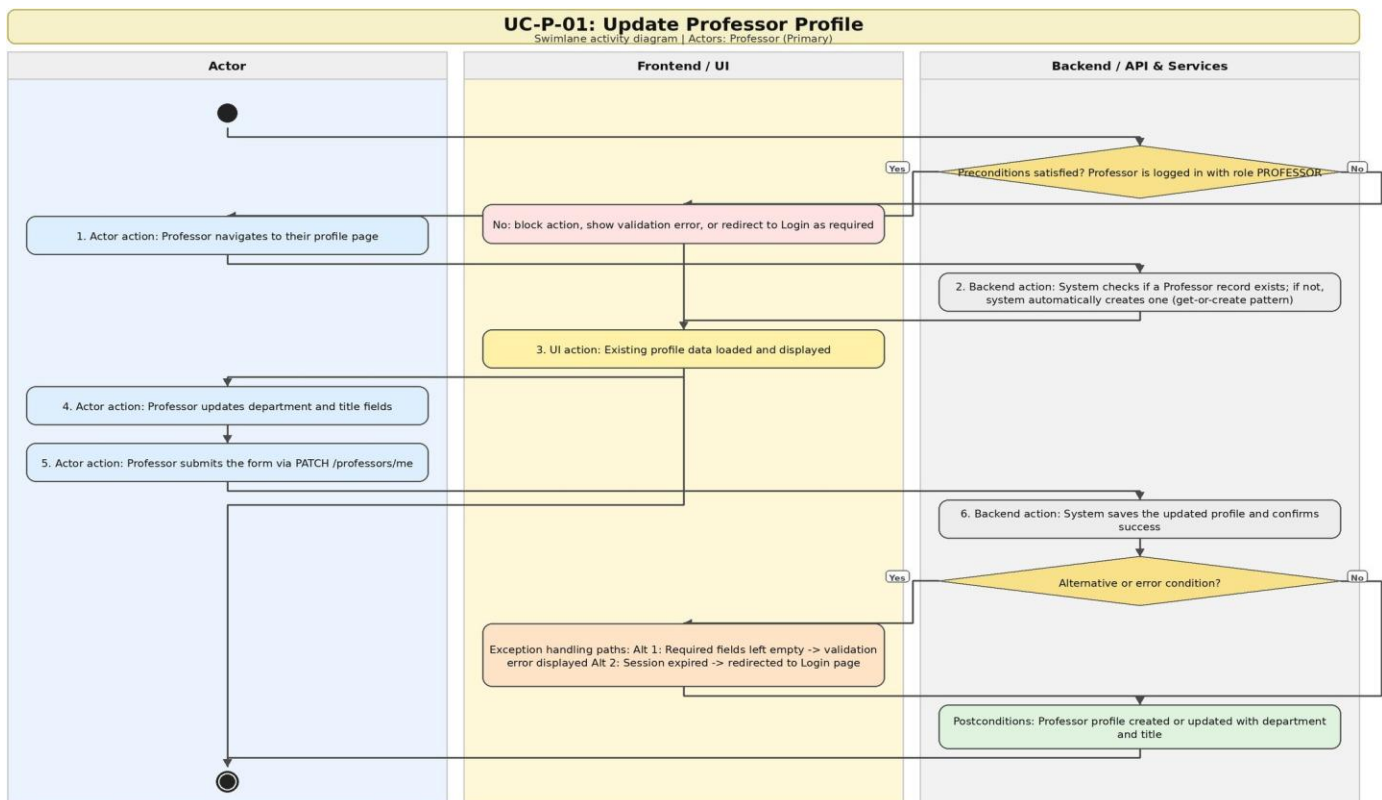
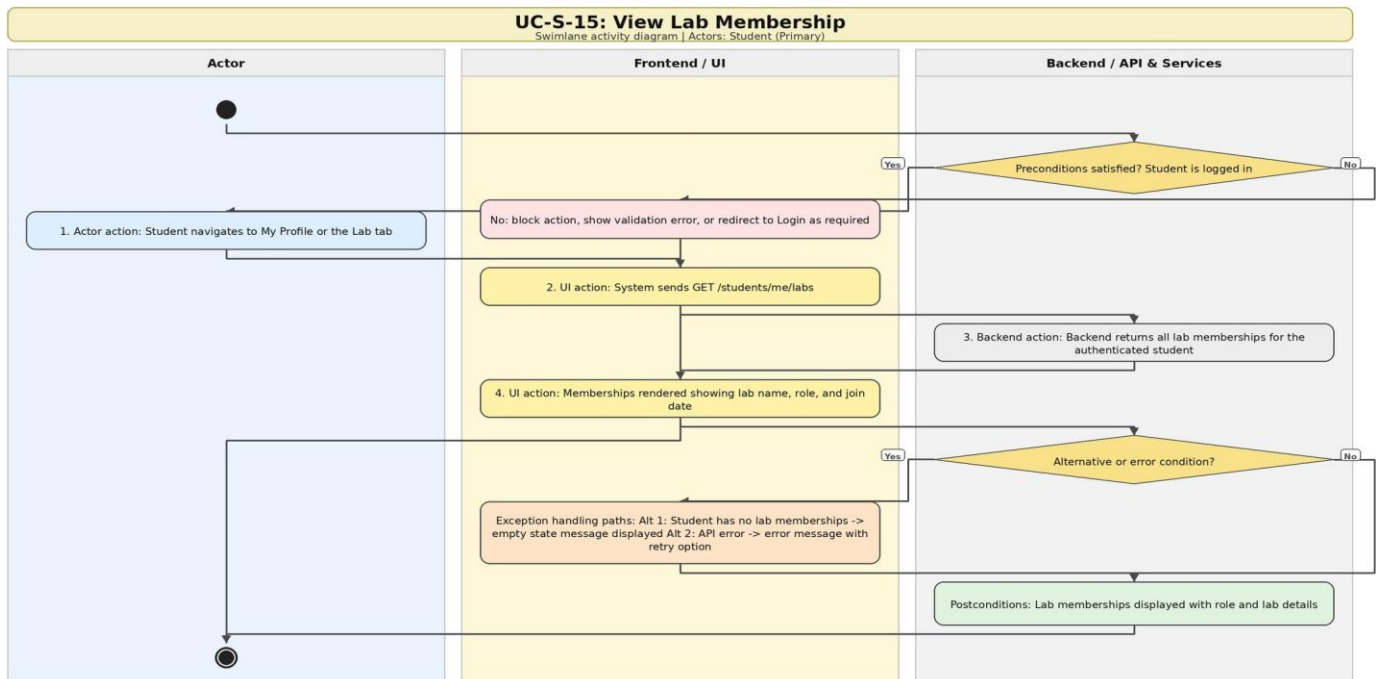
Undergraduate Lab Portal Requirements Specification



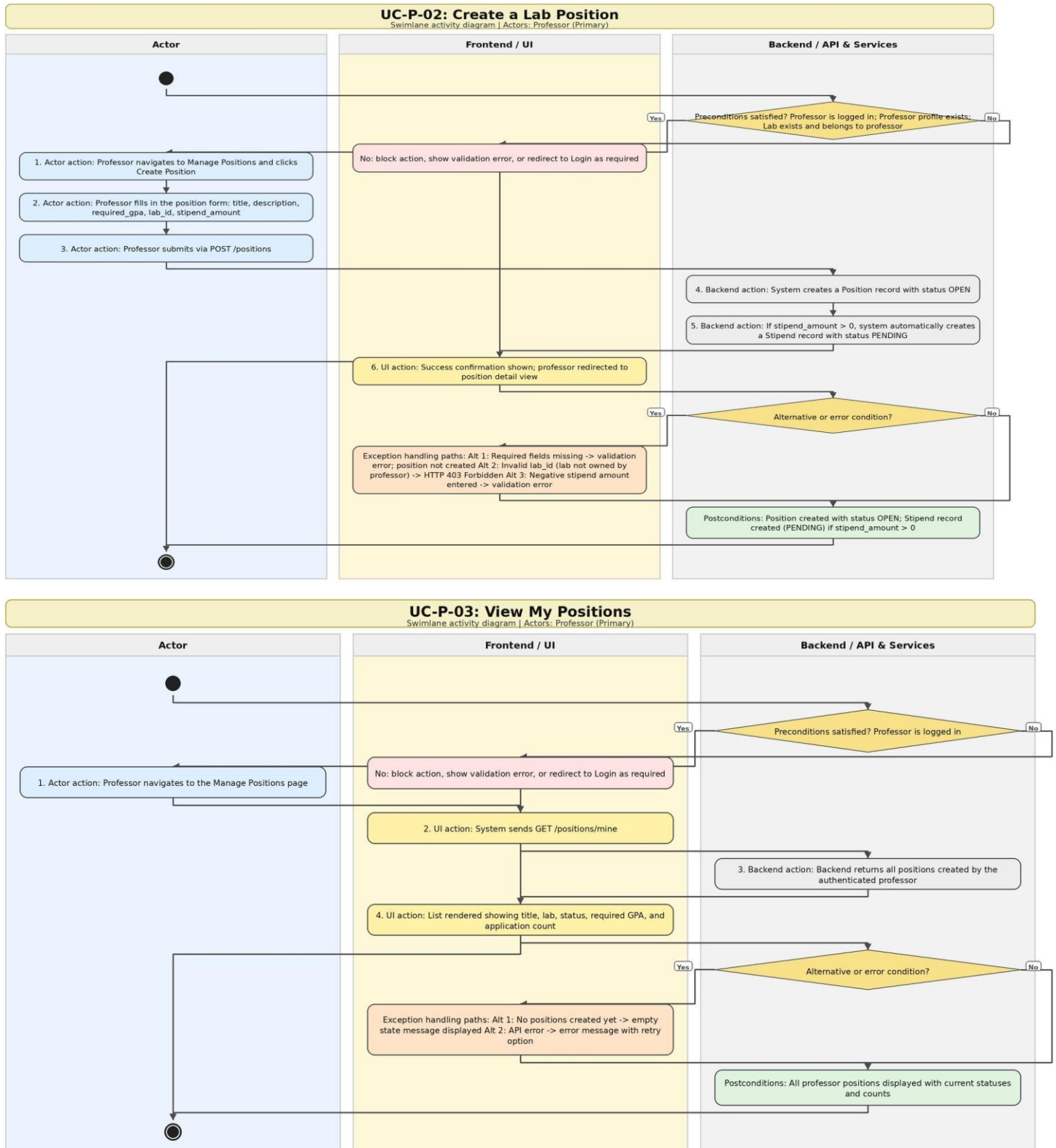
Undergraduate Lab Portal Requirements Specification



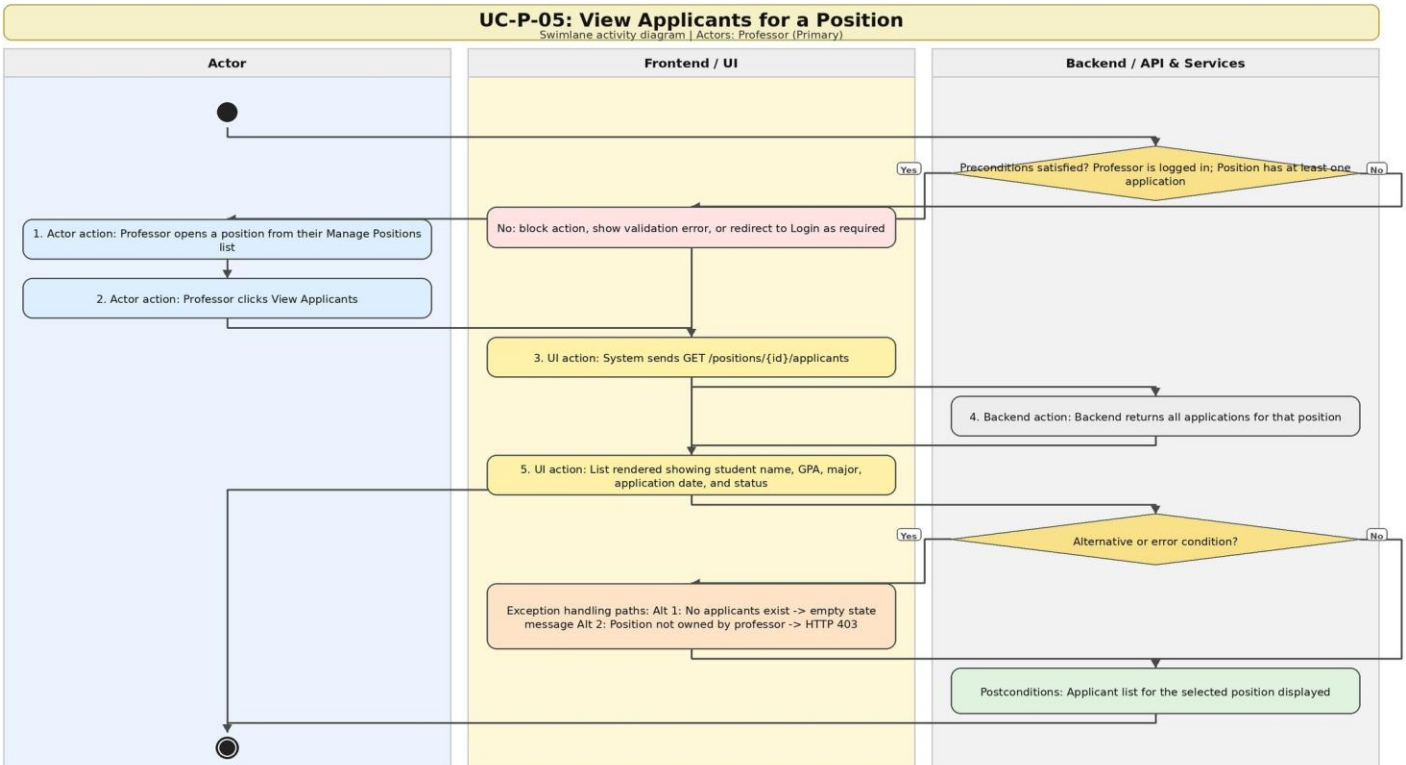
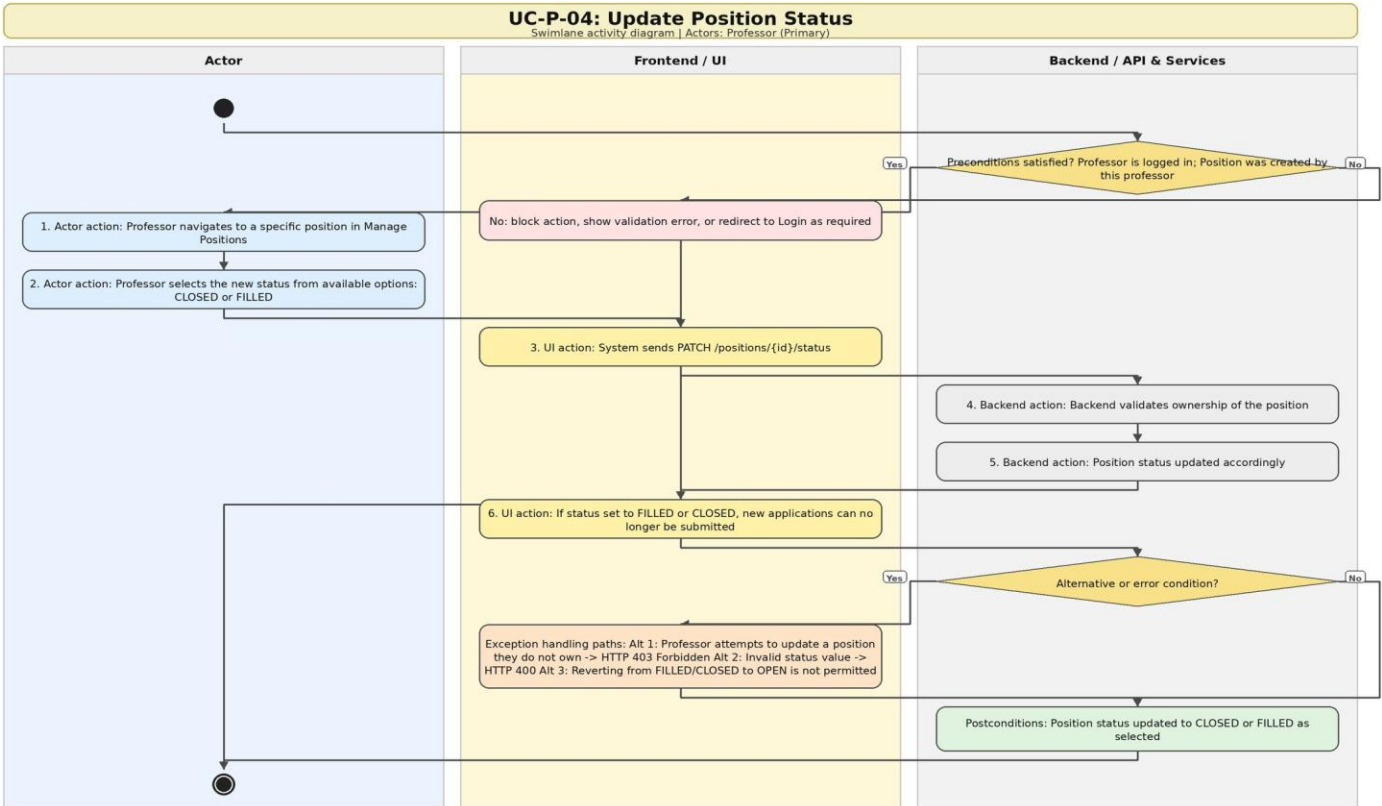
Undergraduate Lab Portal Requirements Specification



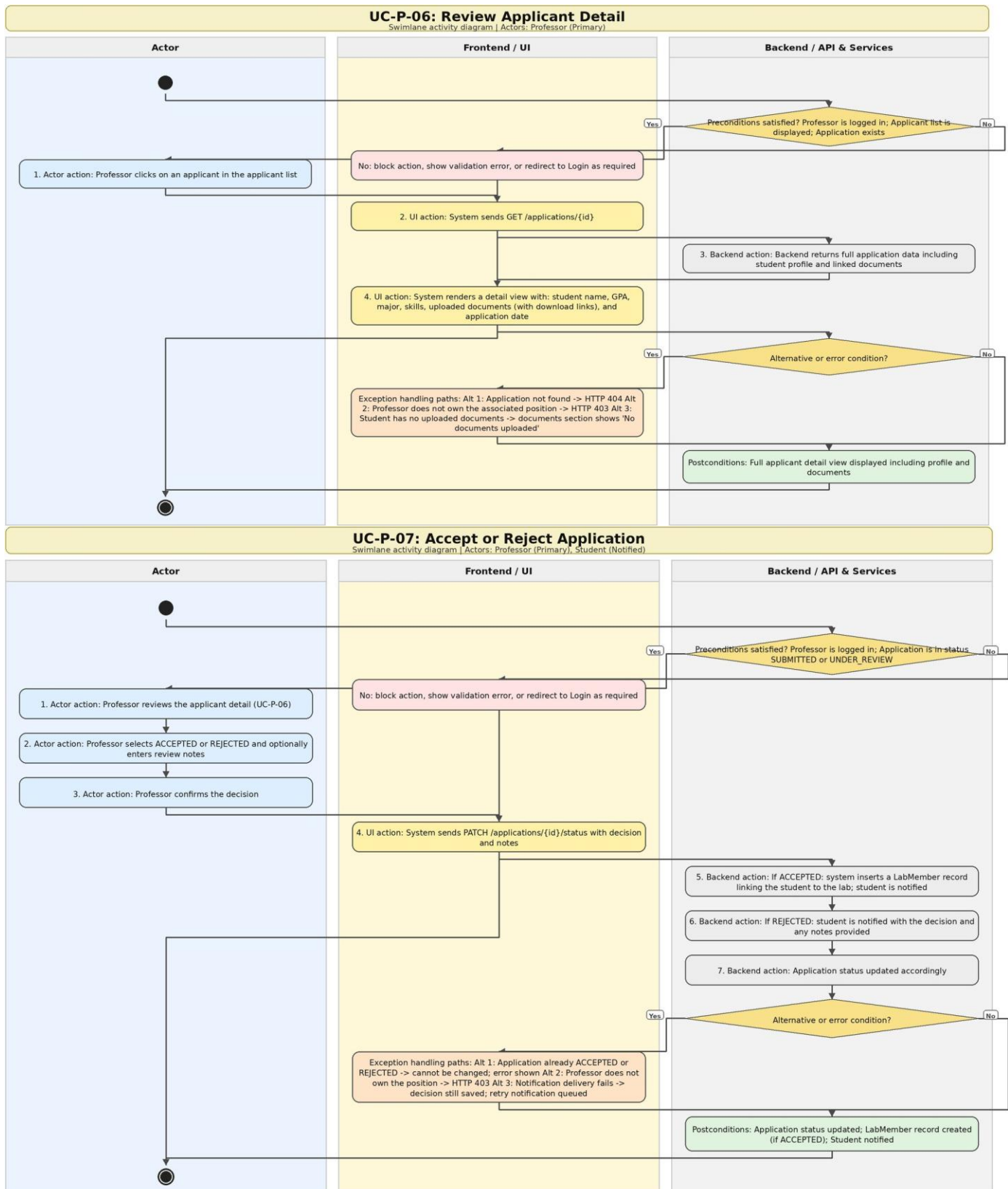
Undergraduate Lab Portal Requirements Specification



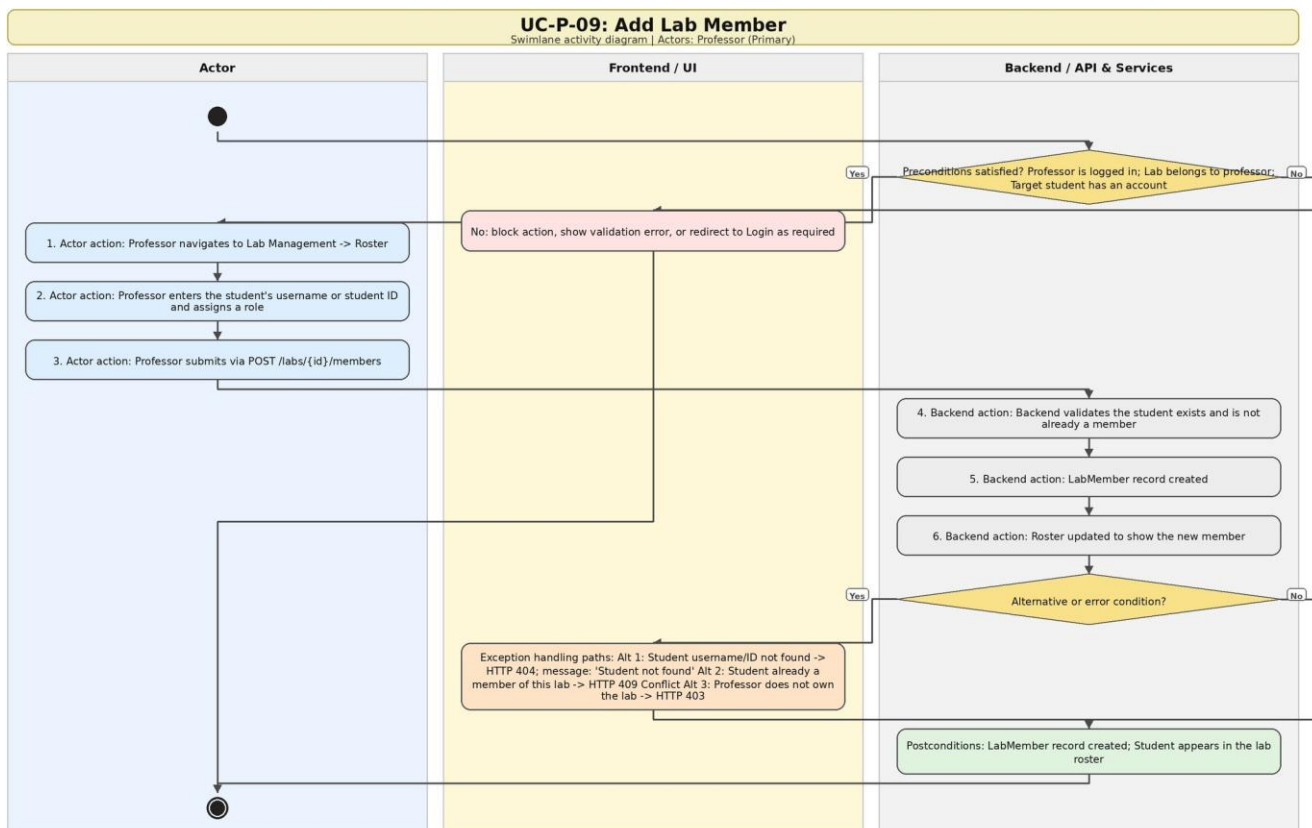
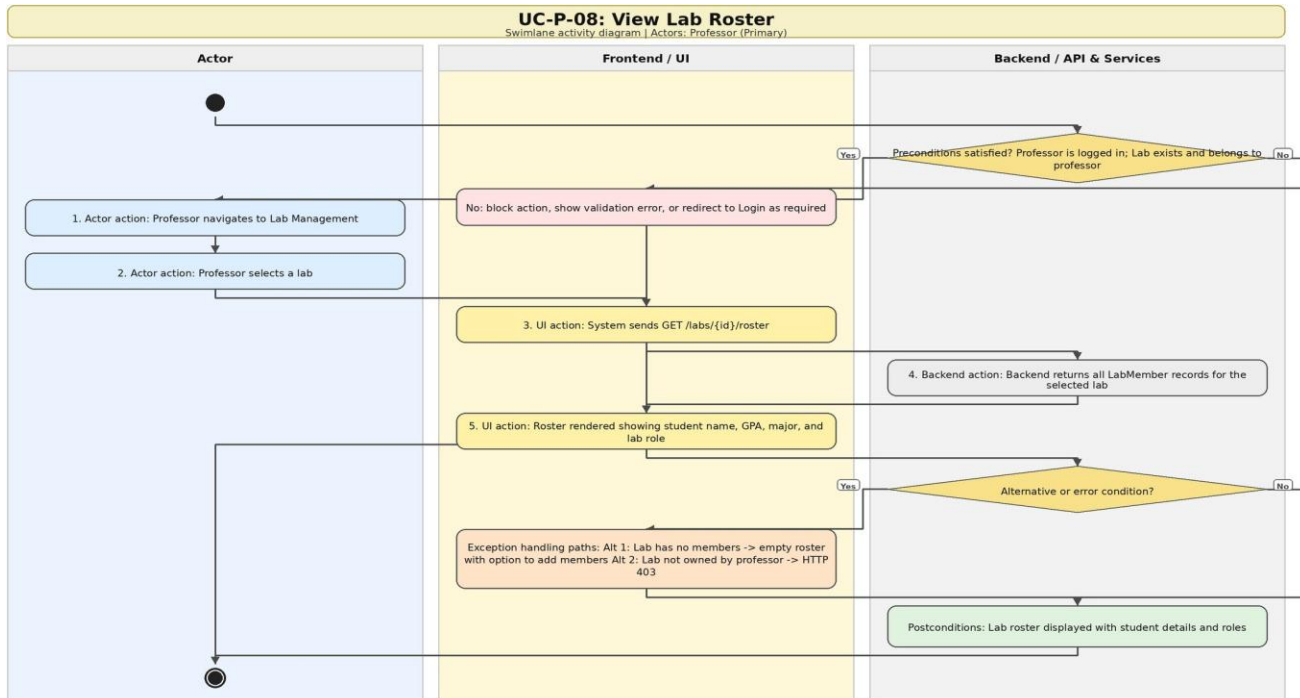
Undergraduate Lab Portal Requirements Specification



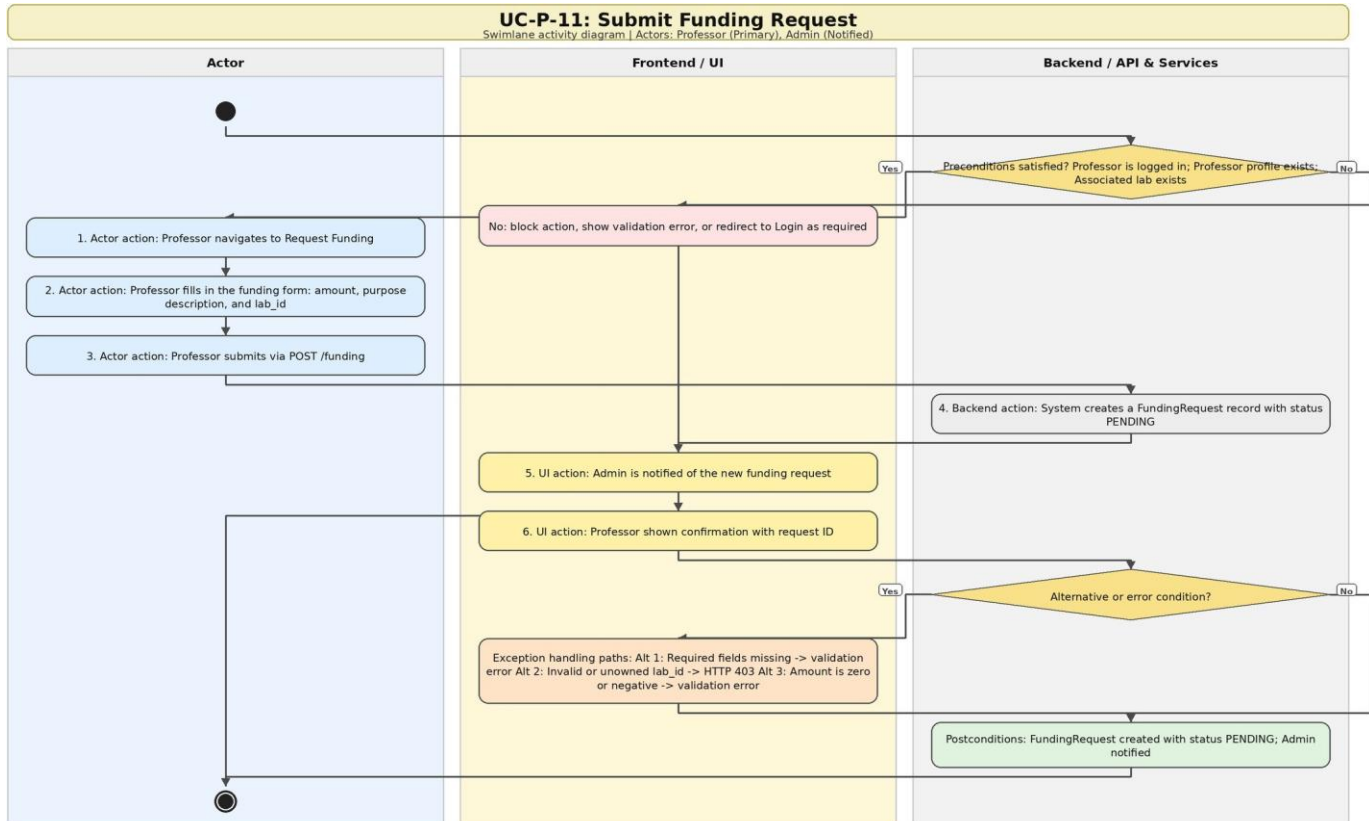
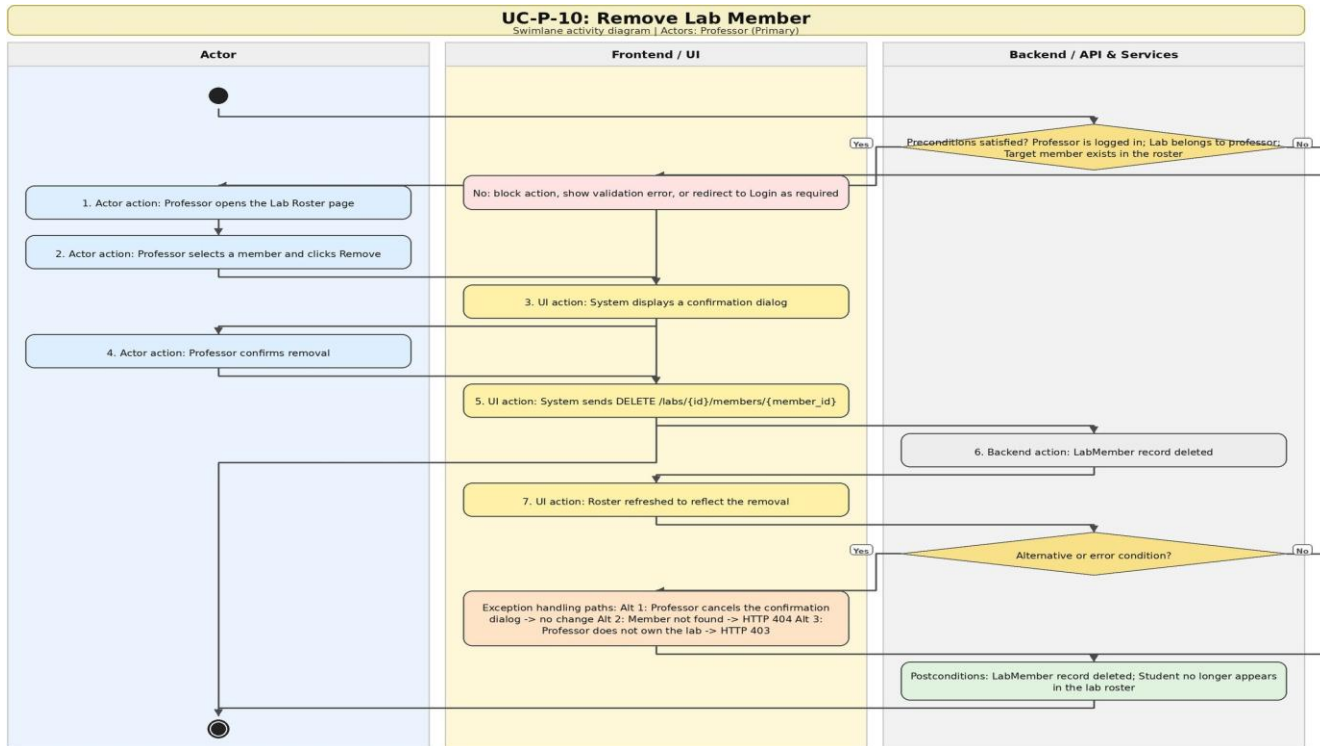
Undergraduate Lab Portal Requirements Specification



Undergraduate Lab Portal Requirements Specification



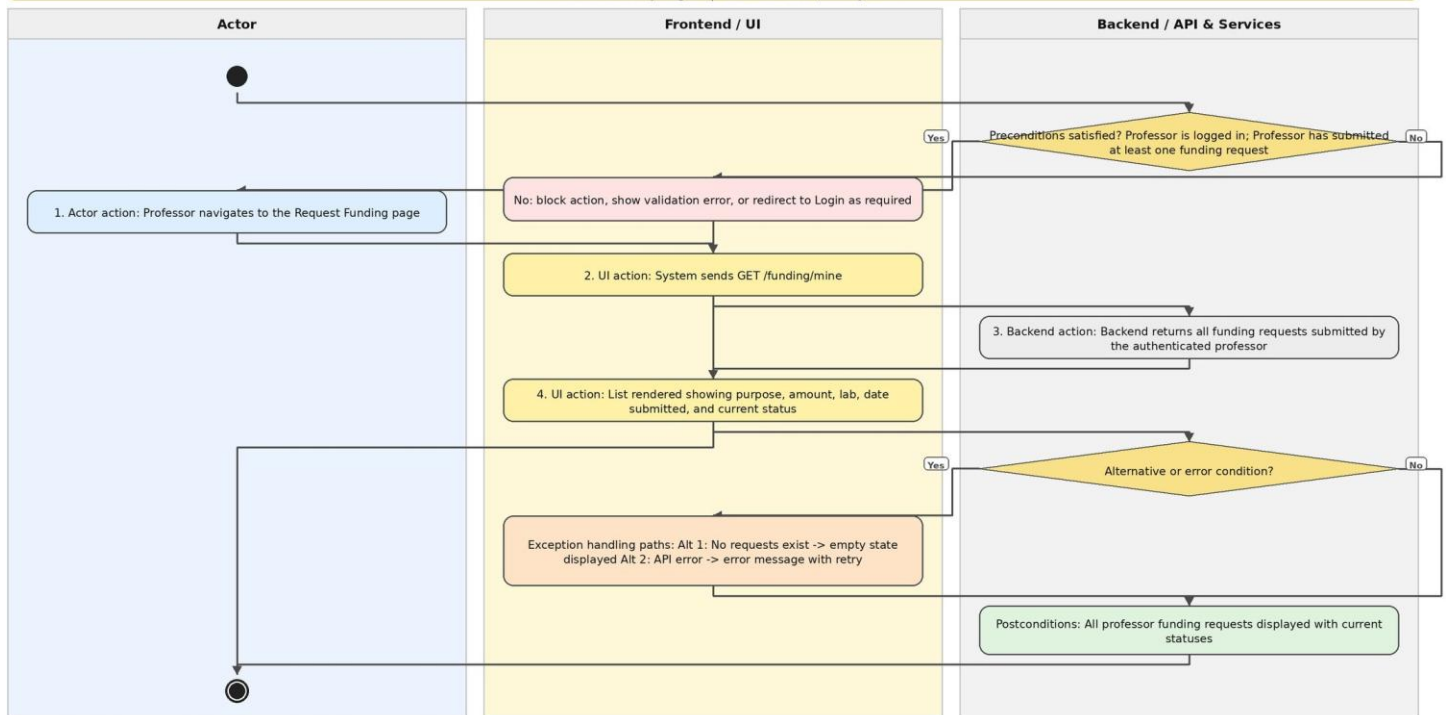
Undergraduate Lab Portal Requirements Specification



Undergraduate Lab Portal Requirements Specification

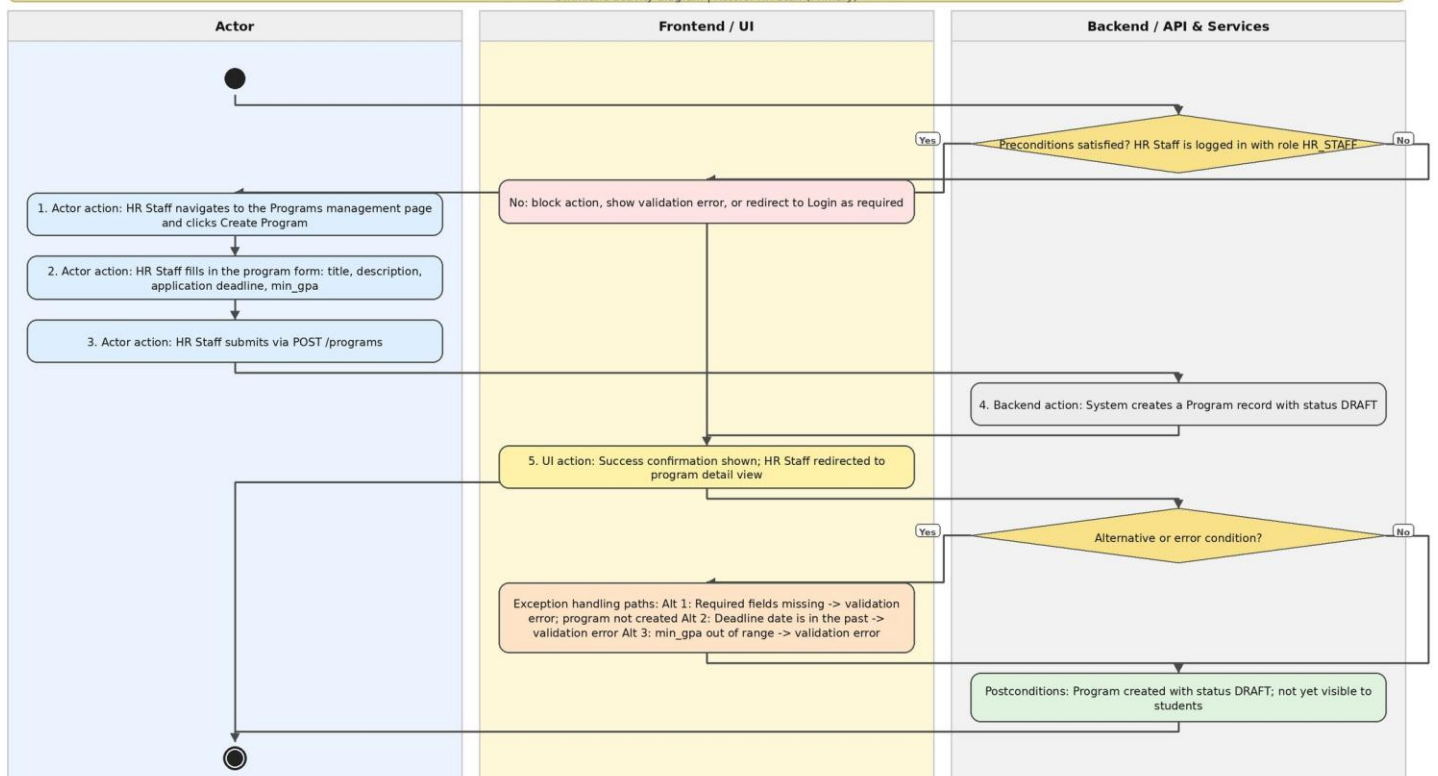
UC-P-12: View My Funding Requests

Swimlane activity diagram | Actors: Professor (Primary)



UC-HR-01: Create International Program

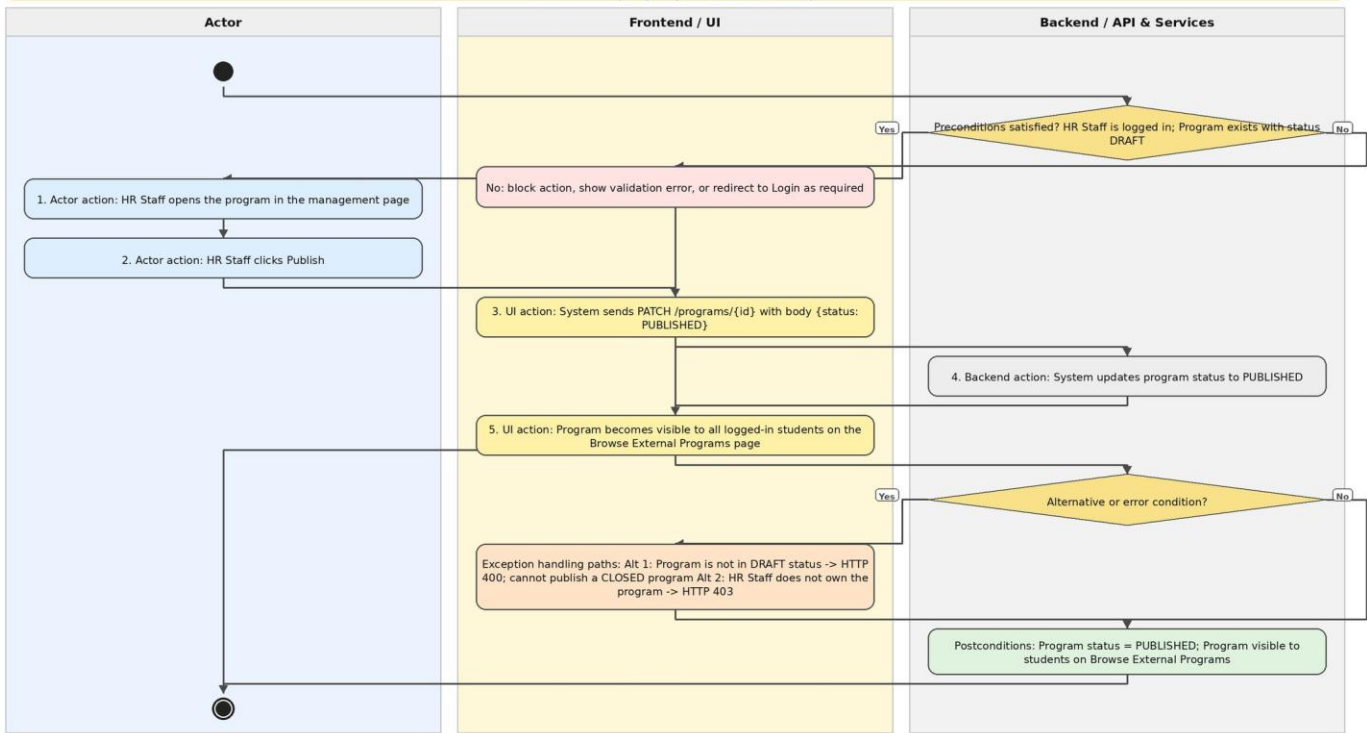
Swimlane activity diagram | Actors: HR Staff (Primary)



Undergraduate Lab Portal Requirements Specification

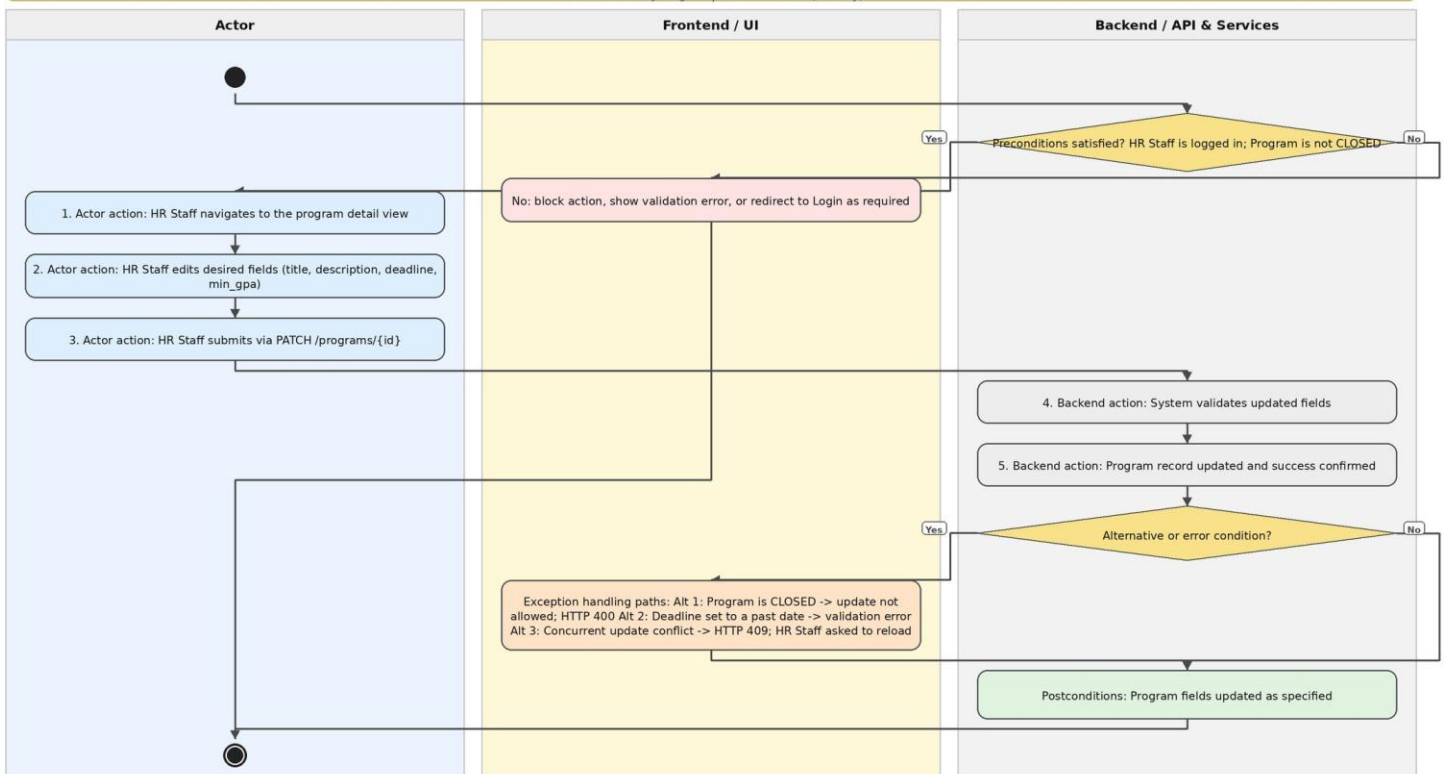
UC-HR-02: Publish a Program

Swimlane activity diagram | Actors: HR Staff (Primary)

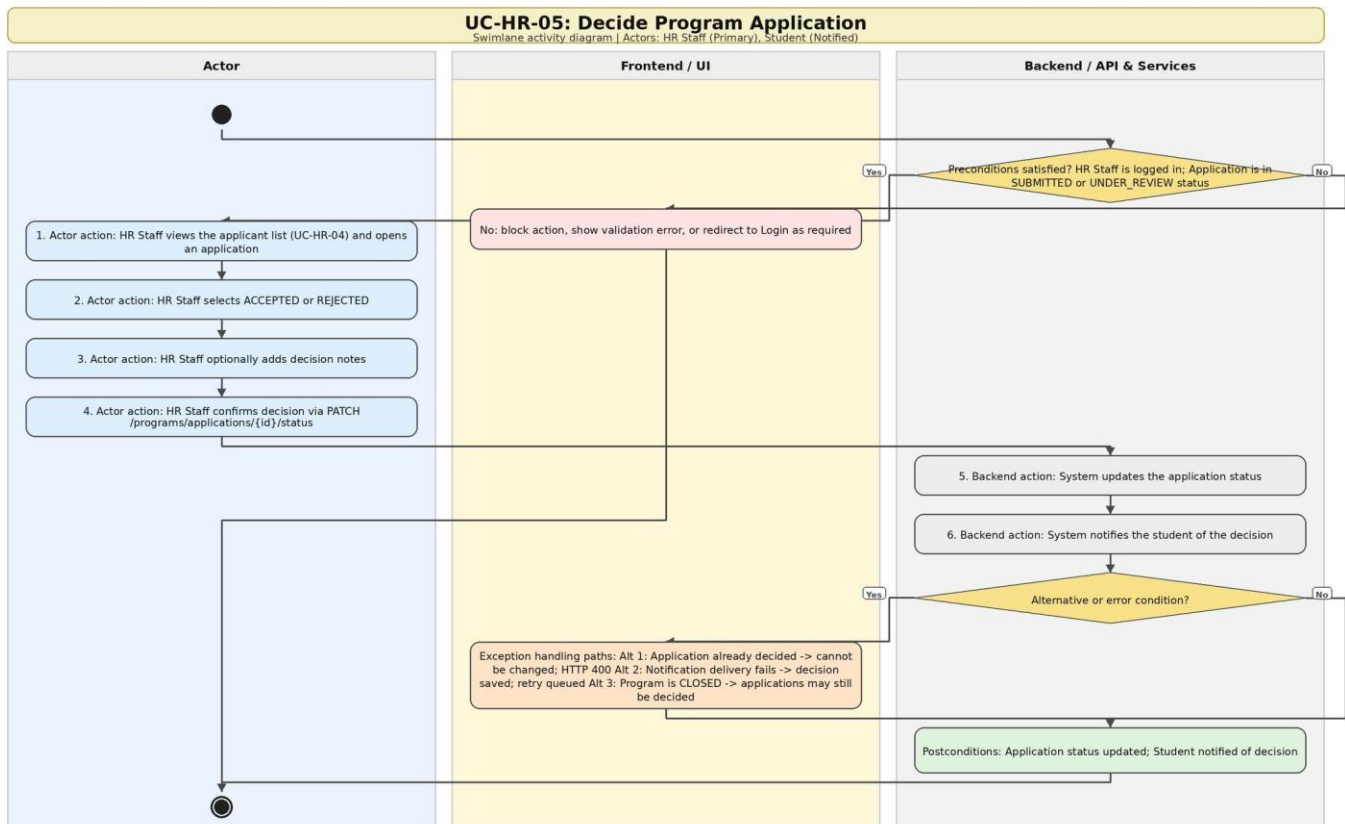
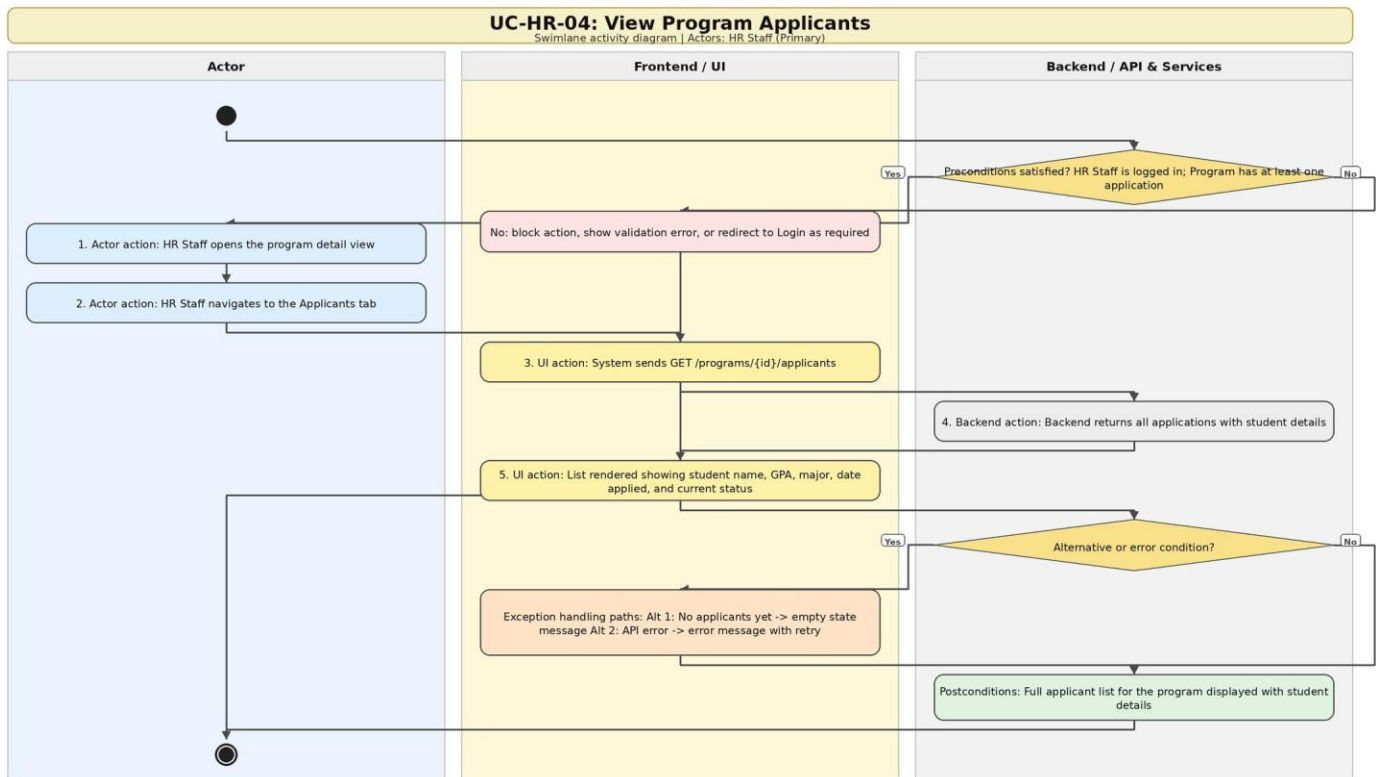


UC-HR-03: Update Program

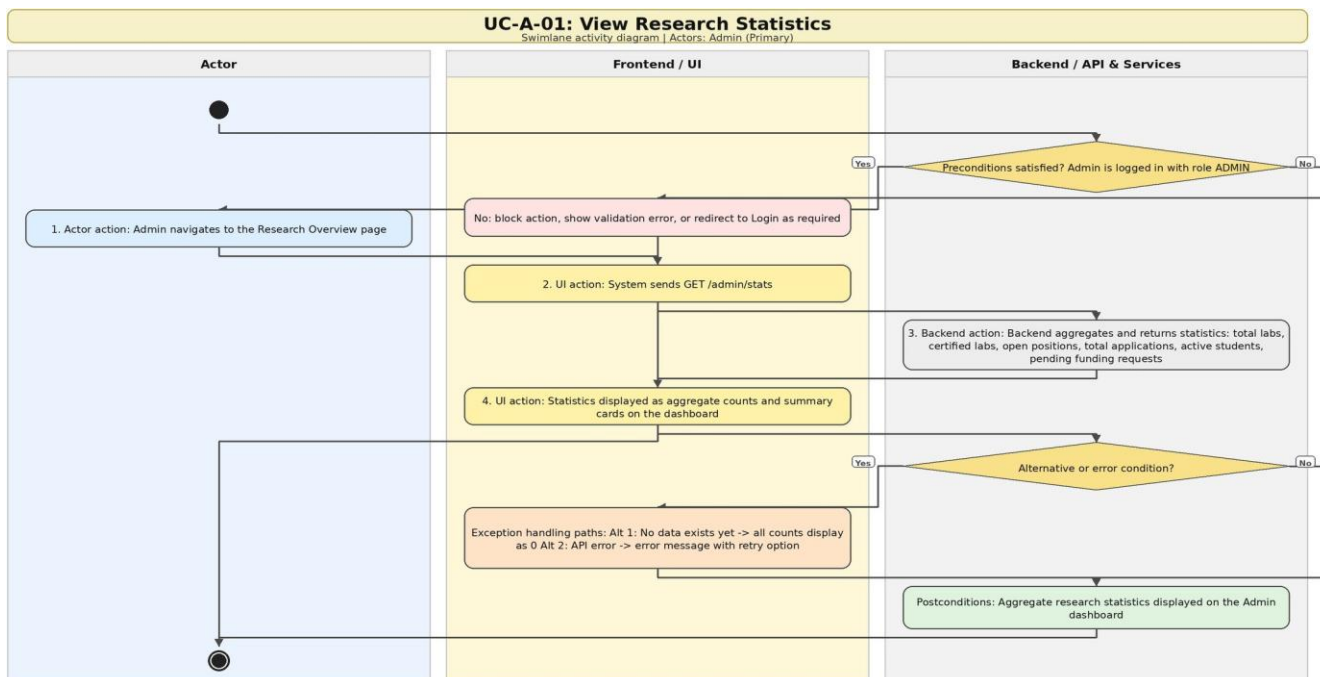
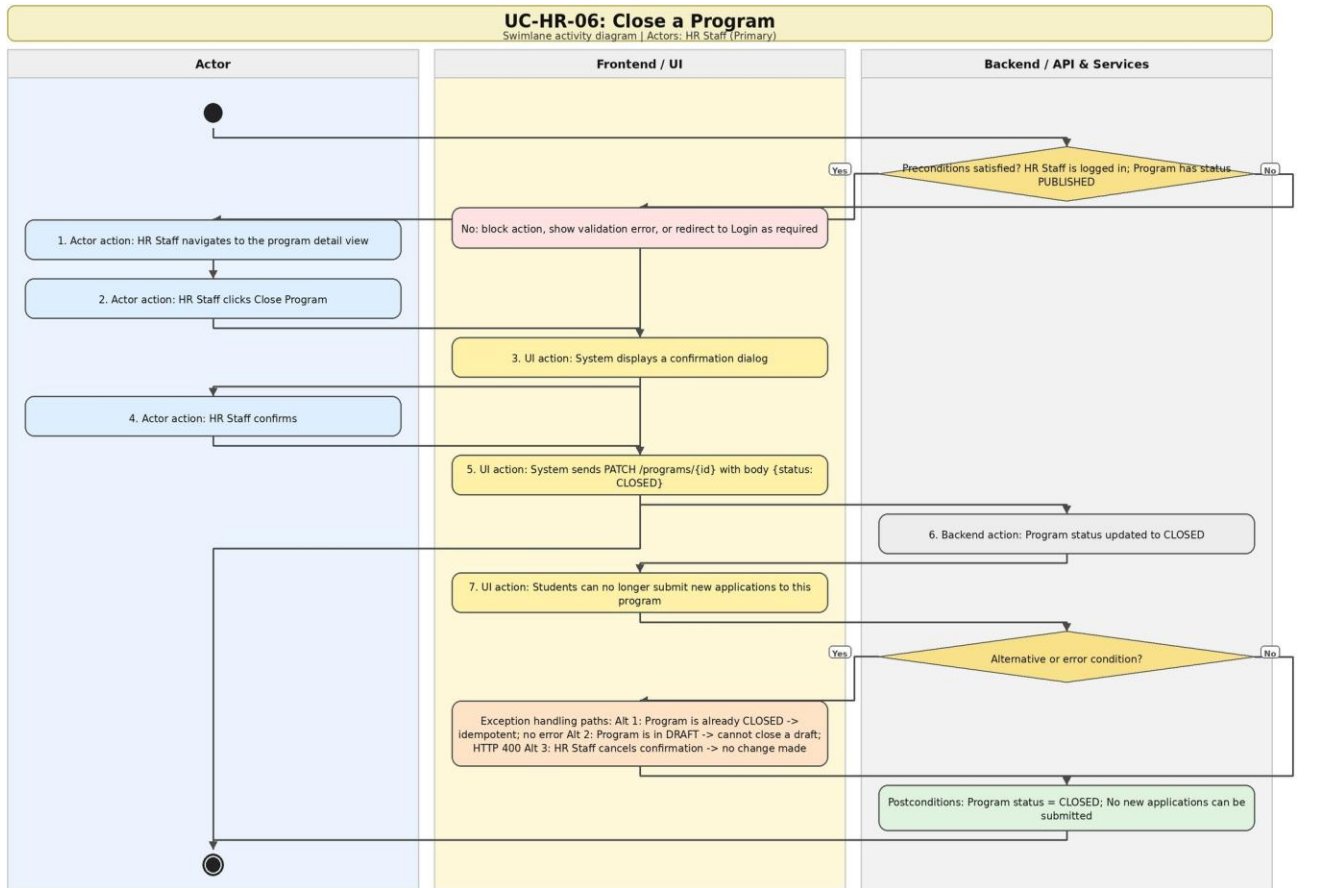
Swimlane activity diagram | Actors: HR Staff (Primary)



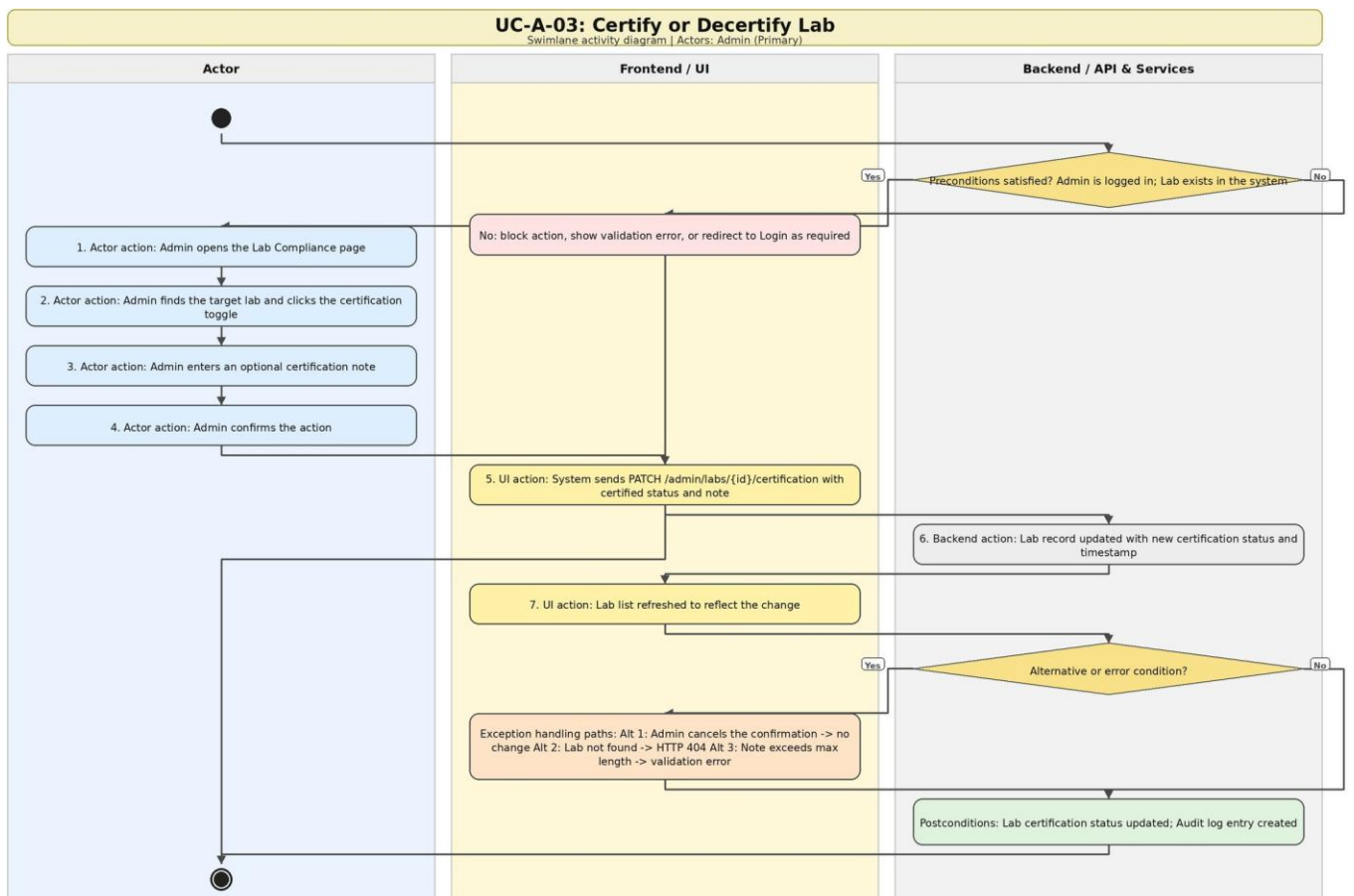
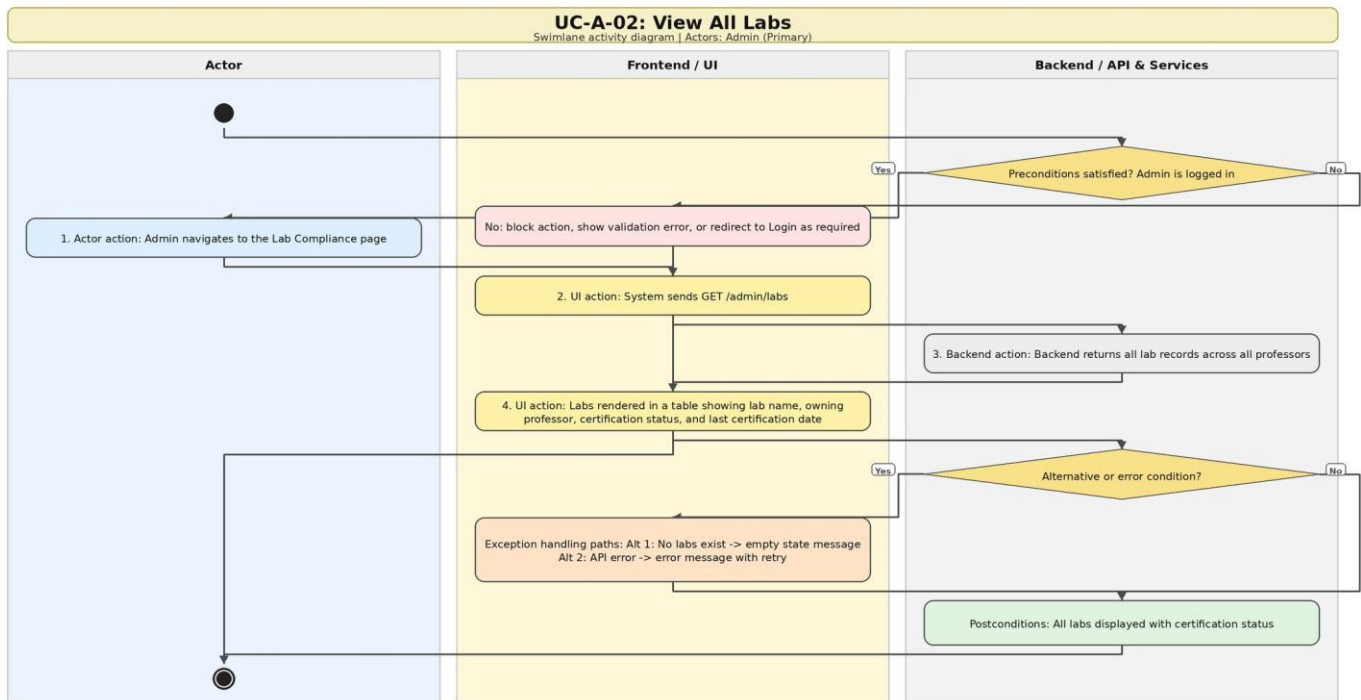
Undergraduate Lab Portal Requirements Specification



Undergraduate Lab Portal Requirements Specification



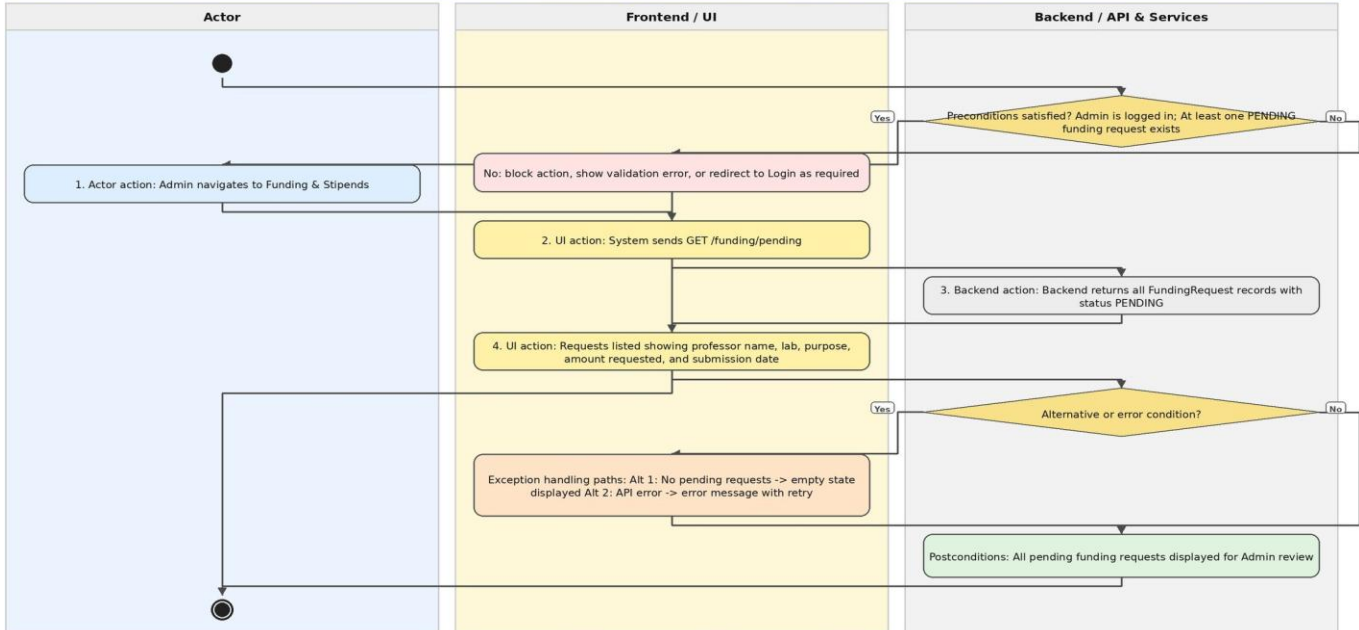
Undergraduate Lab Portal Requirements Specification



Undergraduate Lab Portal Requirements Specification

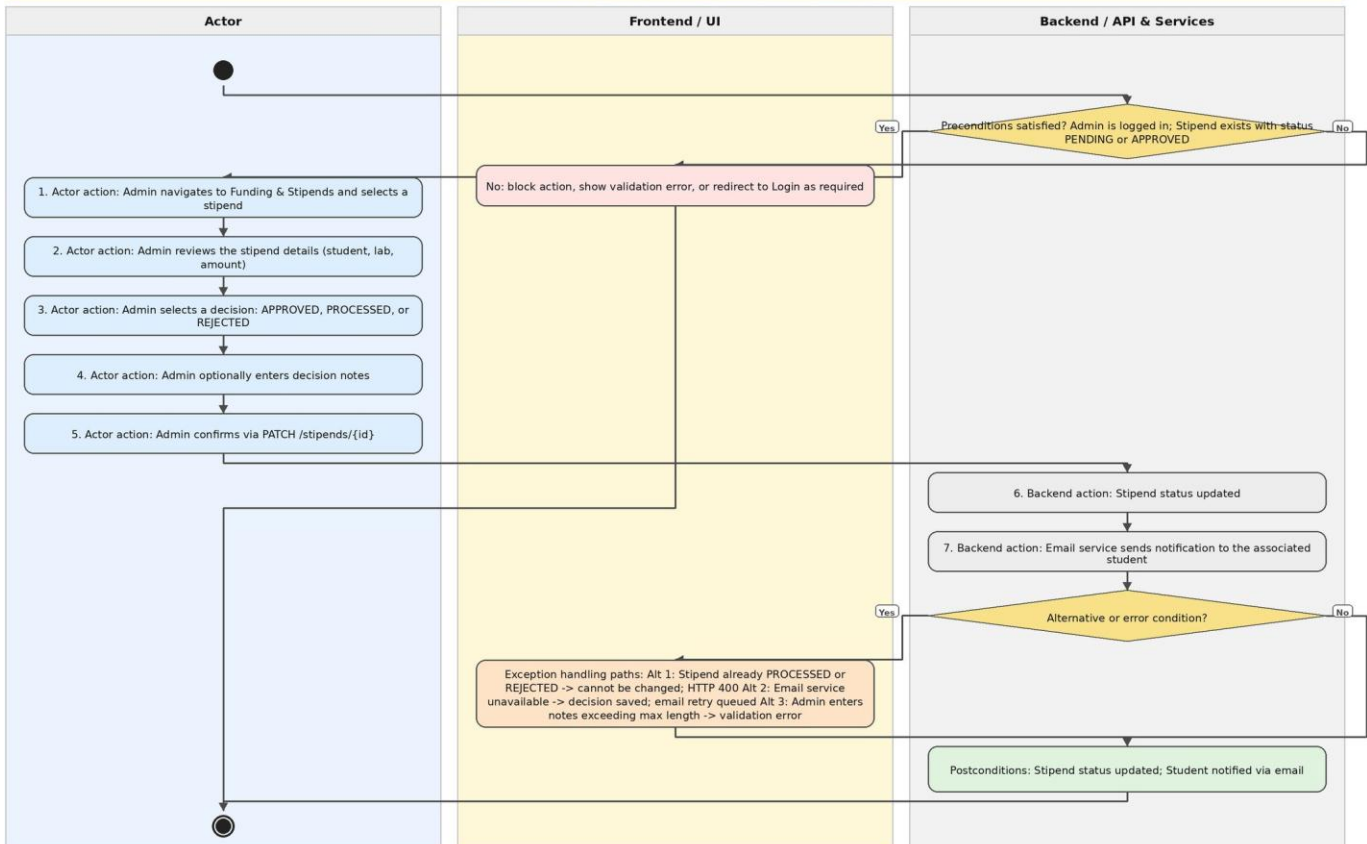
UC-A-04: View Funding Requests

Swimlane activity diagram | Actors: Admin (Primary)

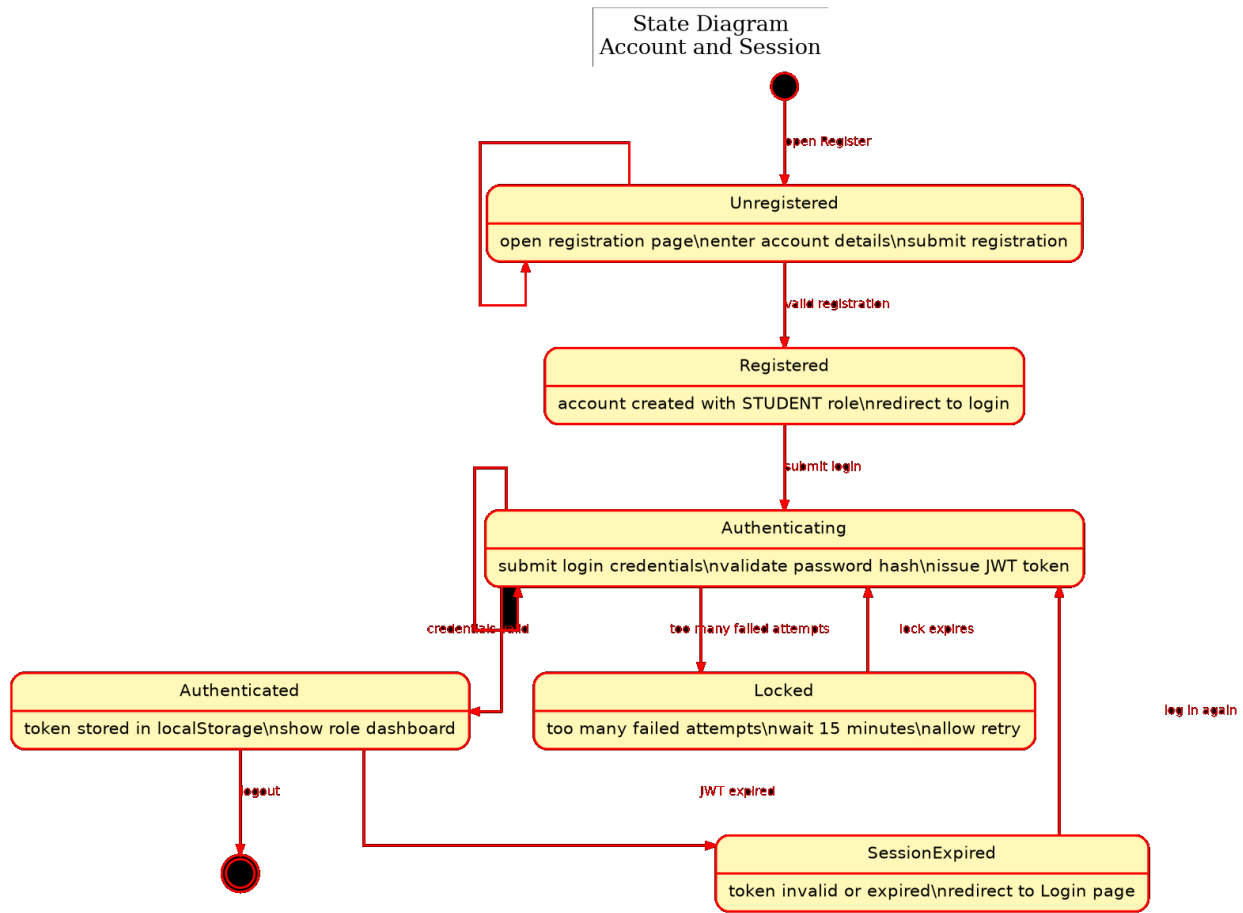


UC-A-05: Process Stipend

Swimlane activity diagram | Actors: Admin (Primary), Student (Notified)

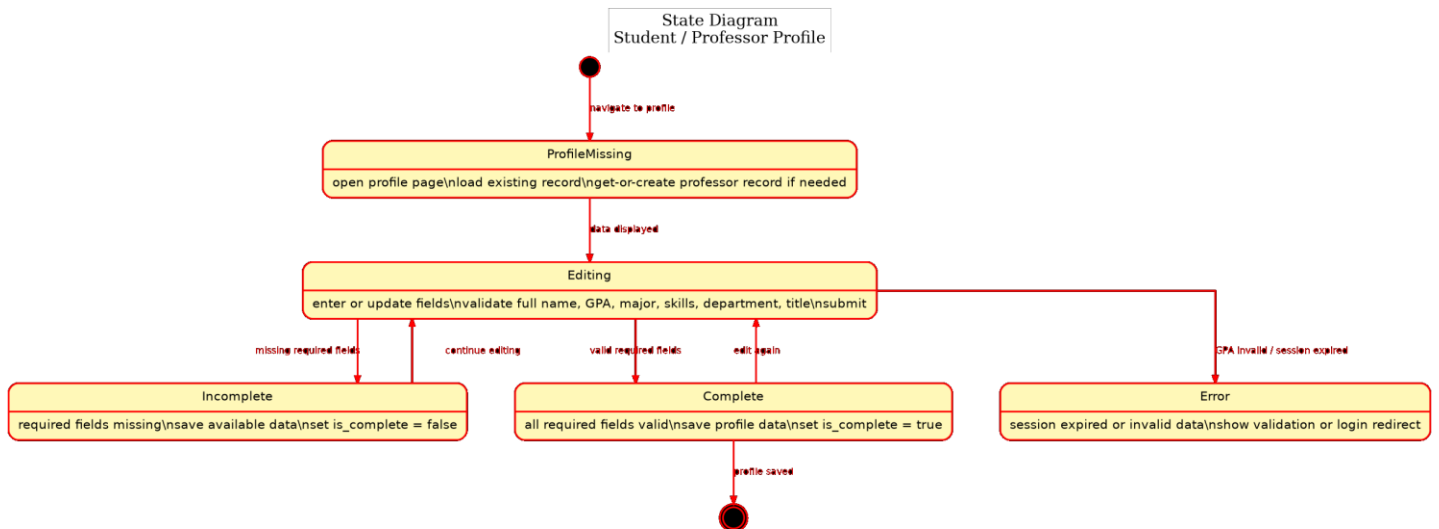


4.2.3 State Diagrams

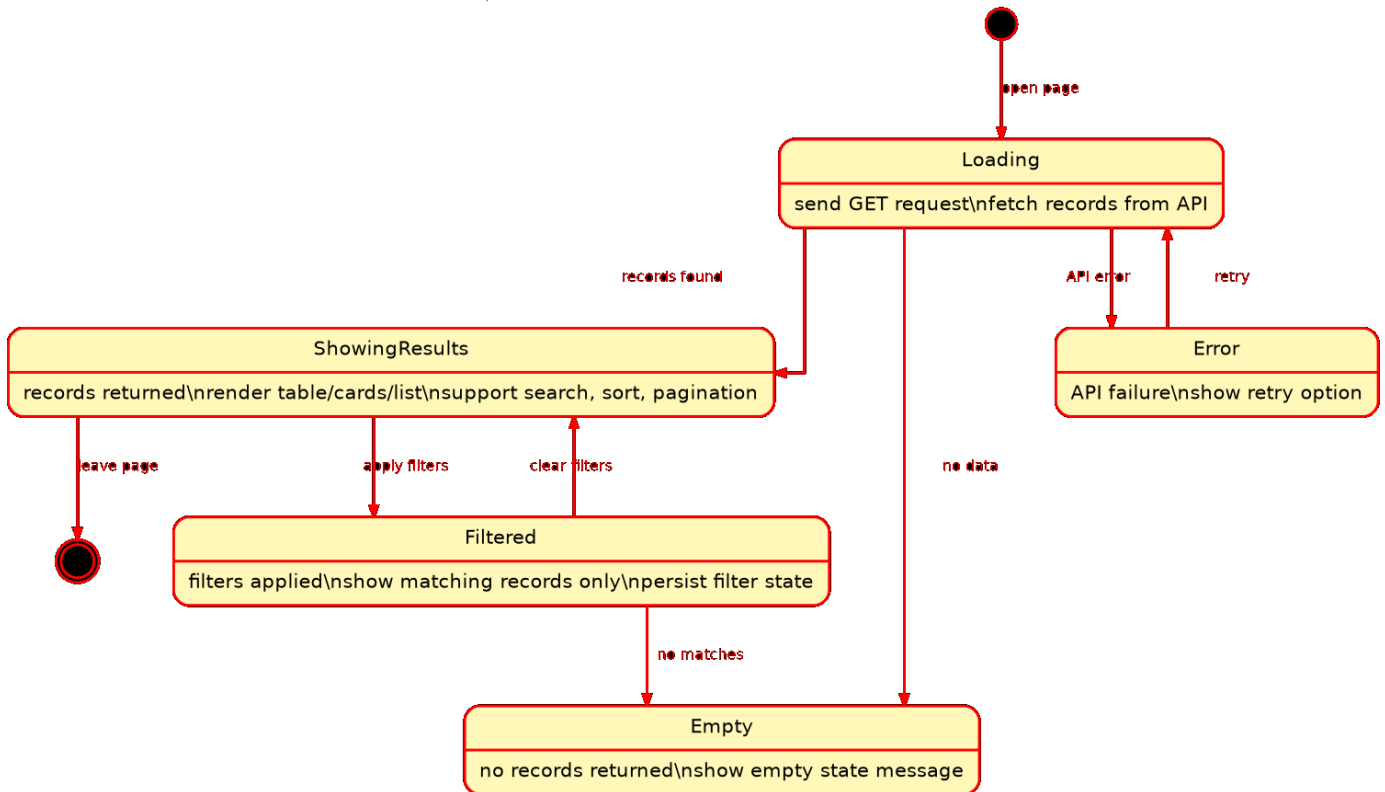


Undergraduate Lab Portal Requirements Specification

State Diagram
Student / Professor Profile

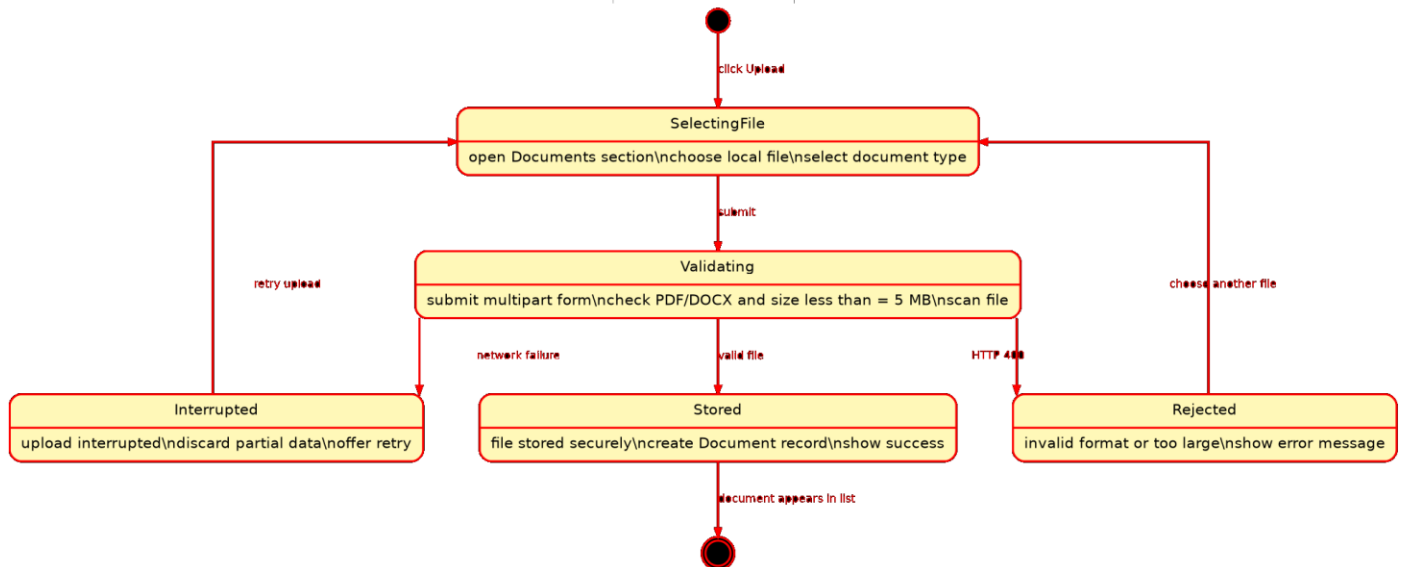


State Diagram
Reusable List / Dashboard View

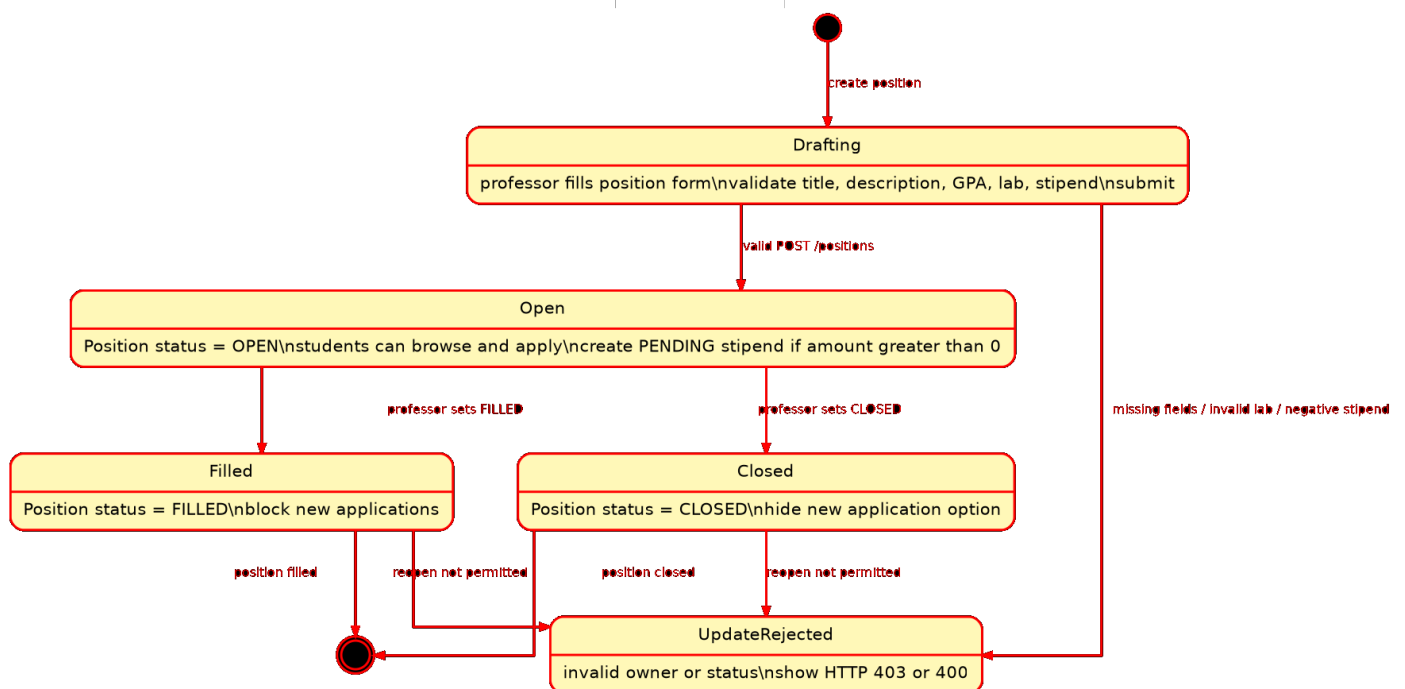


Undergraduate Lab Portal Requirements Specification

State Diagram
Document Upload

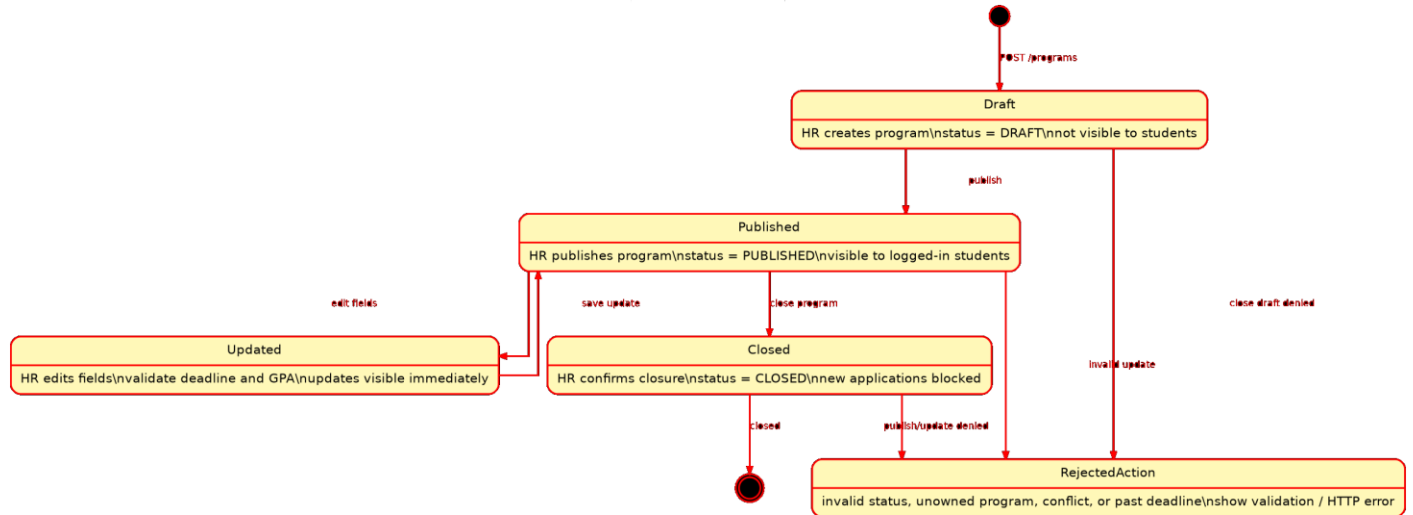


State Diagram
Lab Position

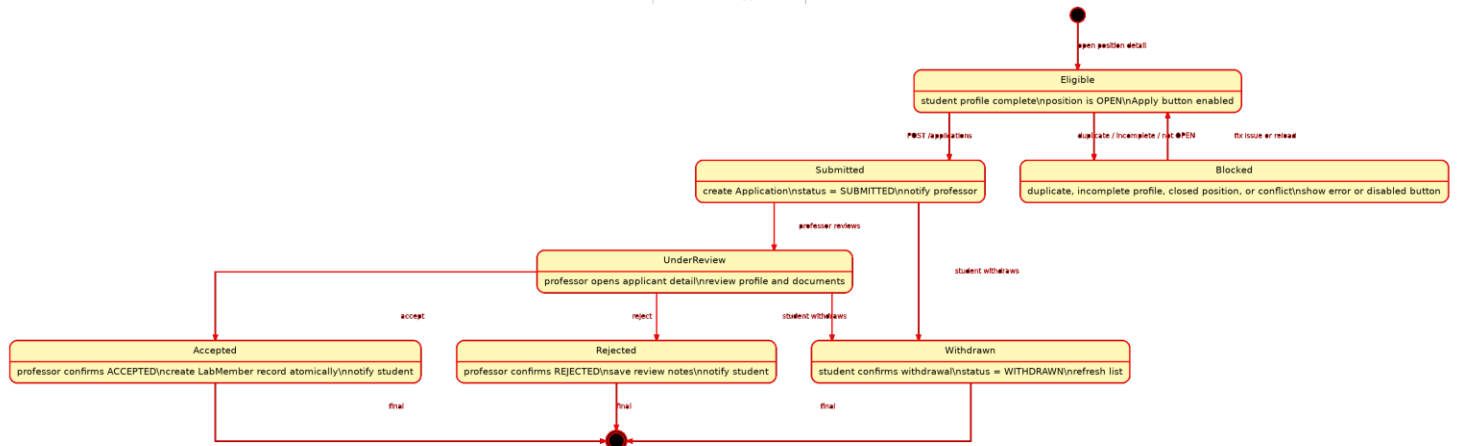


Undergraduate Lab Portal Requirements Specification

State Diagram
External Program

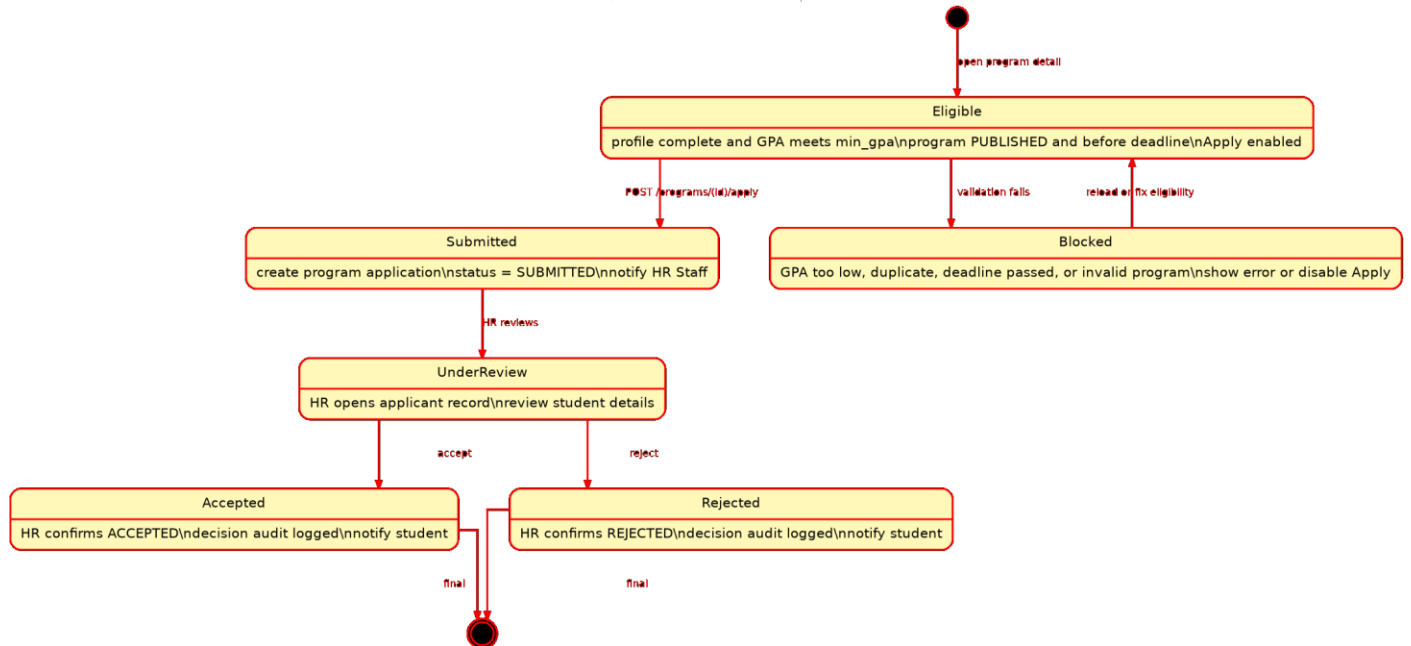


State Diagram
Lab Position Application

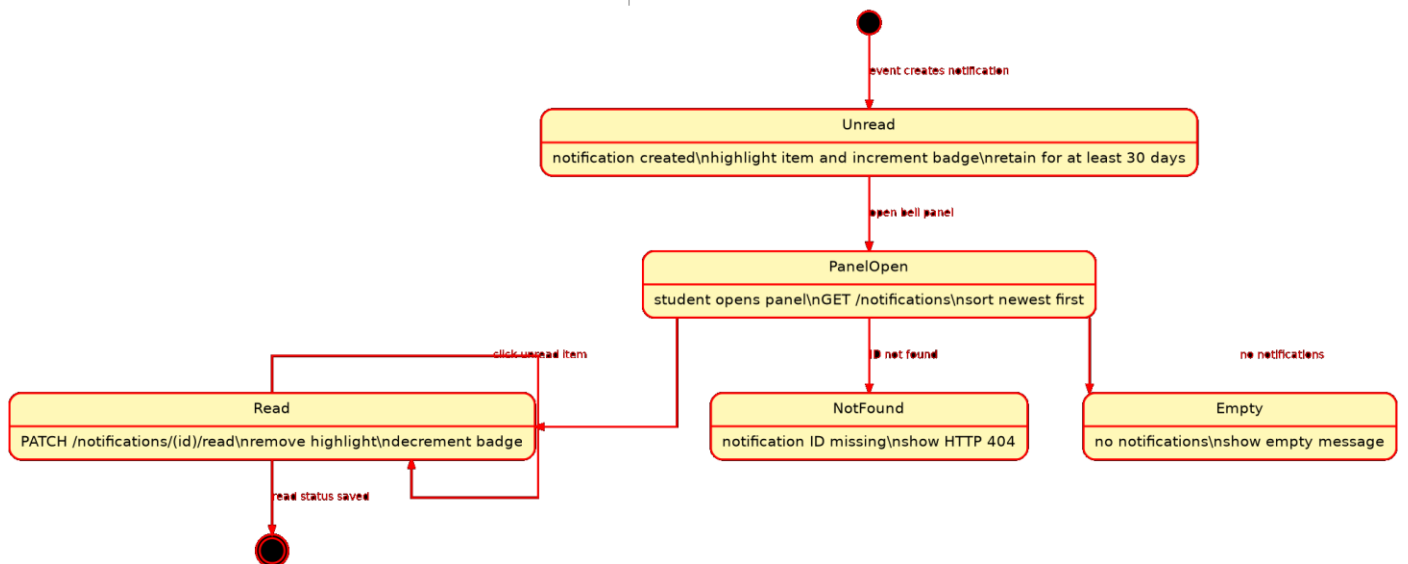


Undergraduate Lab Portal Requirements Specification

State Diagram
Program Application

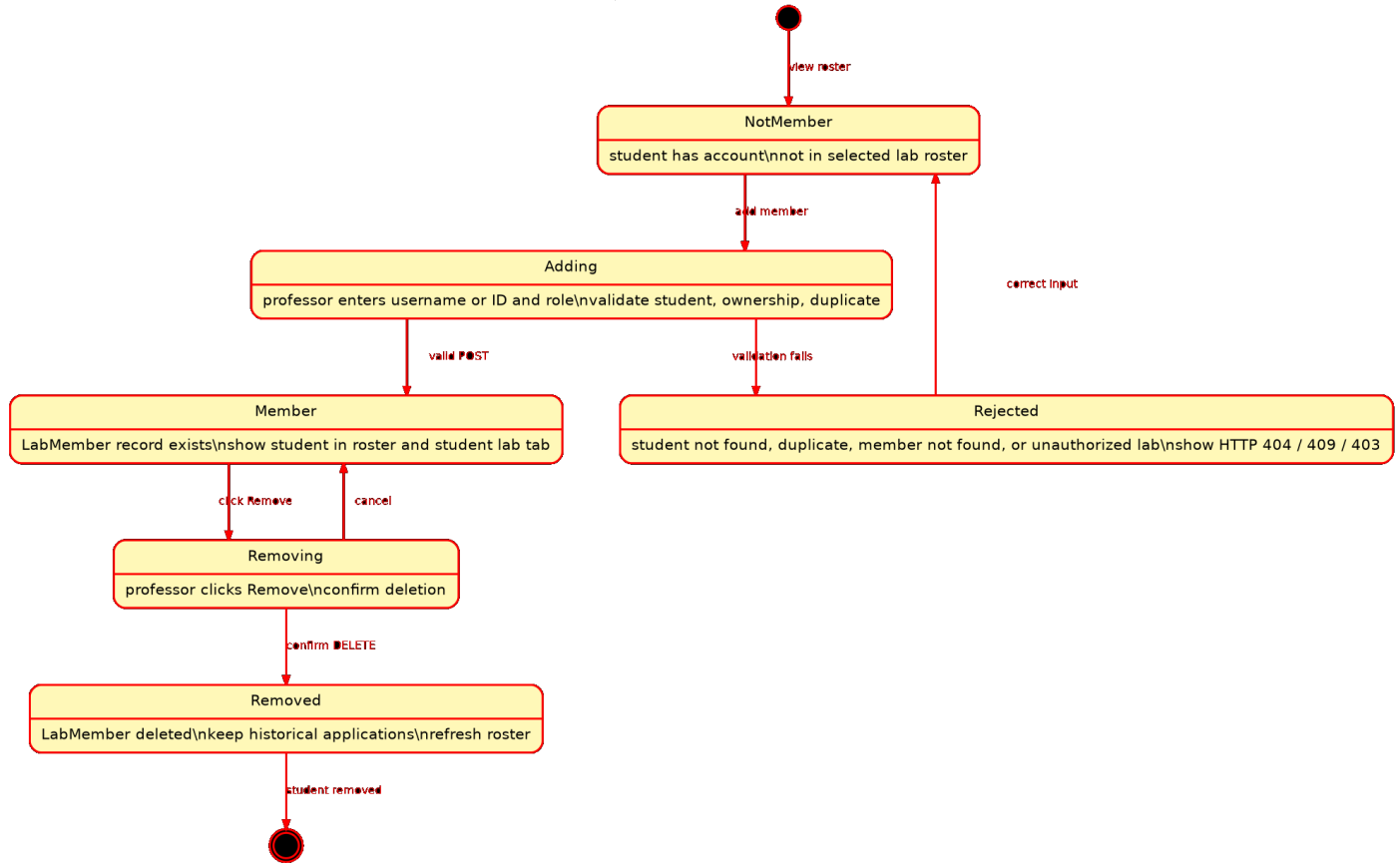


State Diagram
Notification



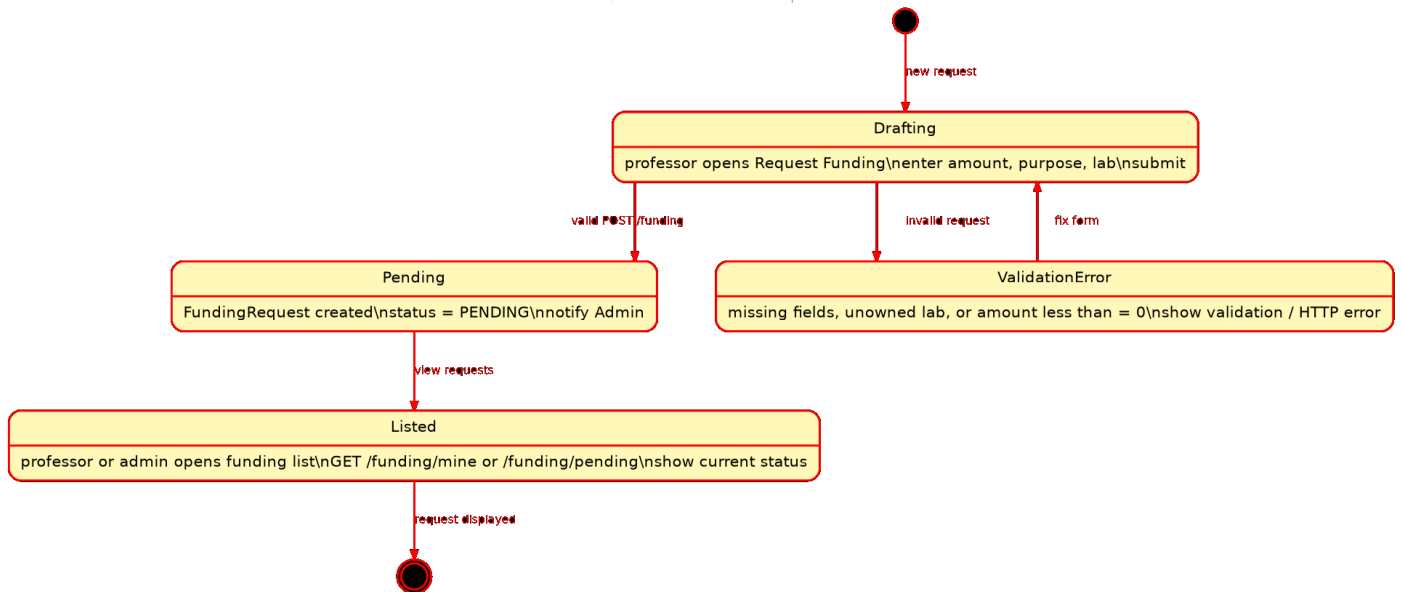
Undergraduate Lab Portal Requirements Specification

State Diagram
Lab Membership

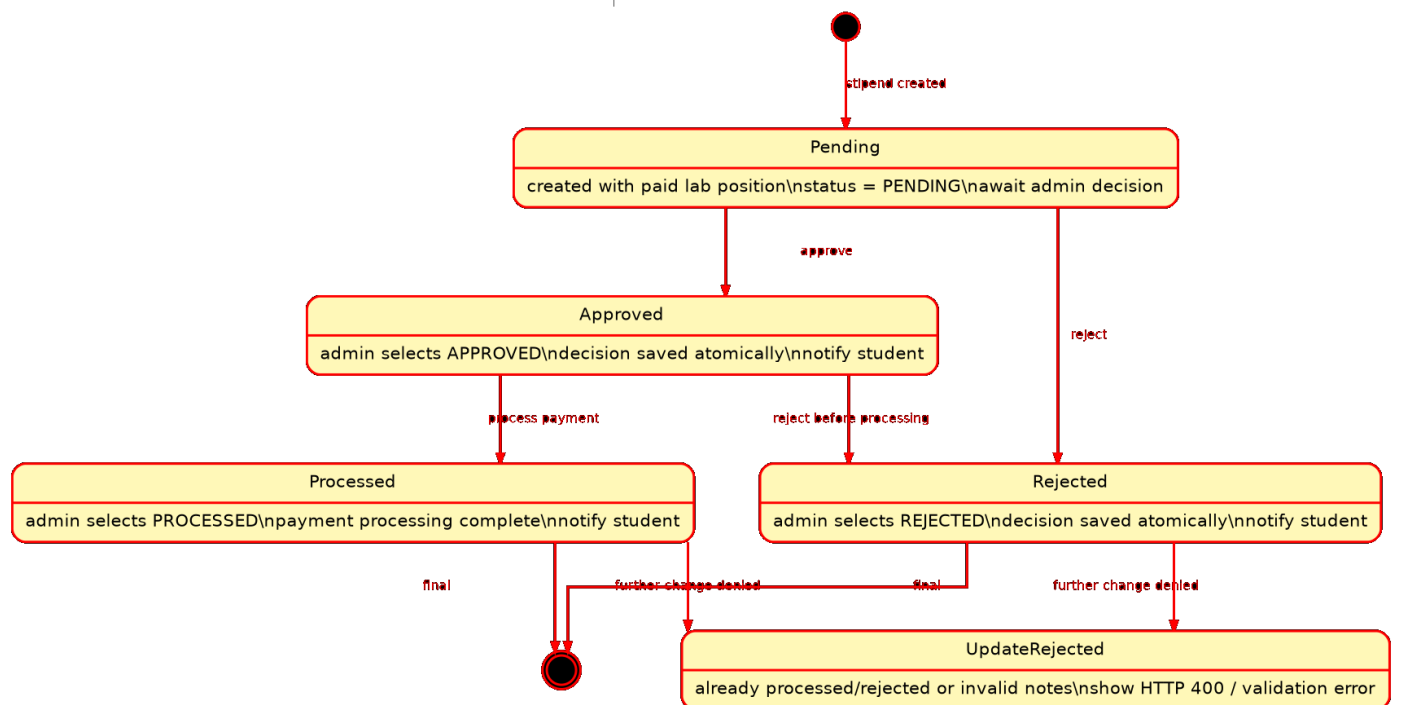


Undergraduate Lab Portal Requirements Specification

State Diagram
Funding Request

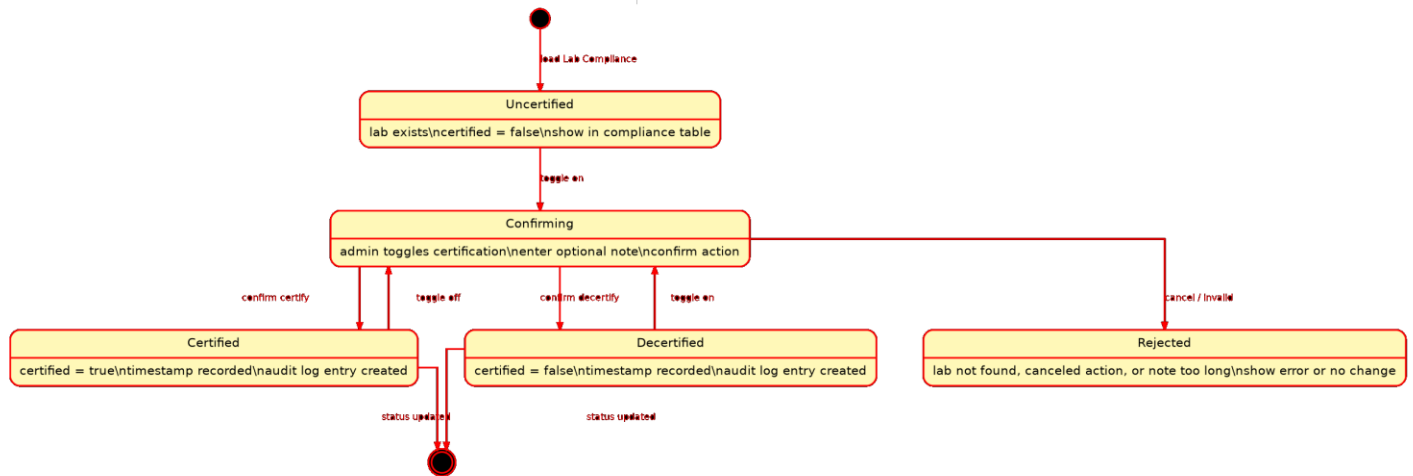


State Diagram
Stipend



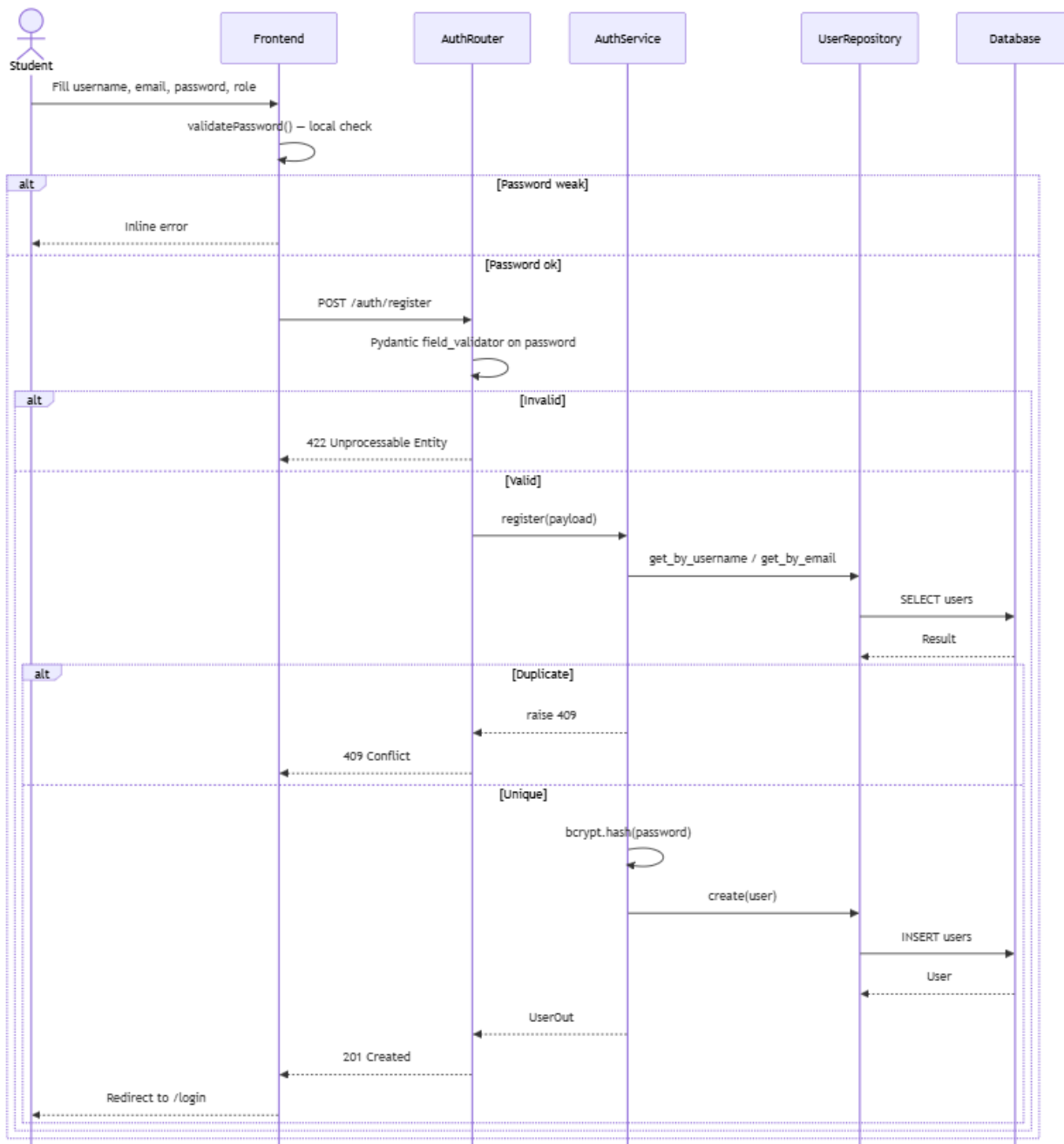
Undergraduate Lab Portal Requirements Specification

State Diagram
Lab Certification



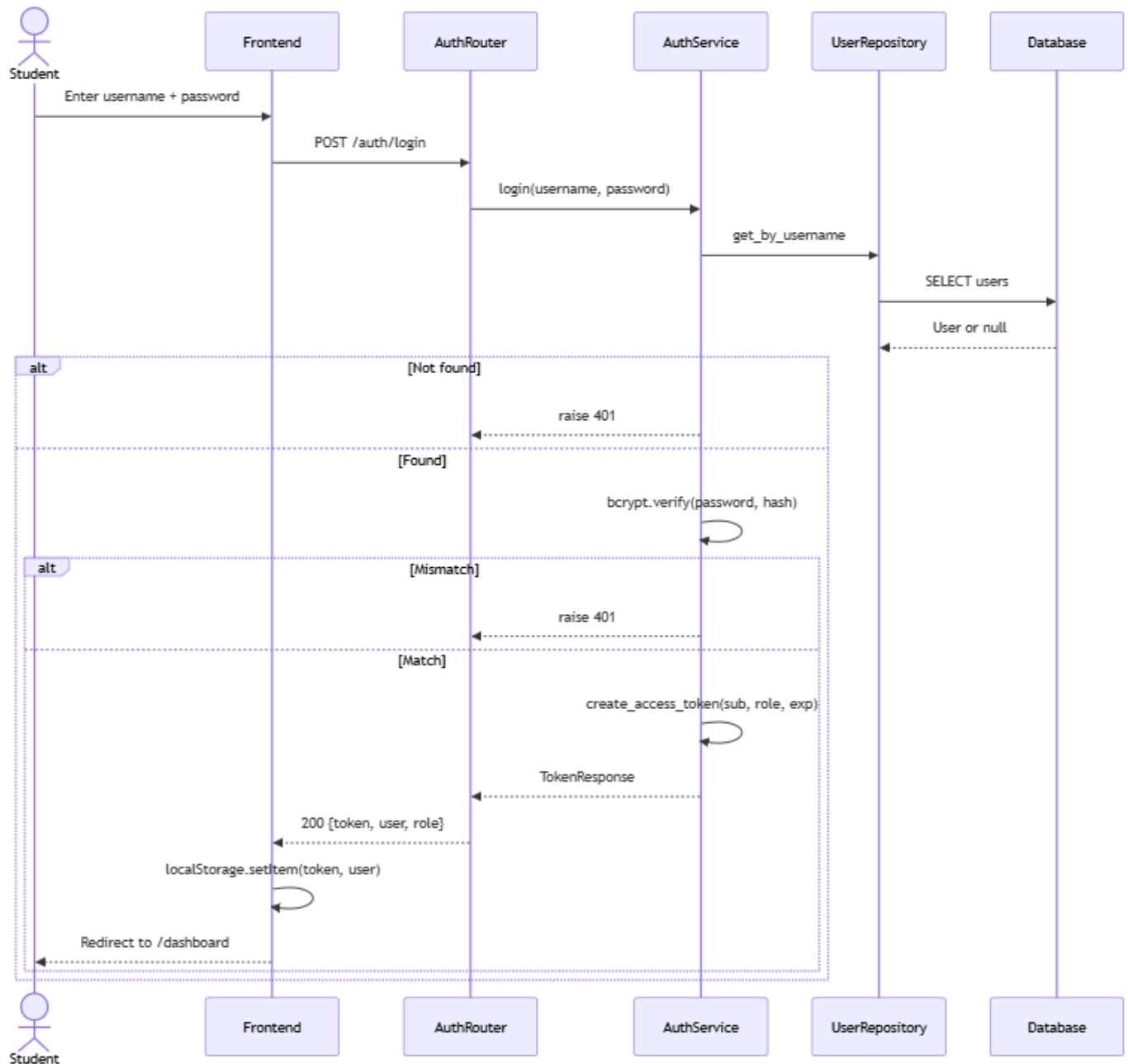
4.3 Sequence Diagrams

SD-S-01: Register an Account



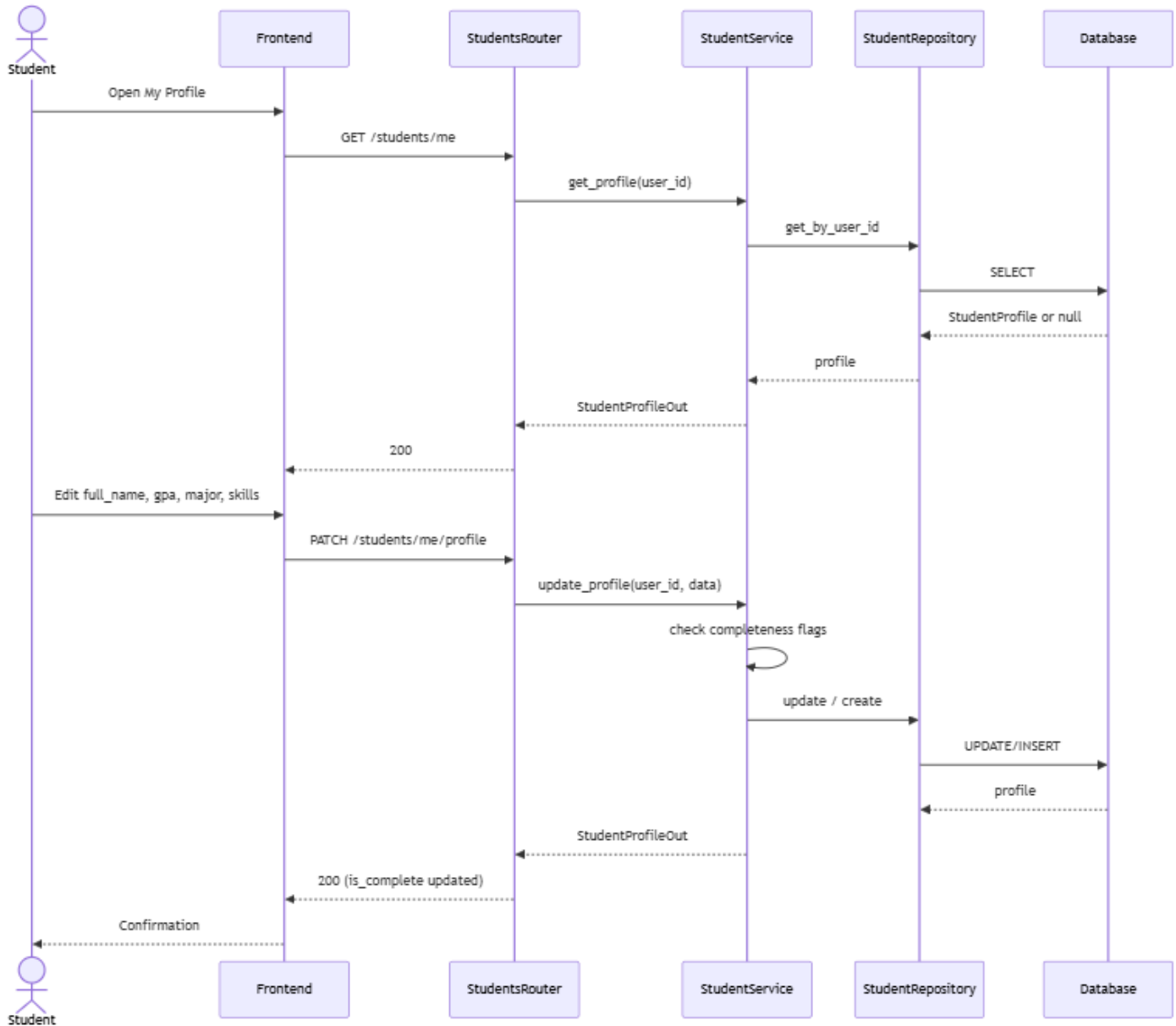
Undergraduate Lab Portal Requirements Specification

SD-S-02: Log In



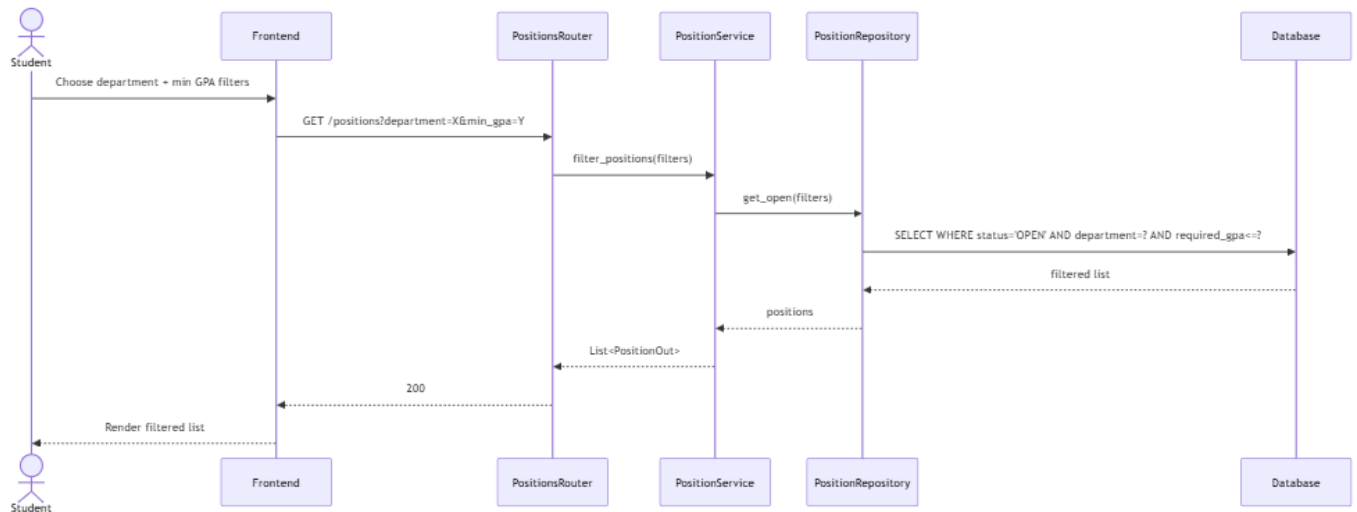
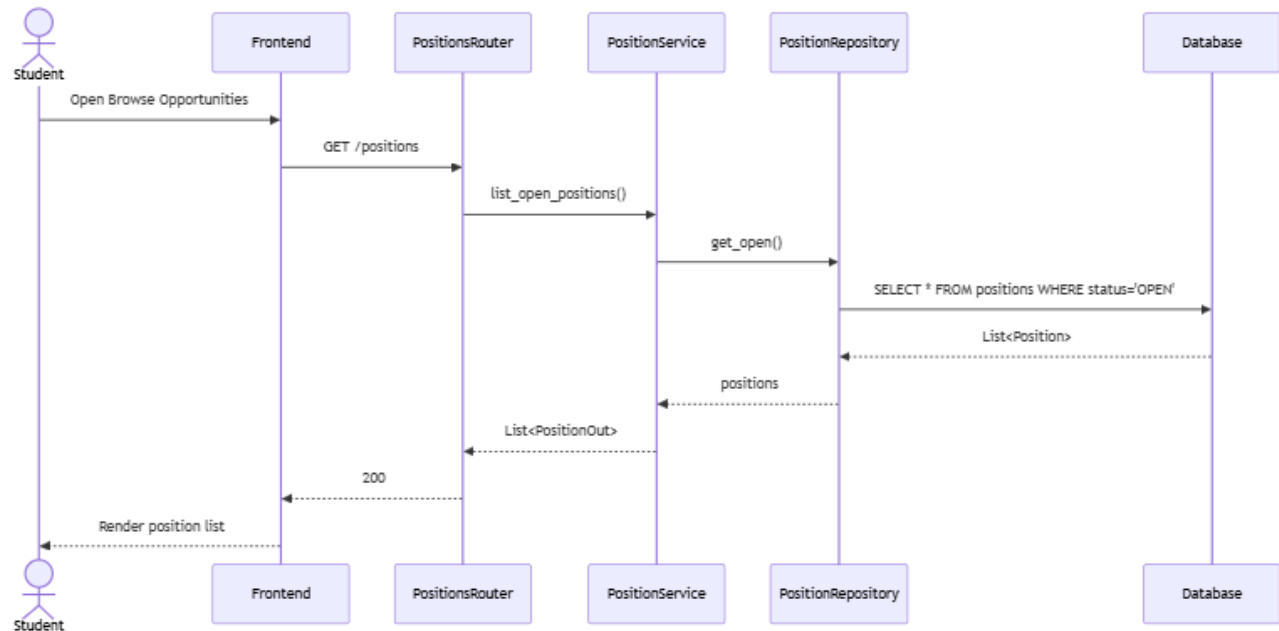
Undergraduate Lab Portal Requirements Specification

SD-S-03: Manage Profile



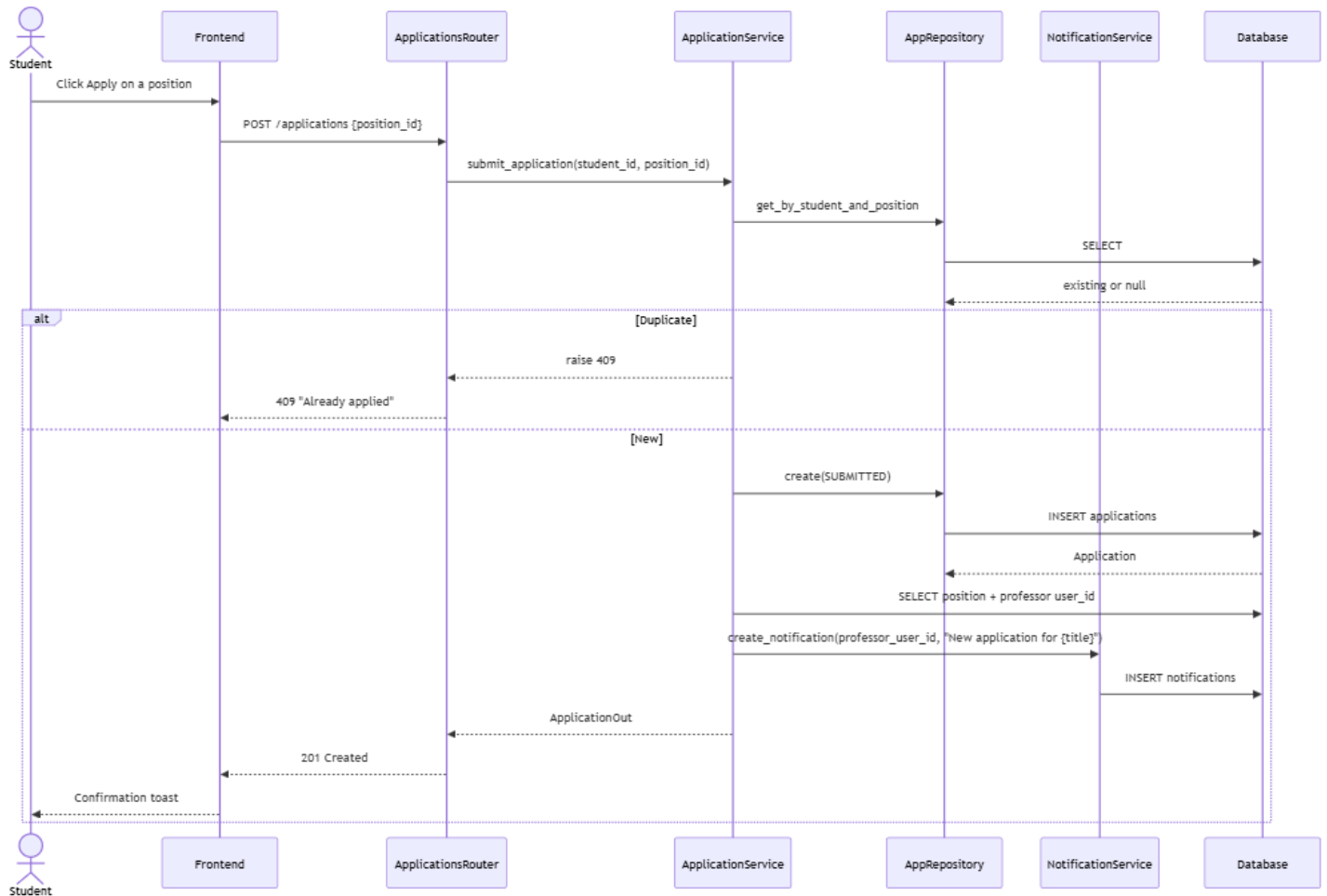
Undergraduate Lab Portal Requirements Specification

SD-S-04: Browse Open Positions



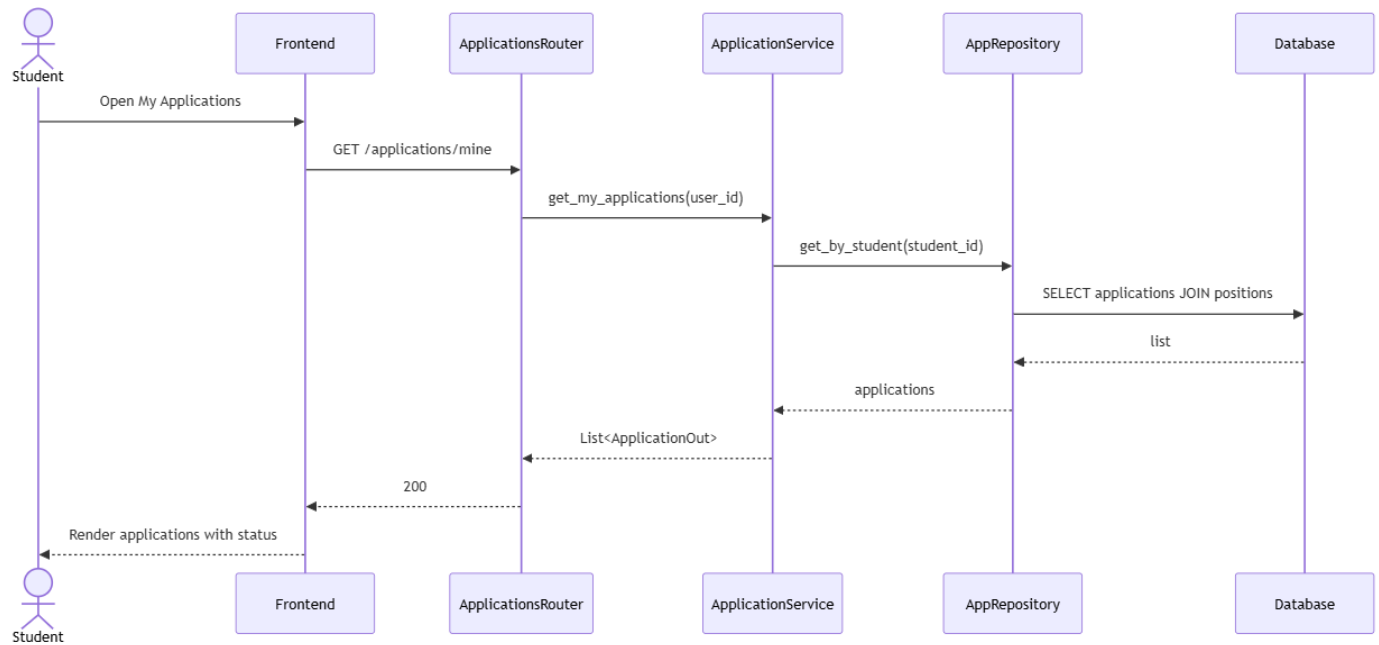
Undergraduate Lab Portal Requirements Specification

SD-S-06: Apply for a Position

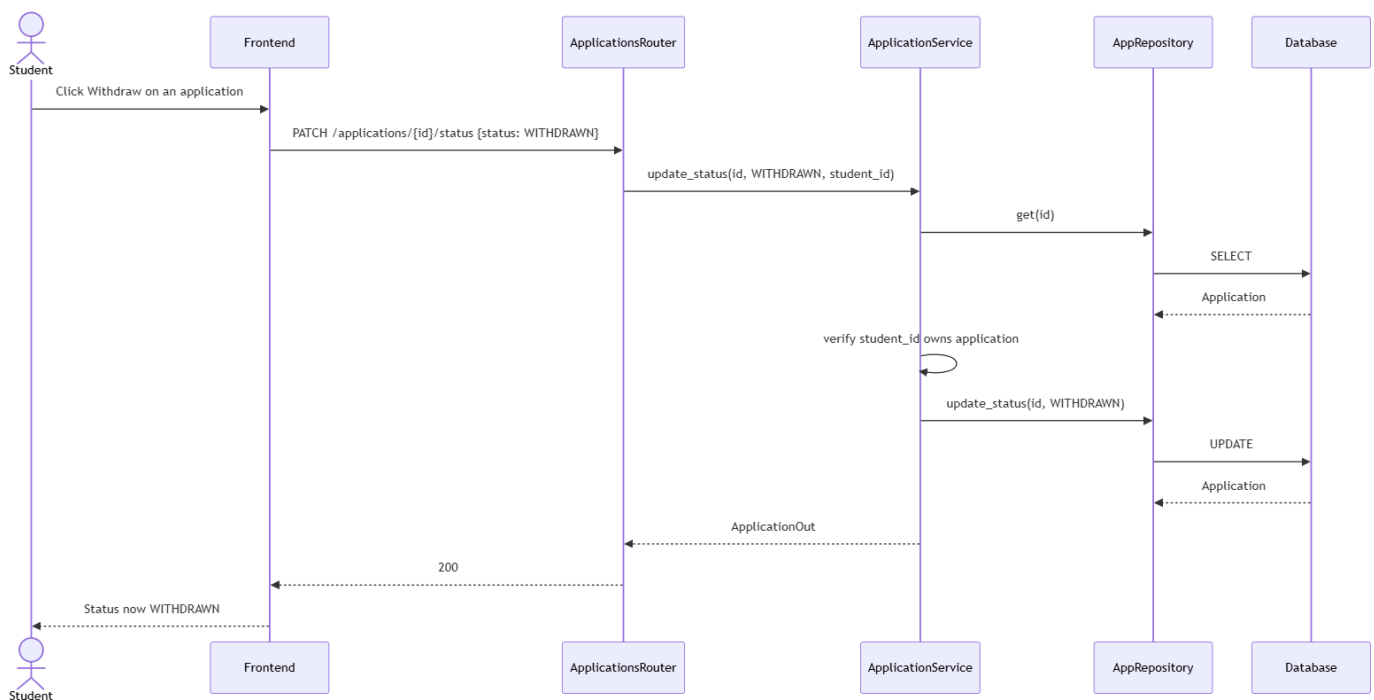


Undergraduate Lab Portal Requirements Specification

SD-S-07: View My Applications

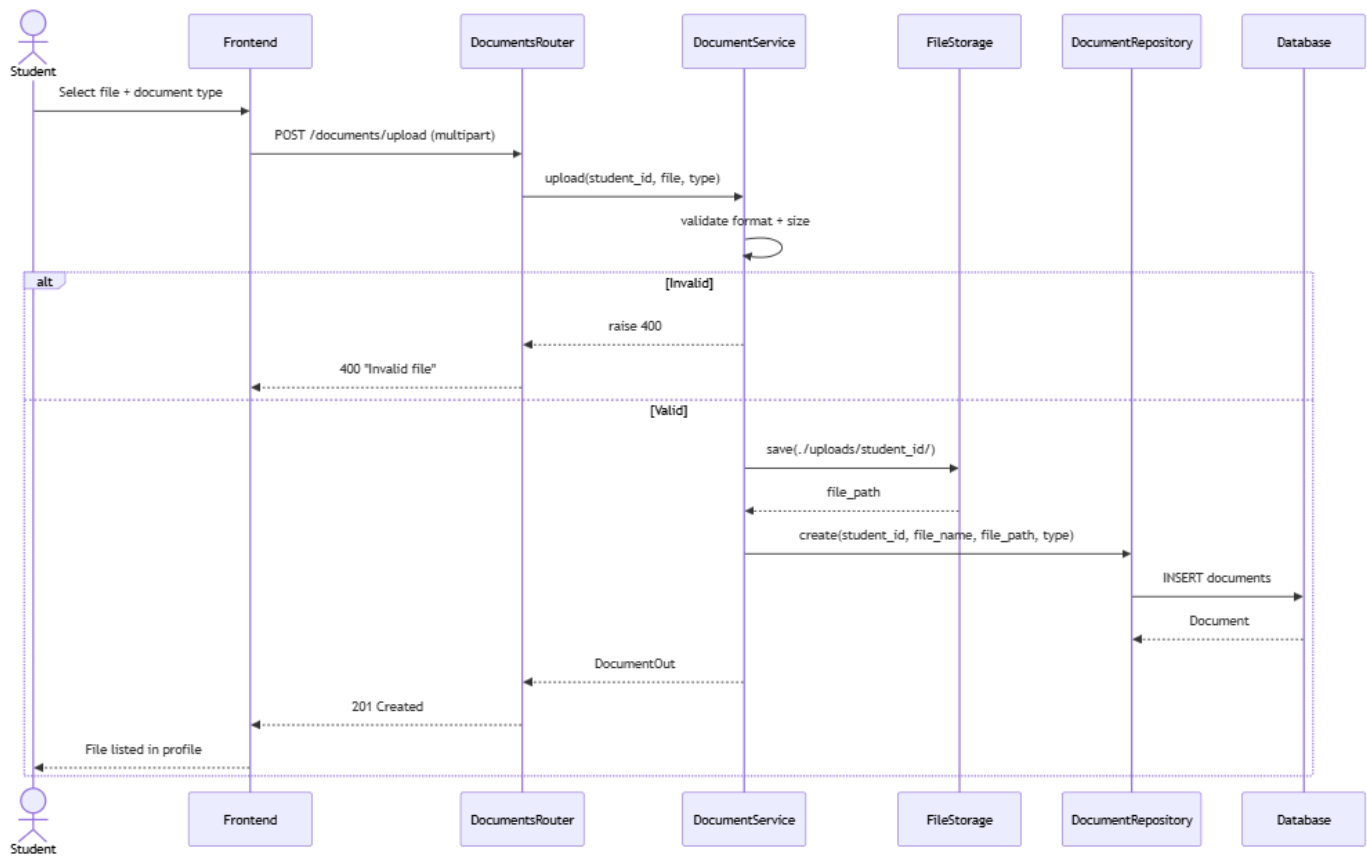


SD-S-08: Withdraw Application



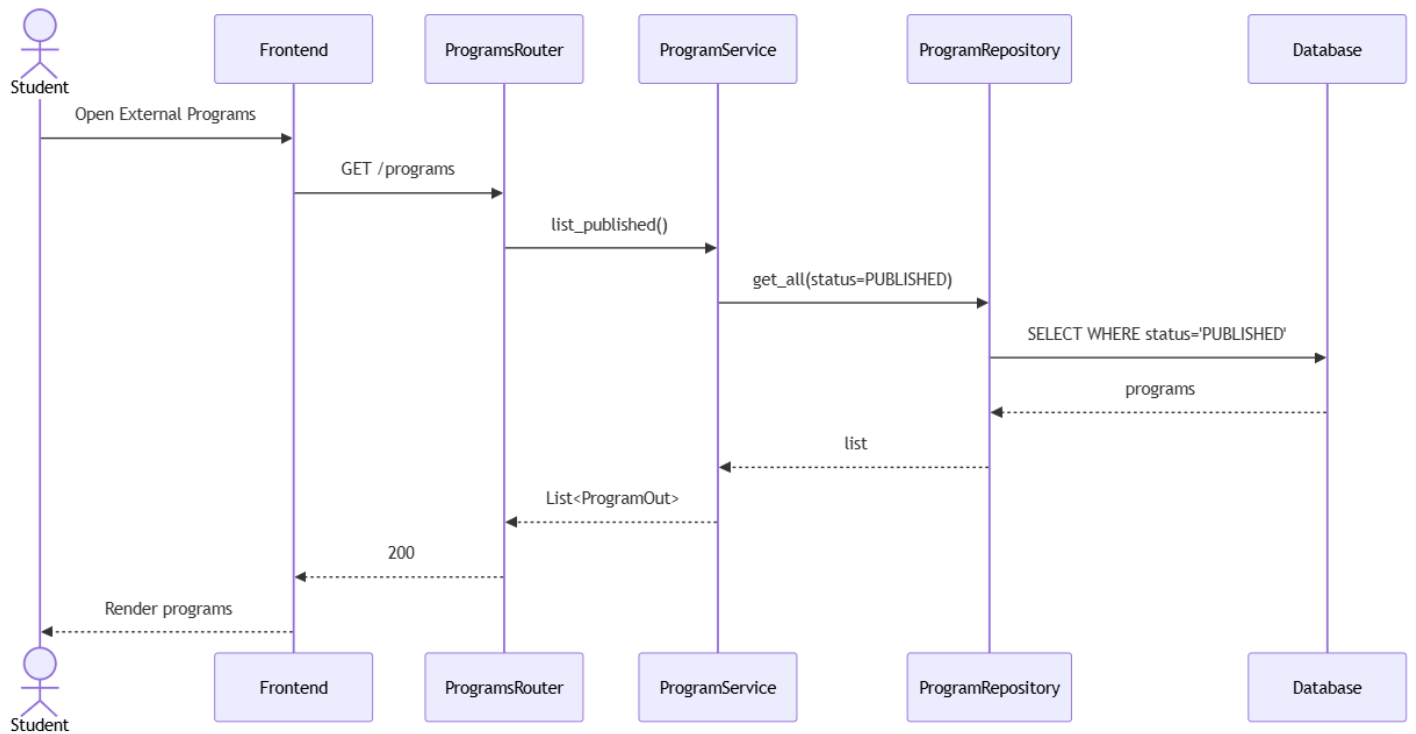
Undergraduate Lab Portal Requirements Specification

SD-S-09: Upload Documents



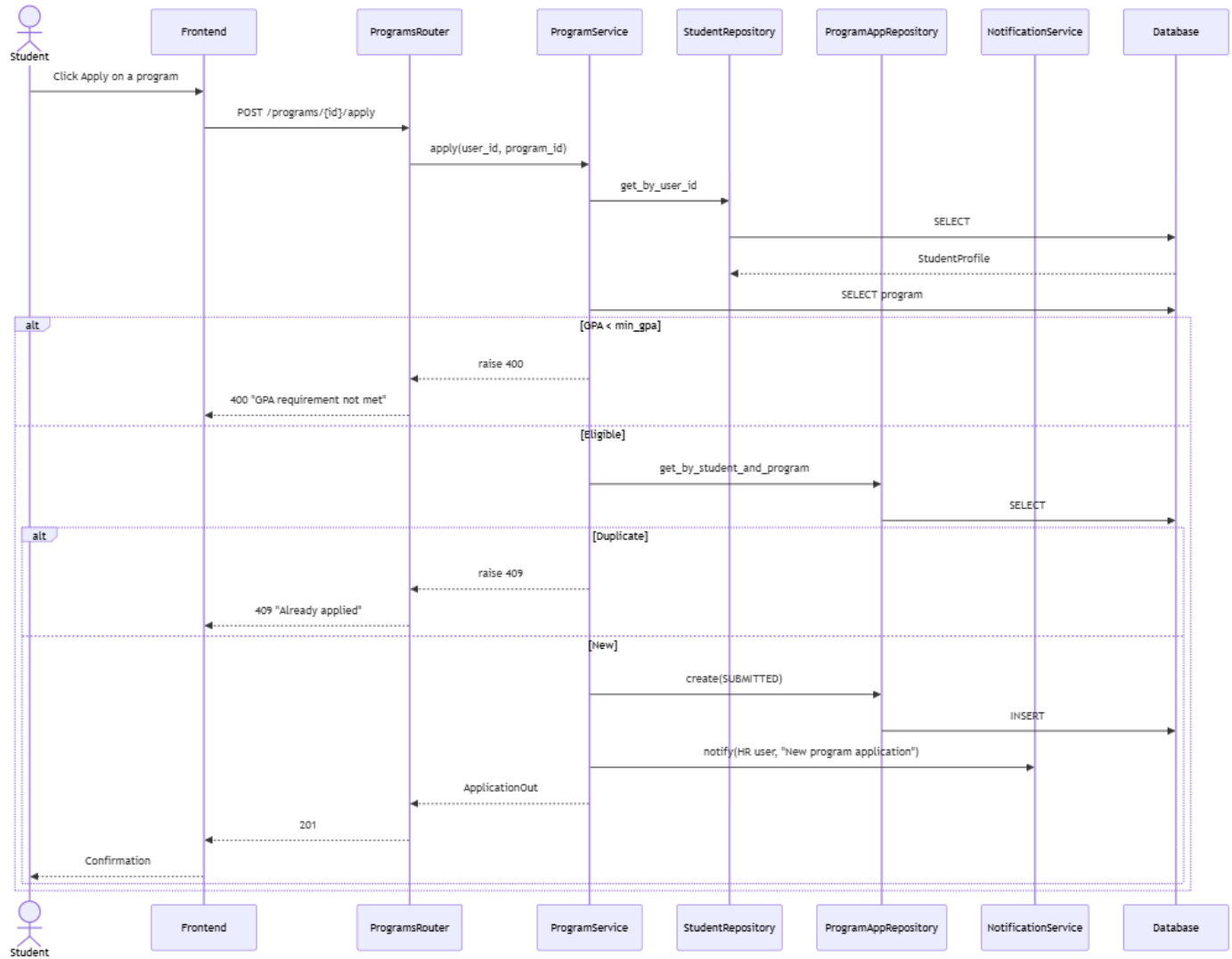
Undergraduate Lab Portal Requirements Specification

SD-S-10: Browse External Programs



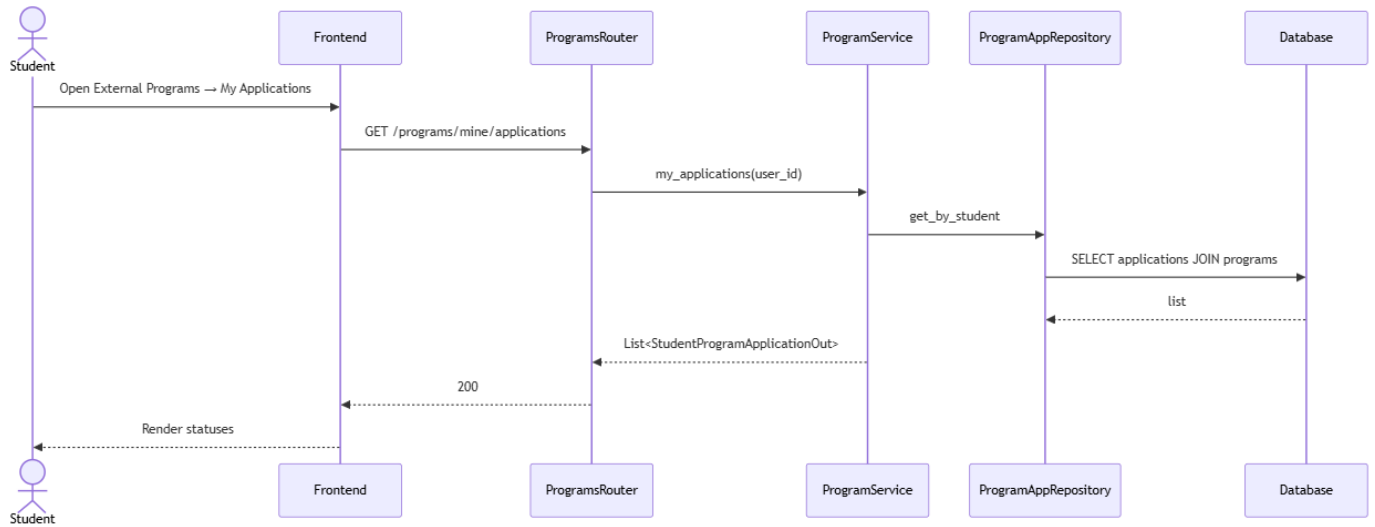
Undergraduate Lab Portal Requirements Specification

SD-S-11: Apply for a Program

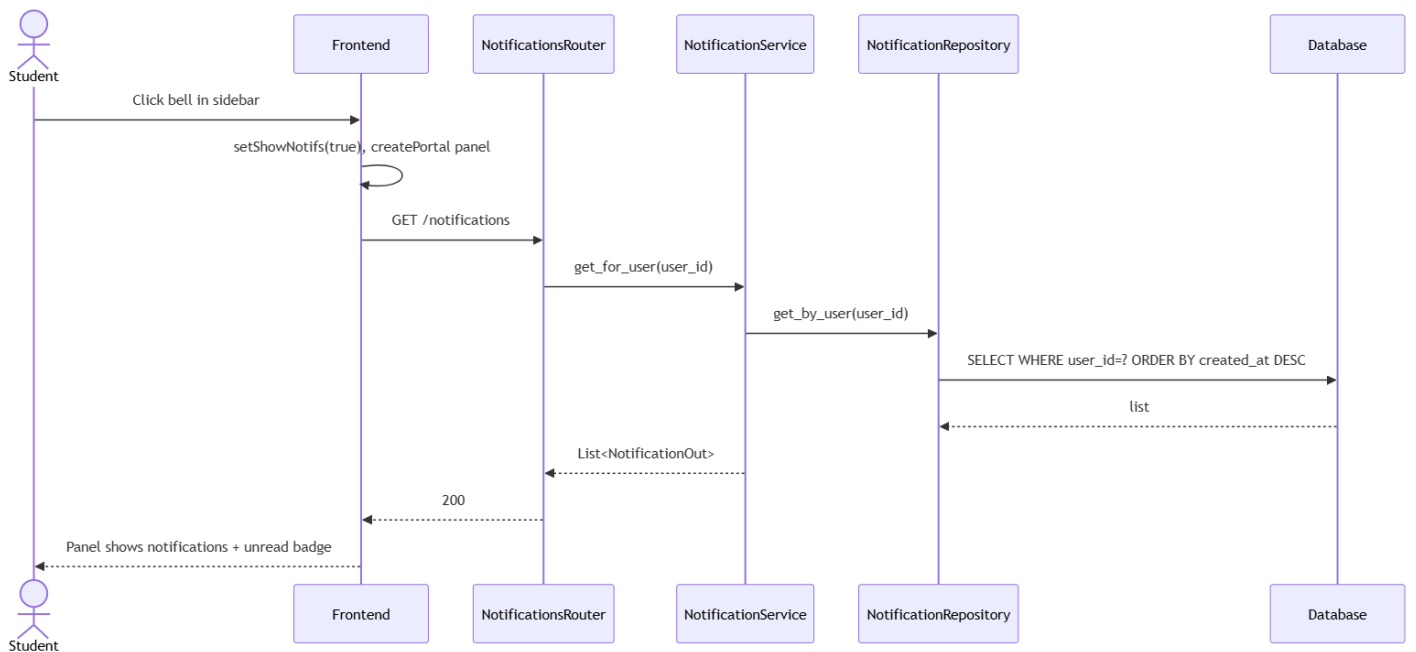


Undergraduate Lab Portal Requirements Specification

SD-S-12: View My Program Applications

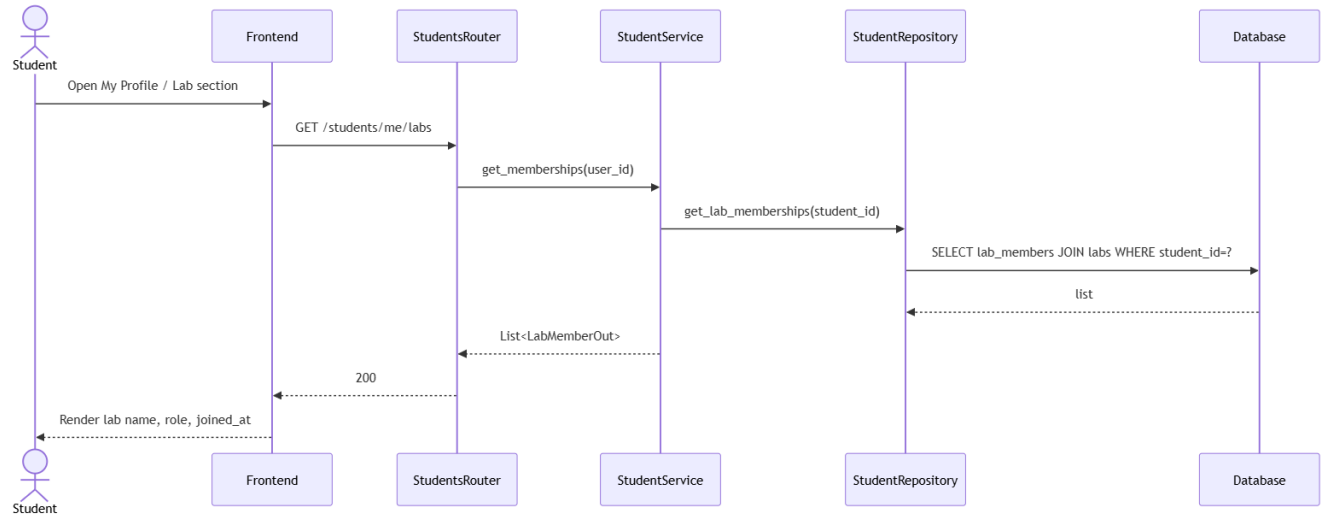


SD-S-13: View Notifications



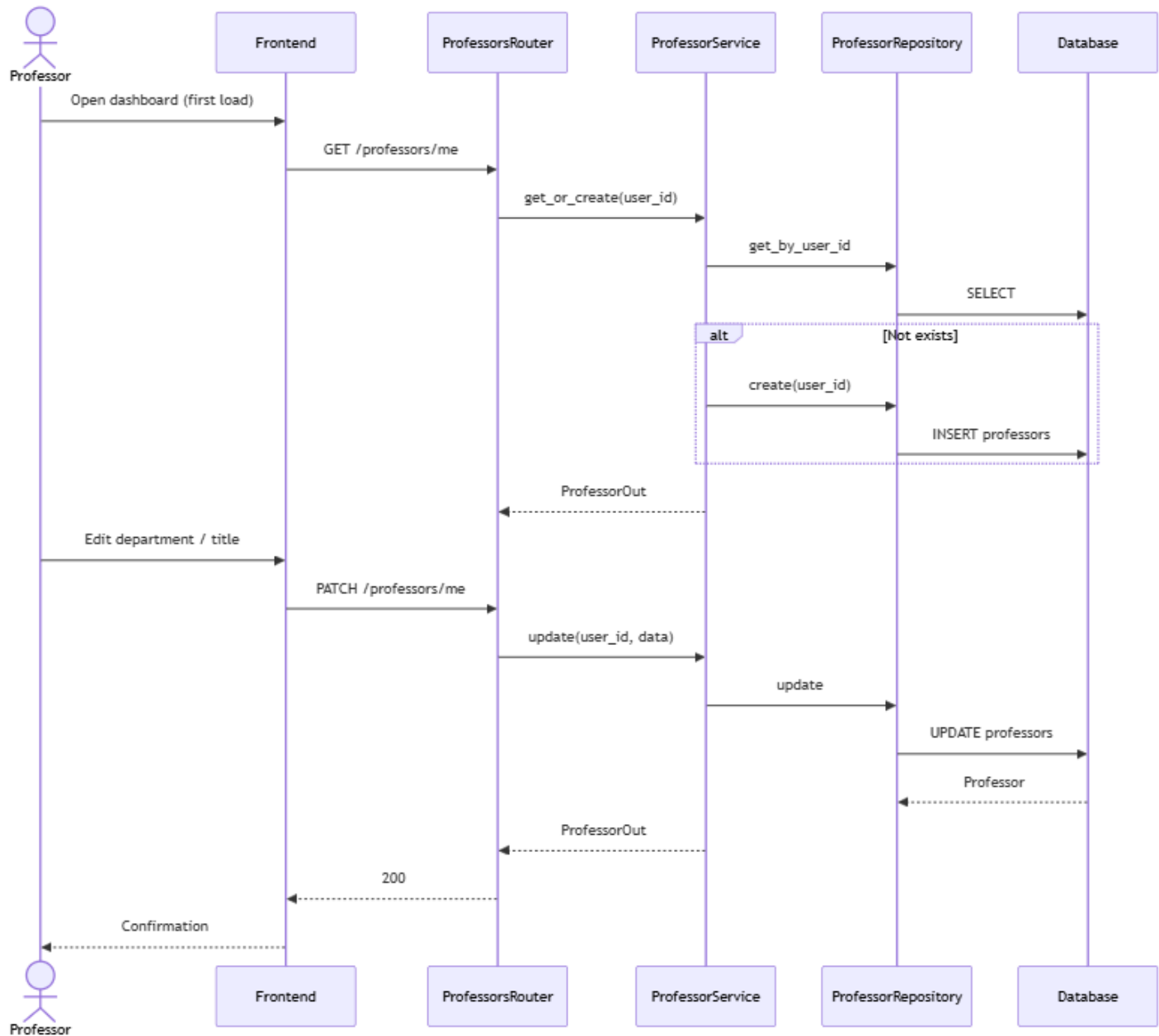
Undergraduate Lab Portal Requirements Specification

SD-S-15: View Lab Membership



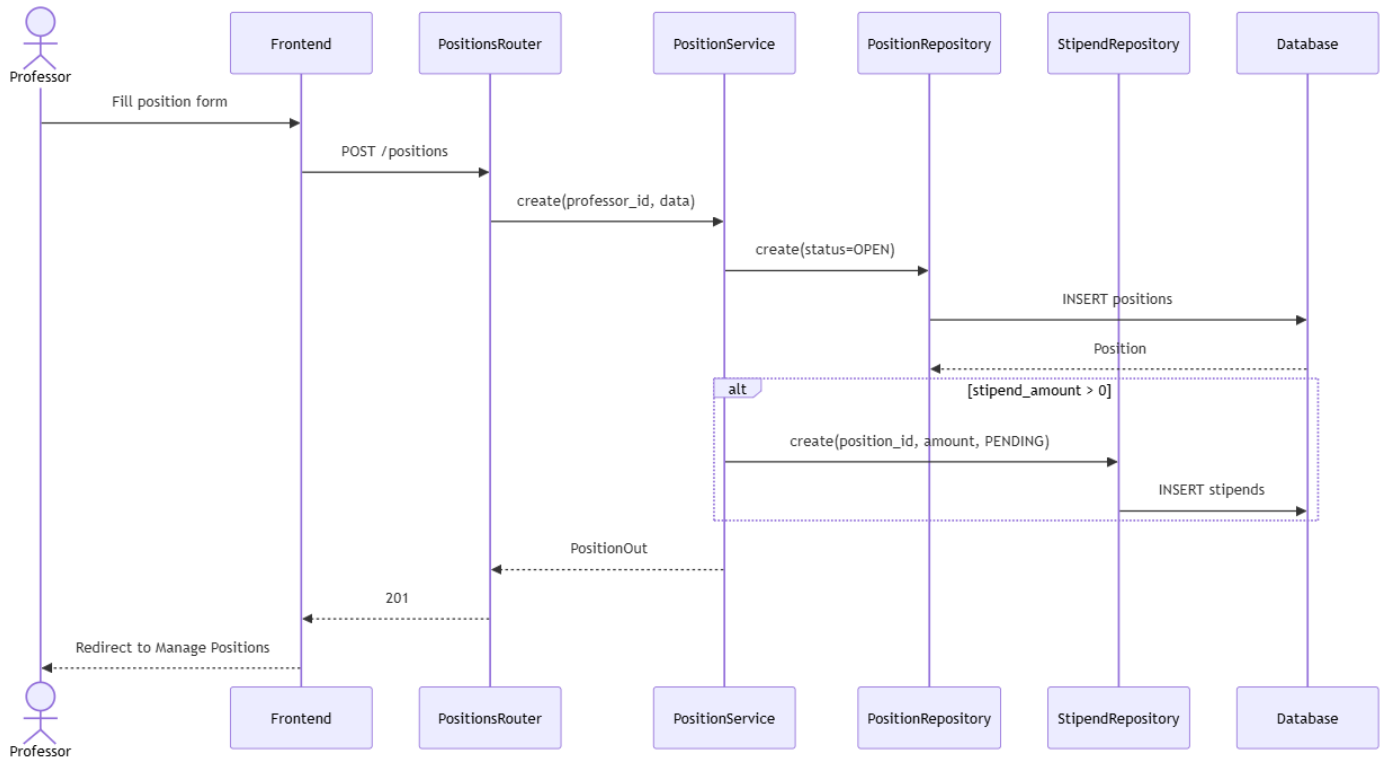
Undergraduate Lab Portal Requirements Specification

SD-P-01: Update Professor Profile



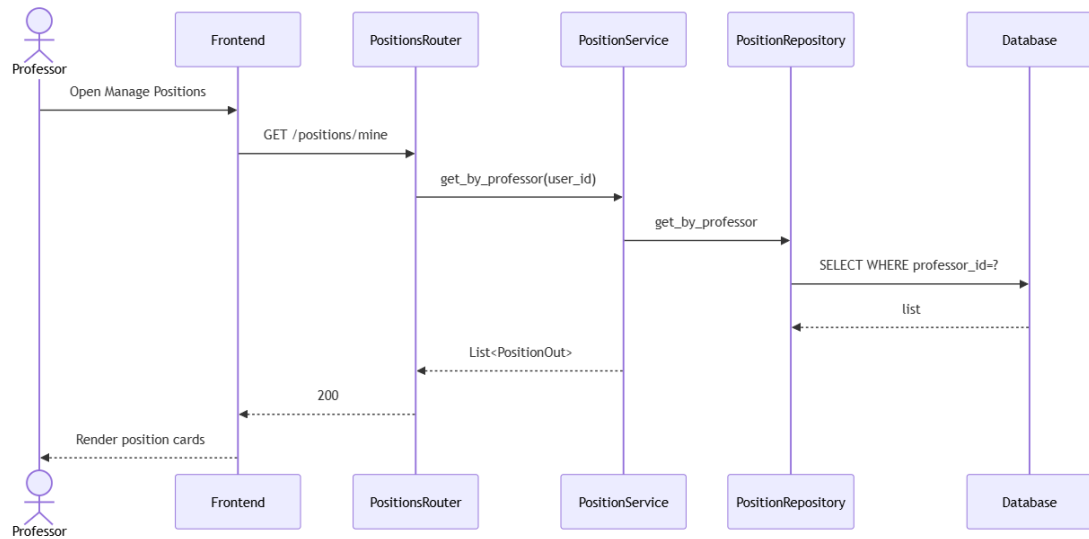
Undergraduate Lab Portal Requirements Specification

SD-P-02: Create a Lab Position

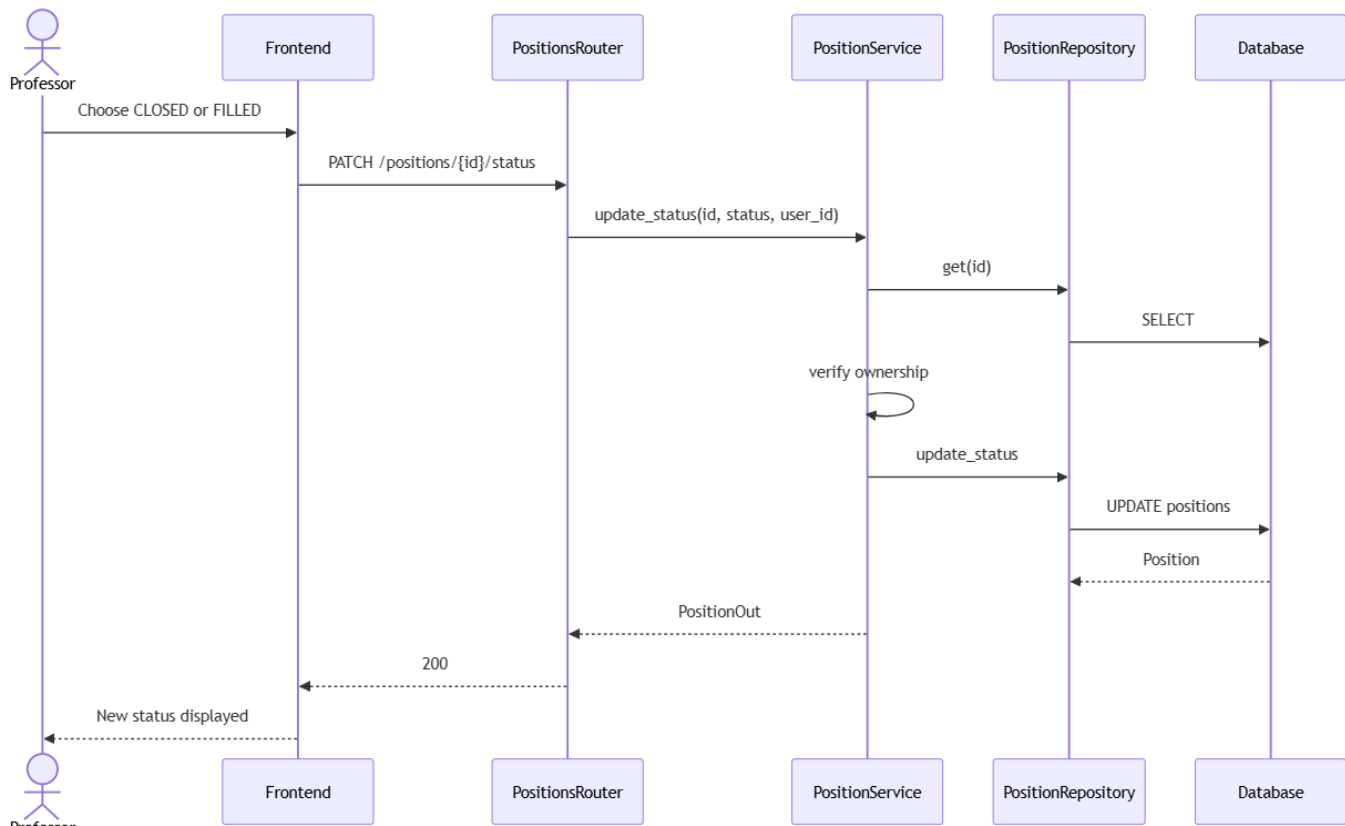


Undergraduate Lab Portal Requirements Specification

SD-P-03: View My Positions

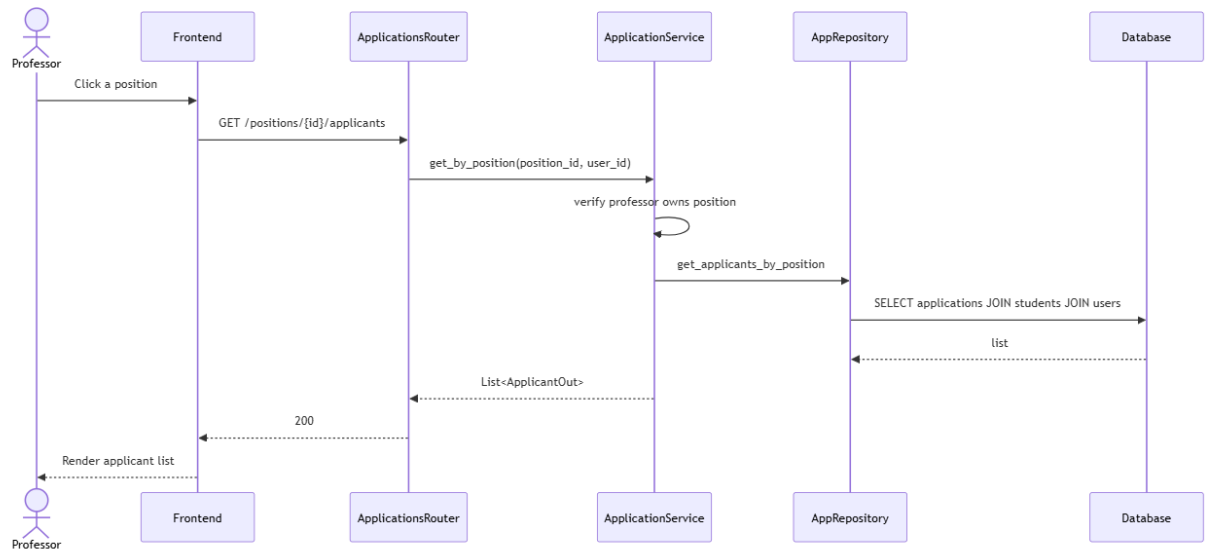


SD-P-04: Update Position Status

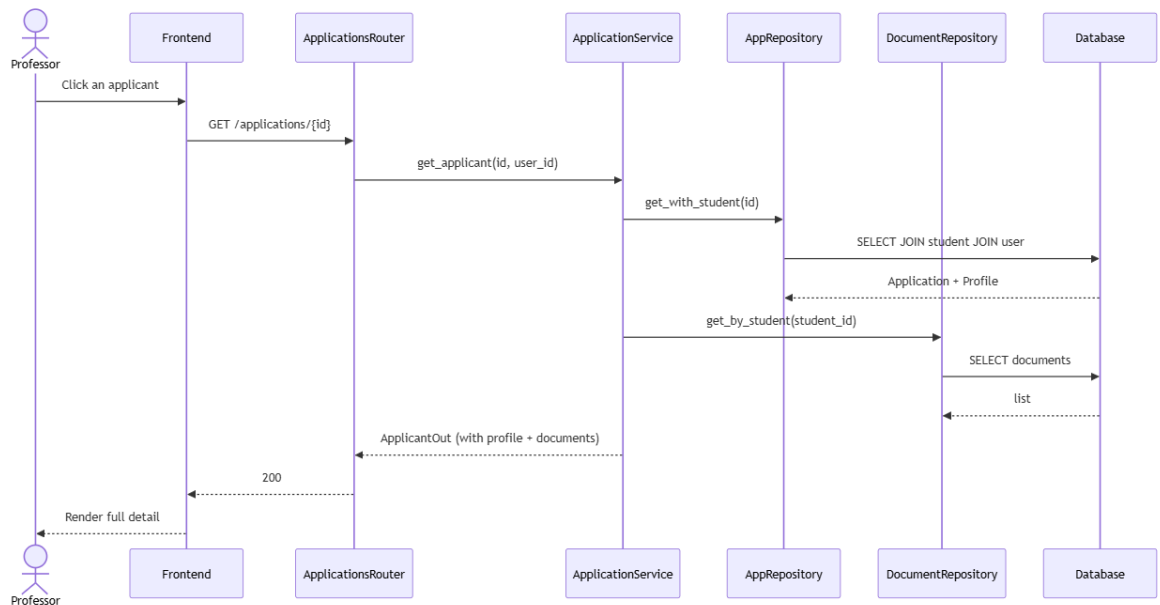


Undergraduate Lab Portal Requirements Specification

SD-P-05: View Applicants for a Position

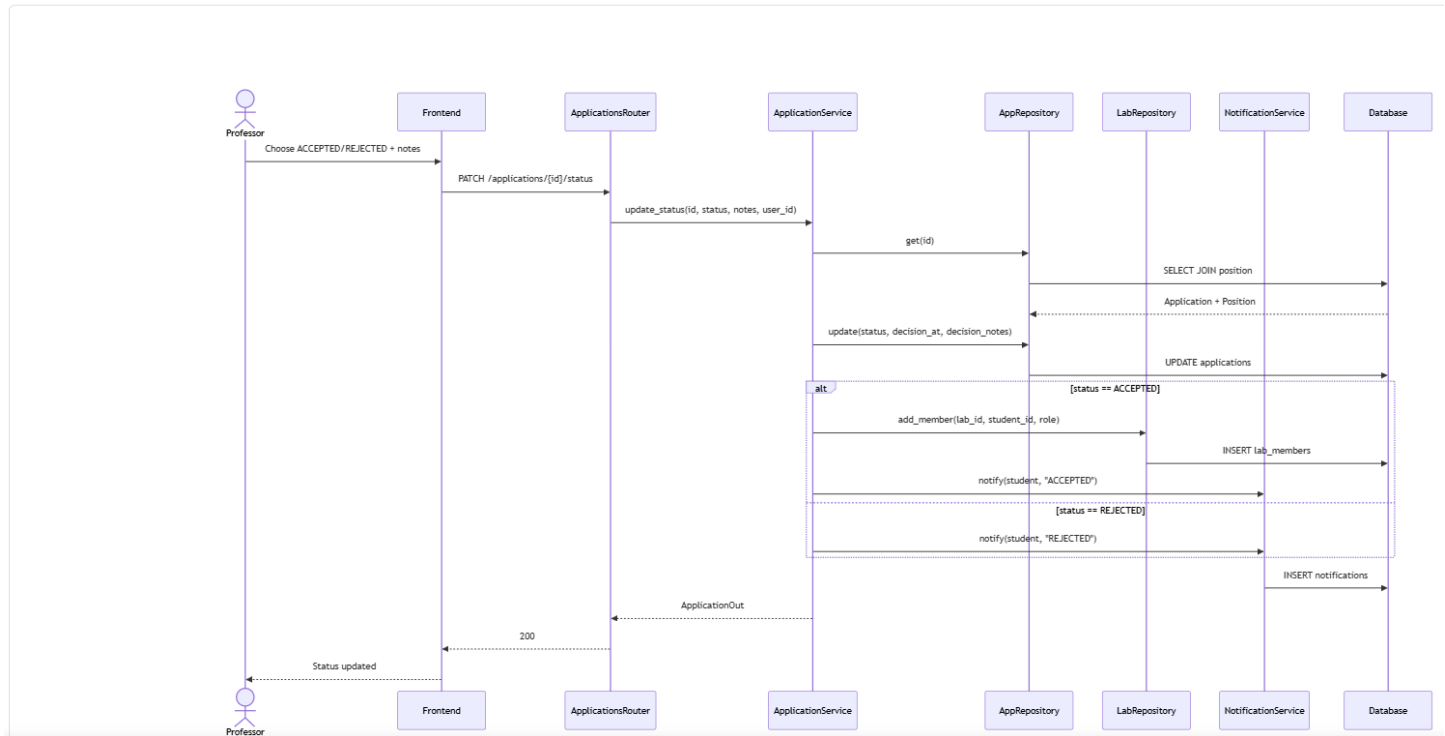


SD-P-06: Review Applicant Detail

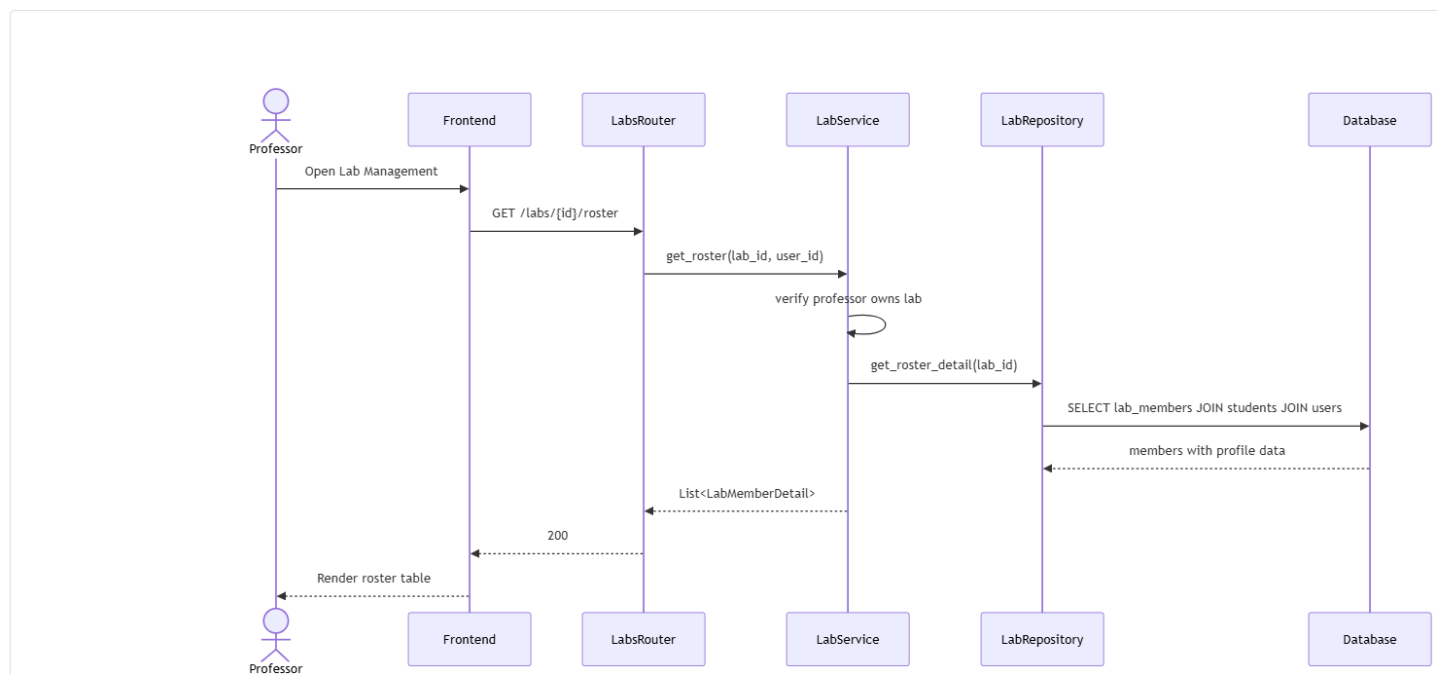


Undergraduate Lab Portal Requirements Specification

SD-P-07: Accept or Reject Application

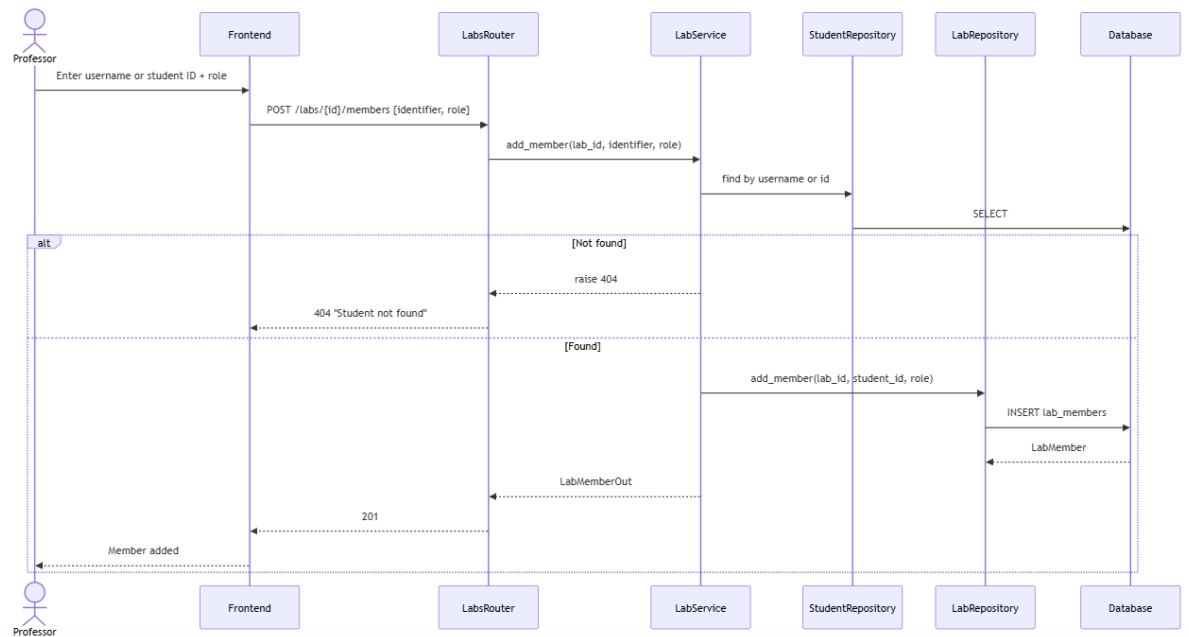


SD-P-08: View Lab Roster

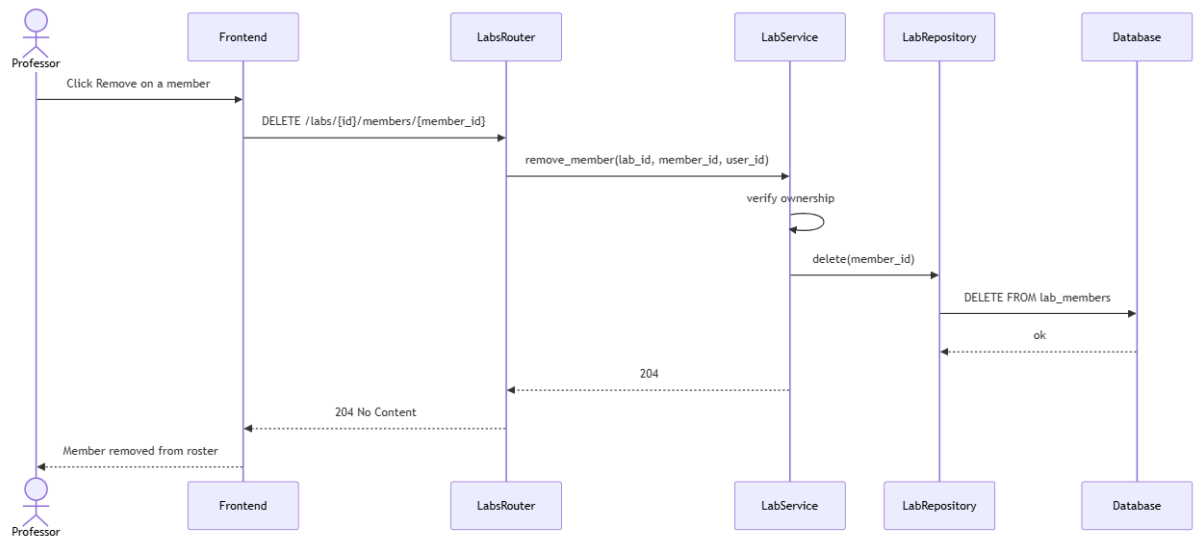


Undergraduate Lab Portal Requirements Specification

SD-P-09: Add Lab Member

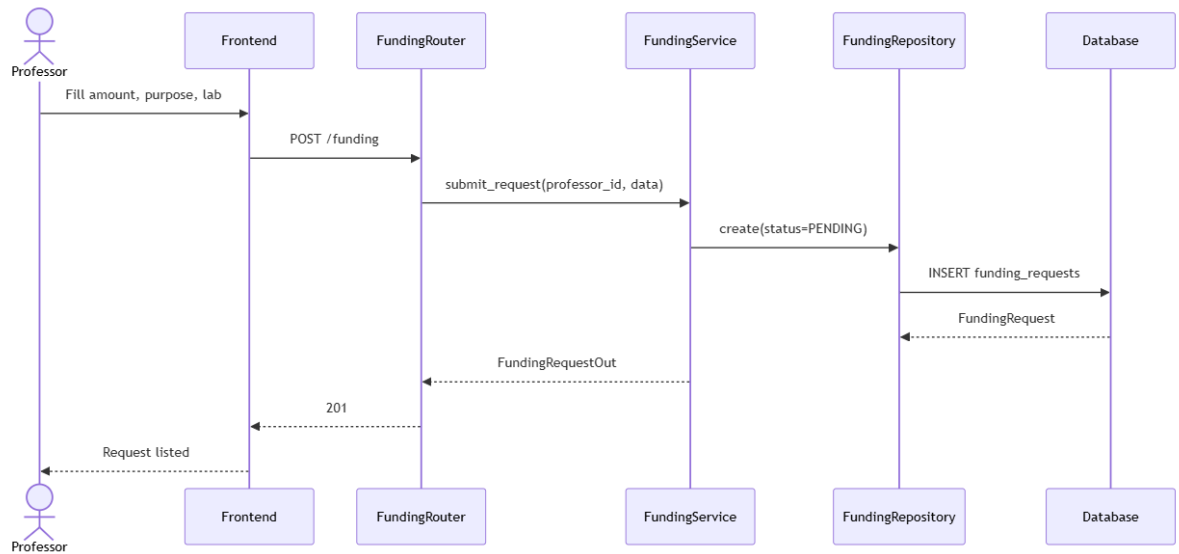


SD-P-10: Remove Lab Member

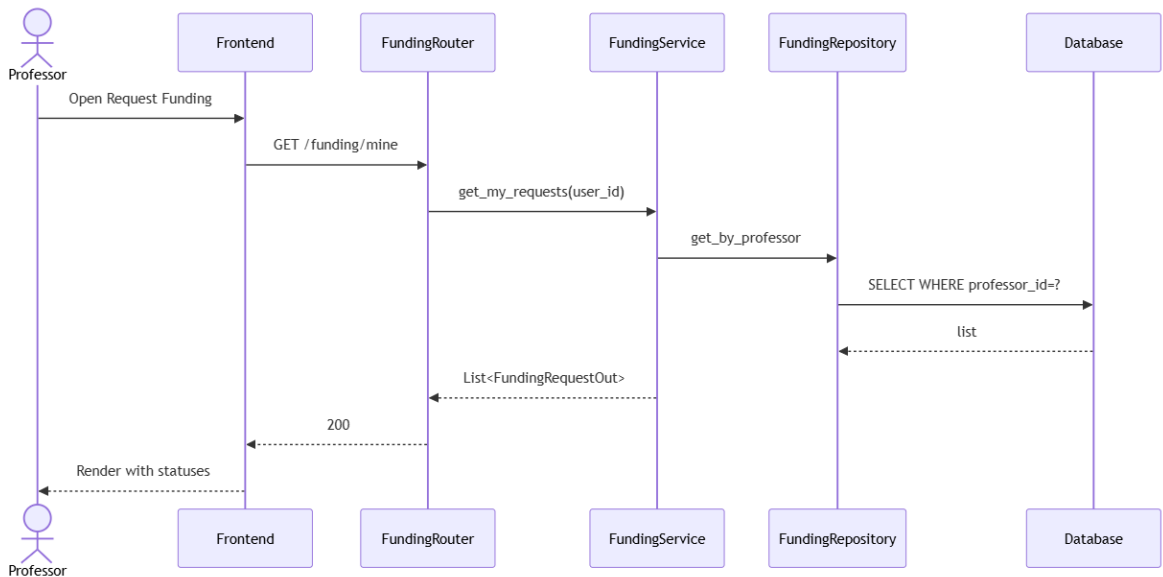


Undergraduate Lab Portal Requirements Specification

SD-P-11: Submit Funding Request

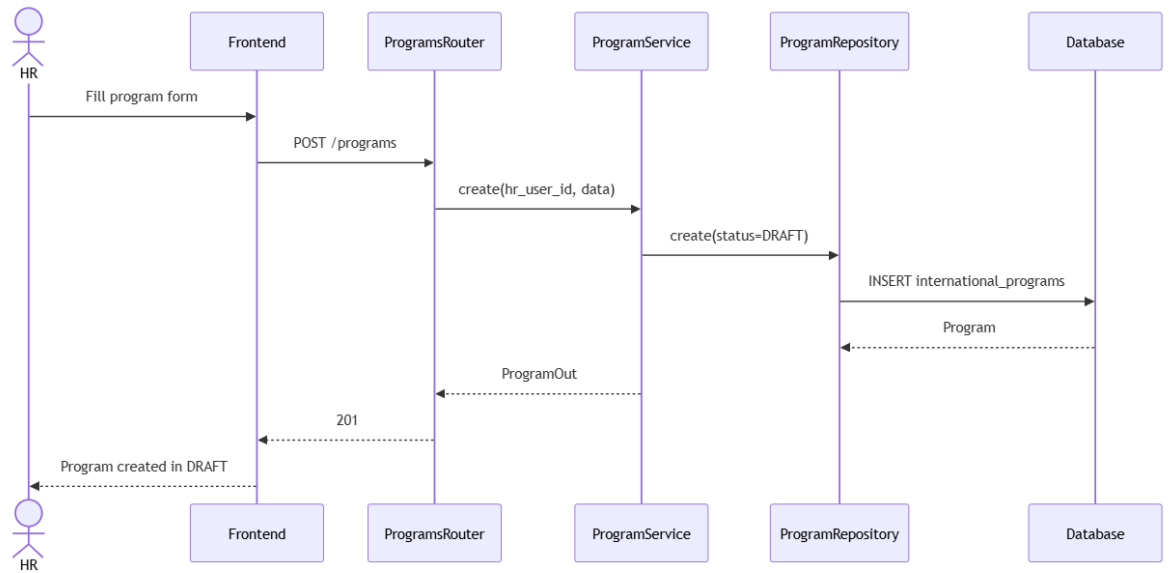


SD-P-12: View My Funding Requests

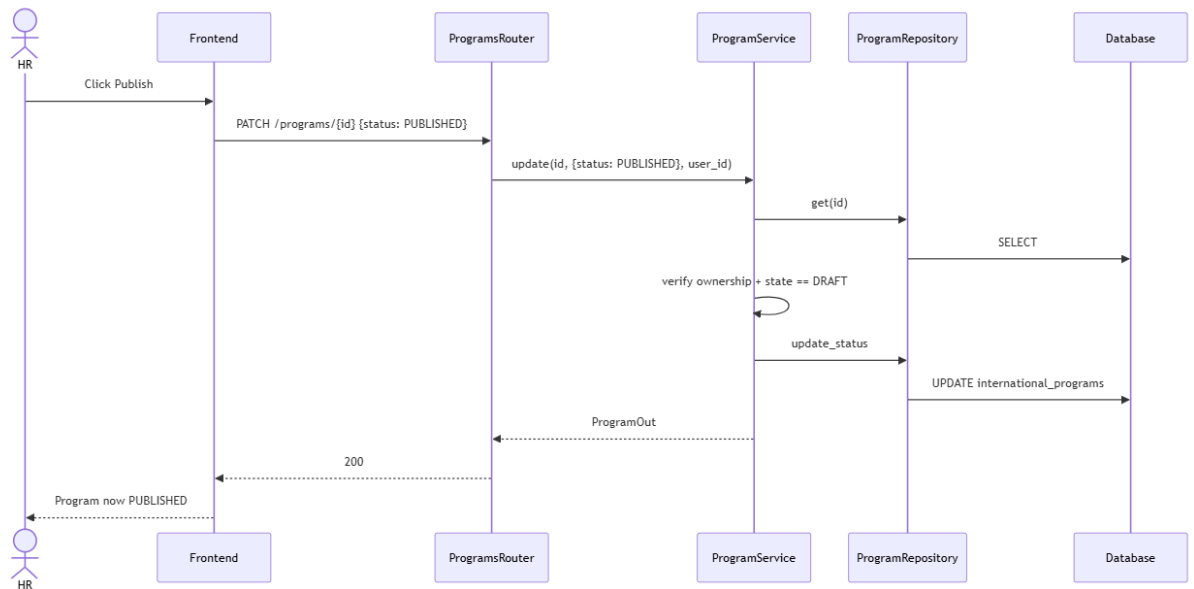


Undergraduate Lab Portal Requirements Specification

SD-HR-01: Create International Program

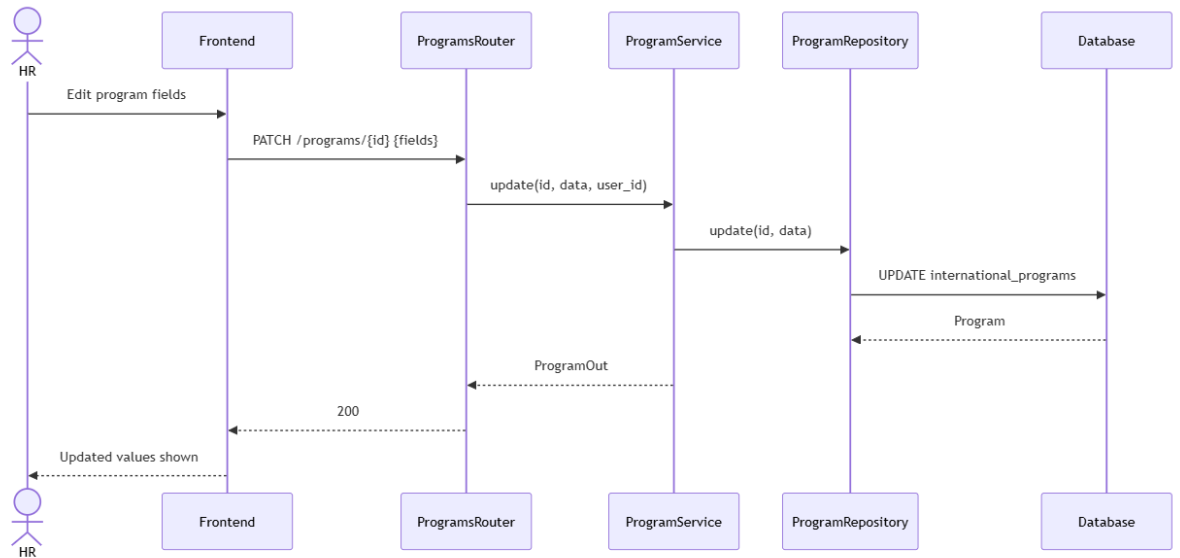


SD-HR-02: Publish a Program

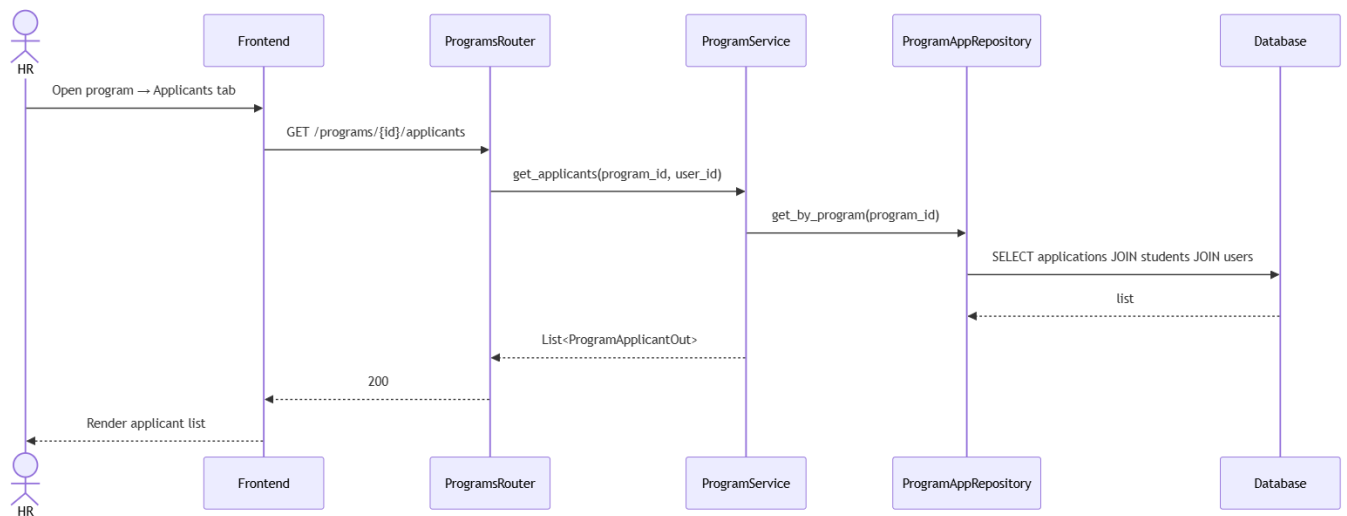


Undergraduate Lab Portal Requirements Specification

SD-HR-03: Update Program

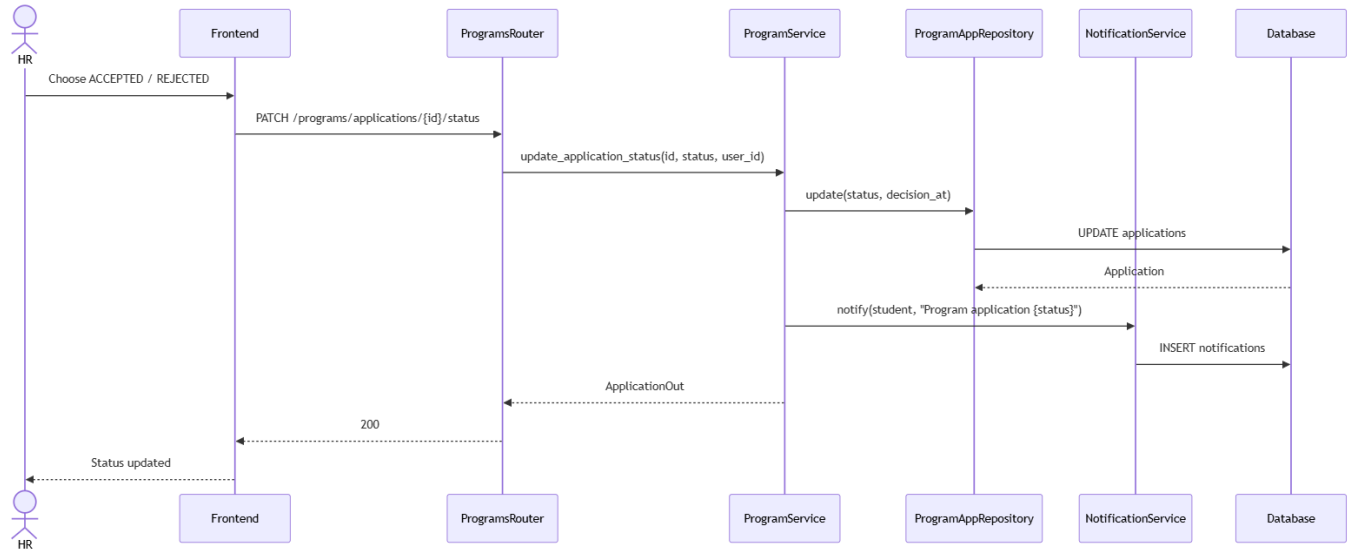


SD-HR-04: View Program Applicants

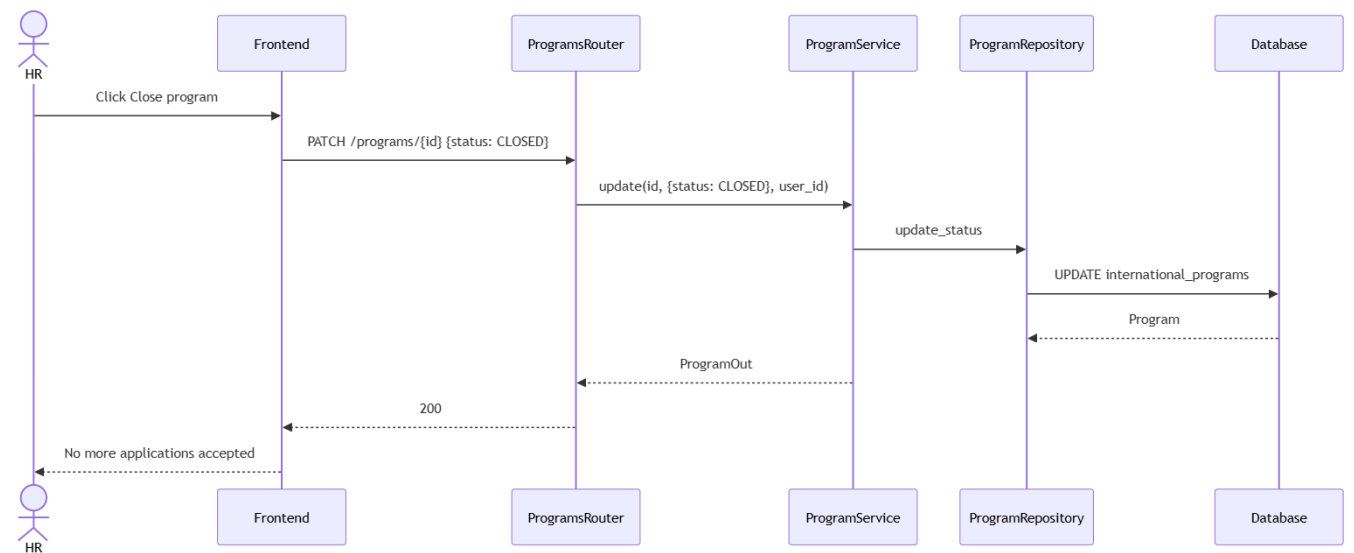


Undergraduate Lab Portal Requirements Specification

SD-HR-05: Decide Program Application

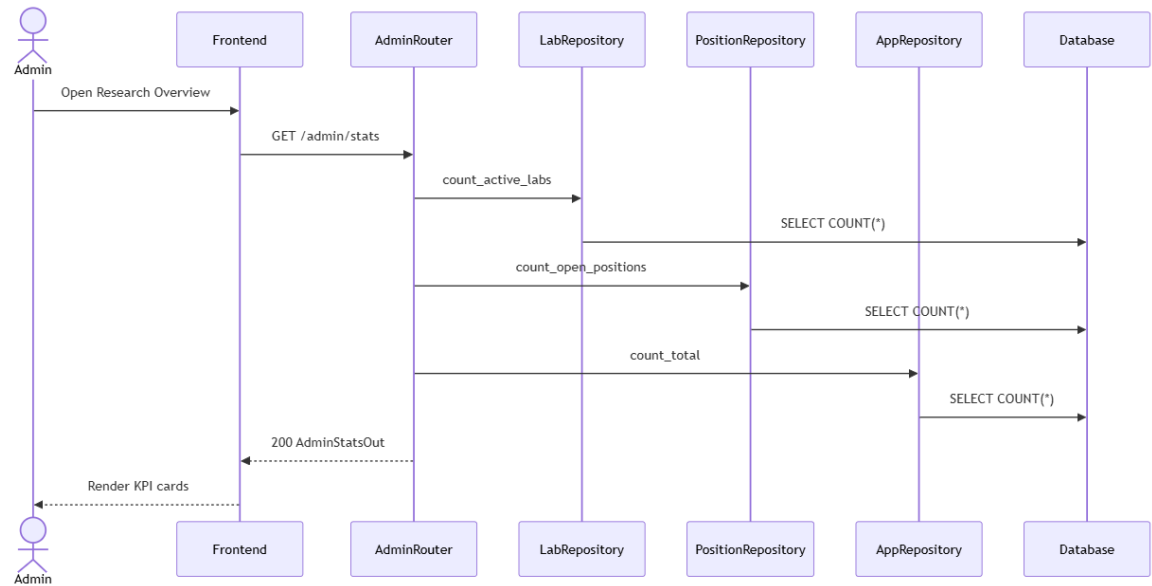


SD-HR-06: Close a Program

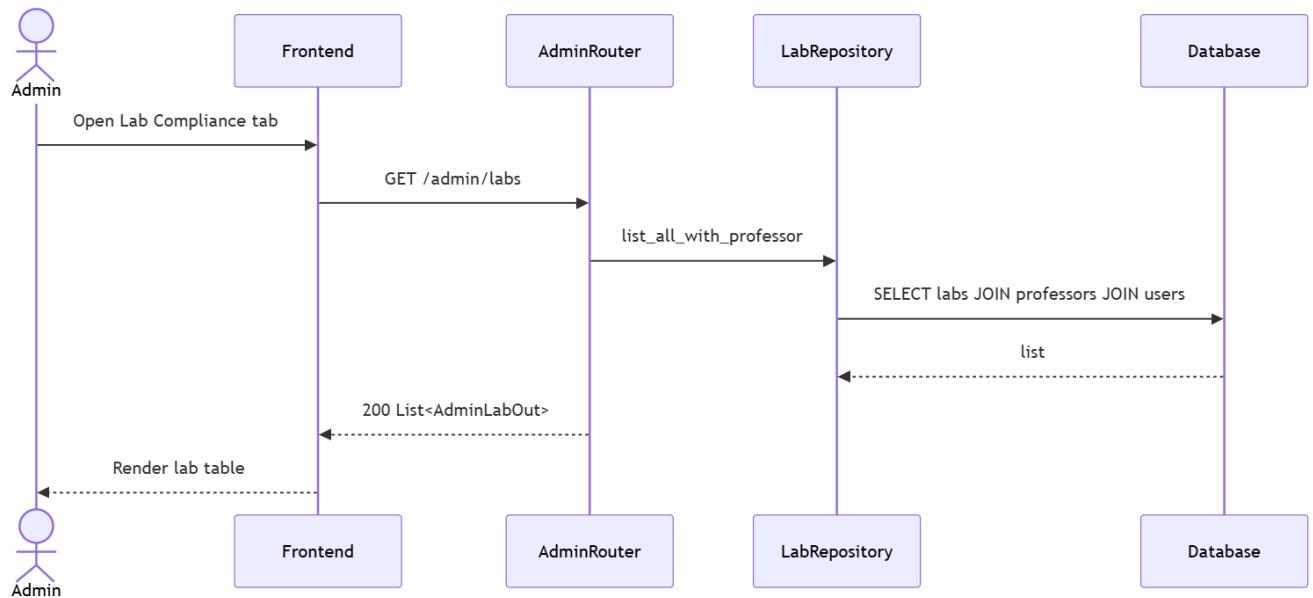


Undergraduate Lab Portal Requirements Specification

SD-A-01: View Research Statistics

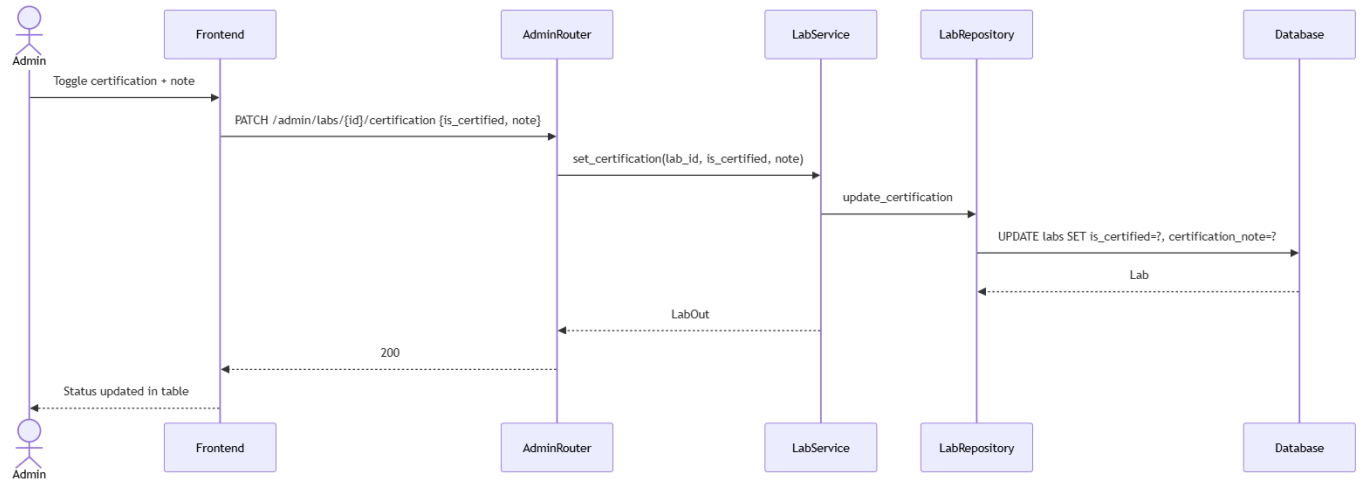


SD-A-02: View All Labs

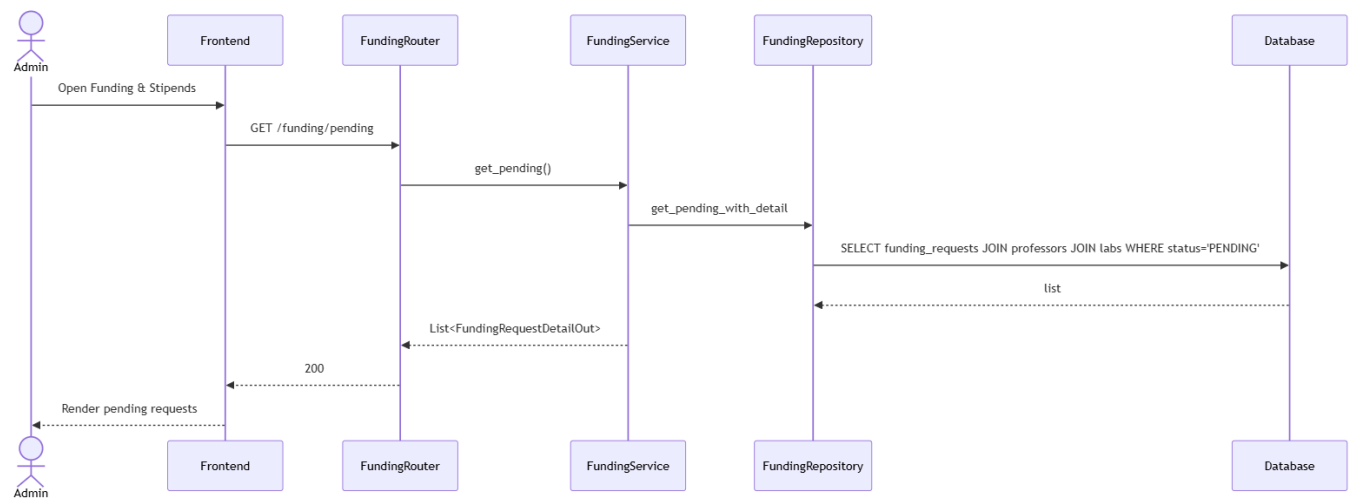


Undergraduate Lab Portal Requirements Specification

SD-A-03: Certify or Decertify Lab

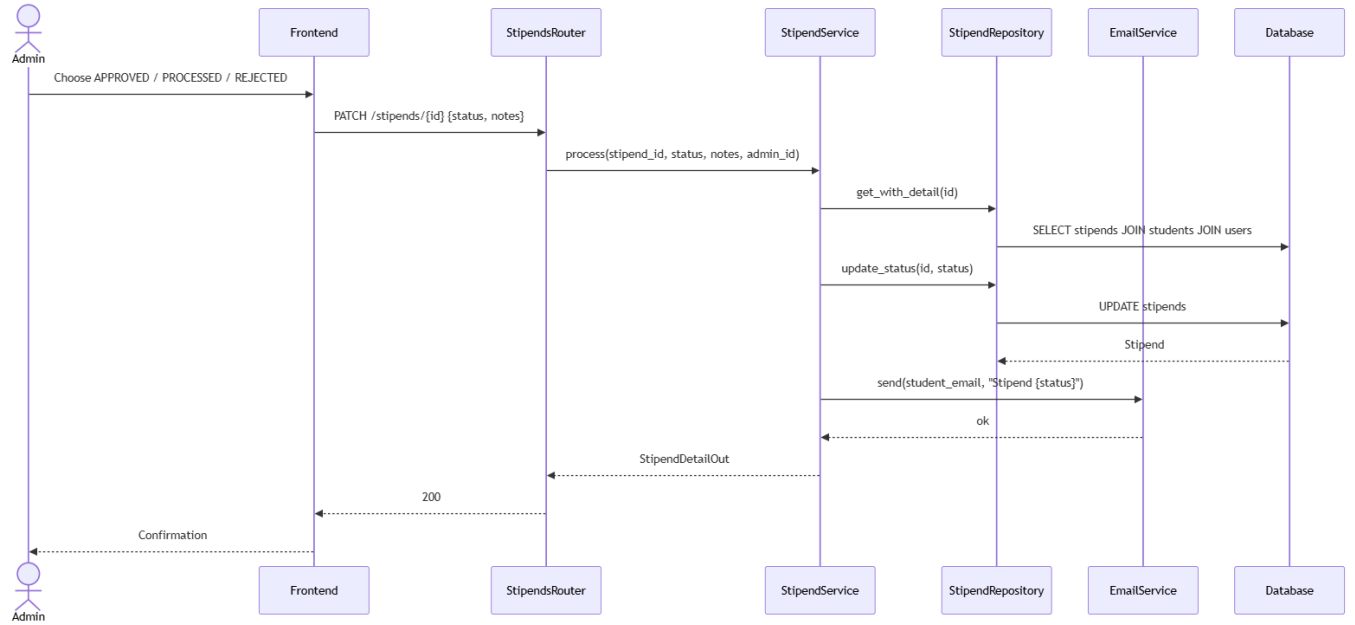


SD-A-04: View Funding Requests



Undergraduate Lab Portal Requirements Specification

SD-A-05: Process Stipend



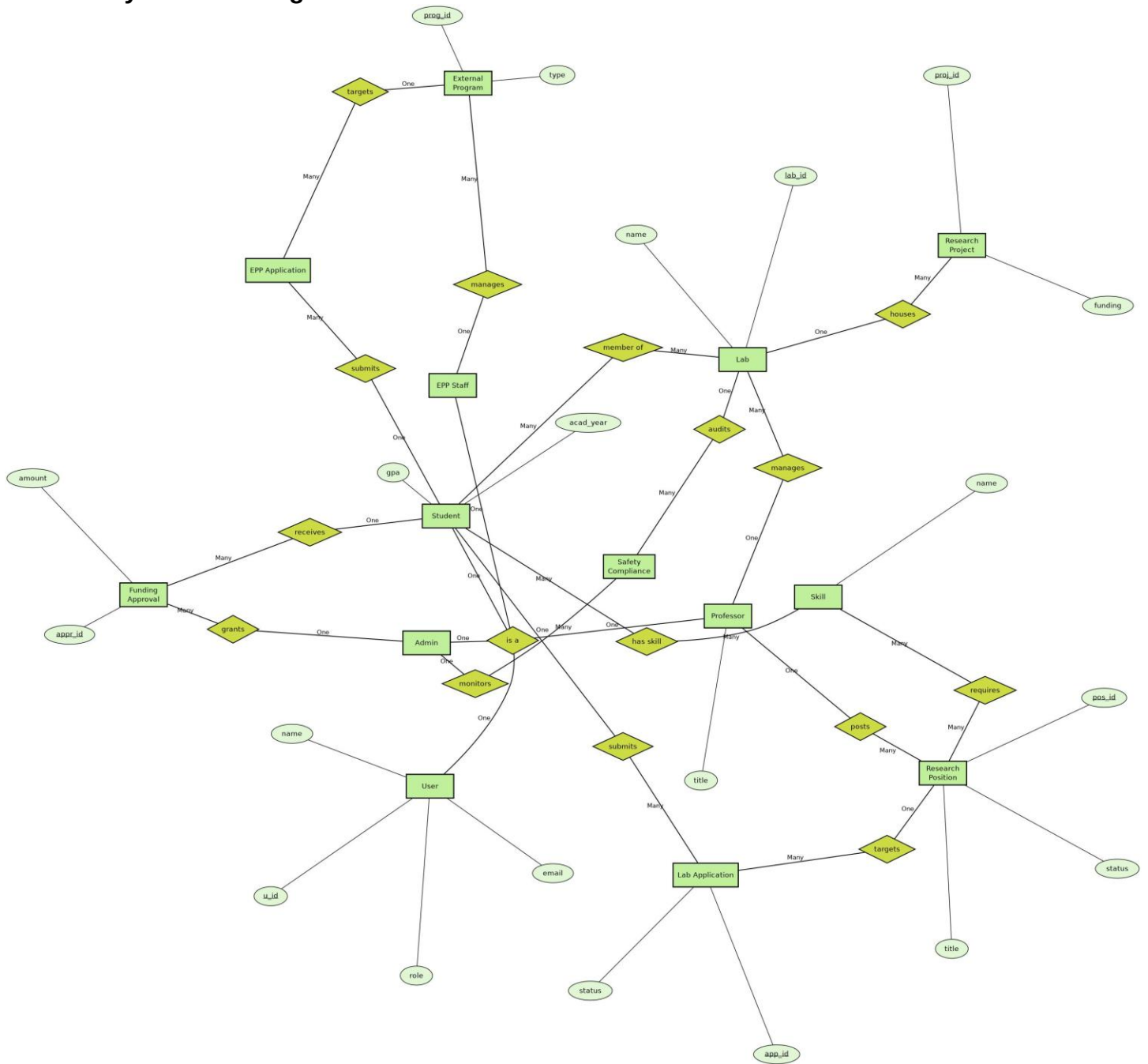
Undergraduate Lab Portal Requirements Specification

4.4 Entity Relation

4.4.1 Database Schema Design

Undergraduate Lab Portal Requirements Specification

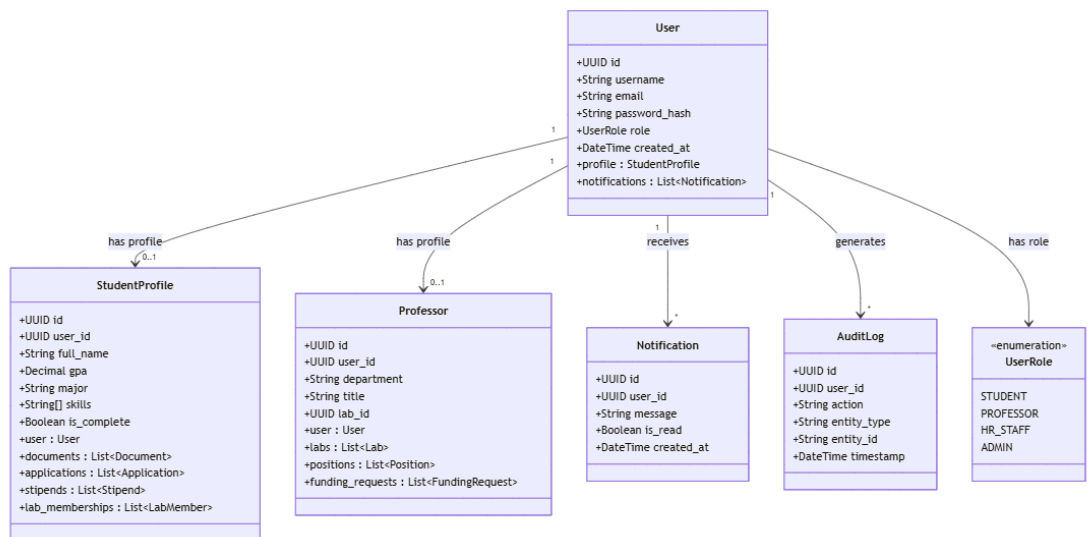
4.4.2.Entity Relation Diagram



4.5 Structural Diagrams

4.5.1 Class Diagrams

Diagram 1 — Core User & Profile Domain (Entities)



Undergraduate Lab Portal Requirements Specification

Diagram 2 — Lab, Position & Membership Domain

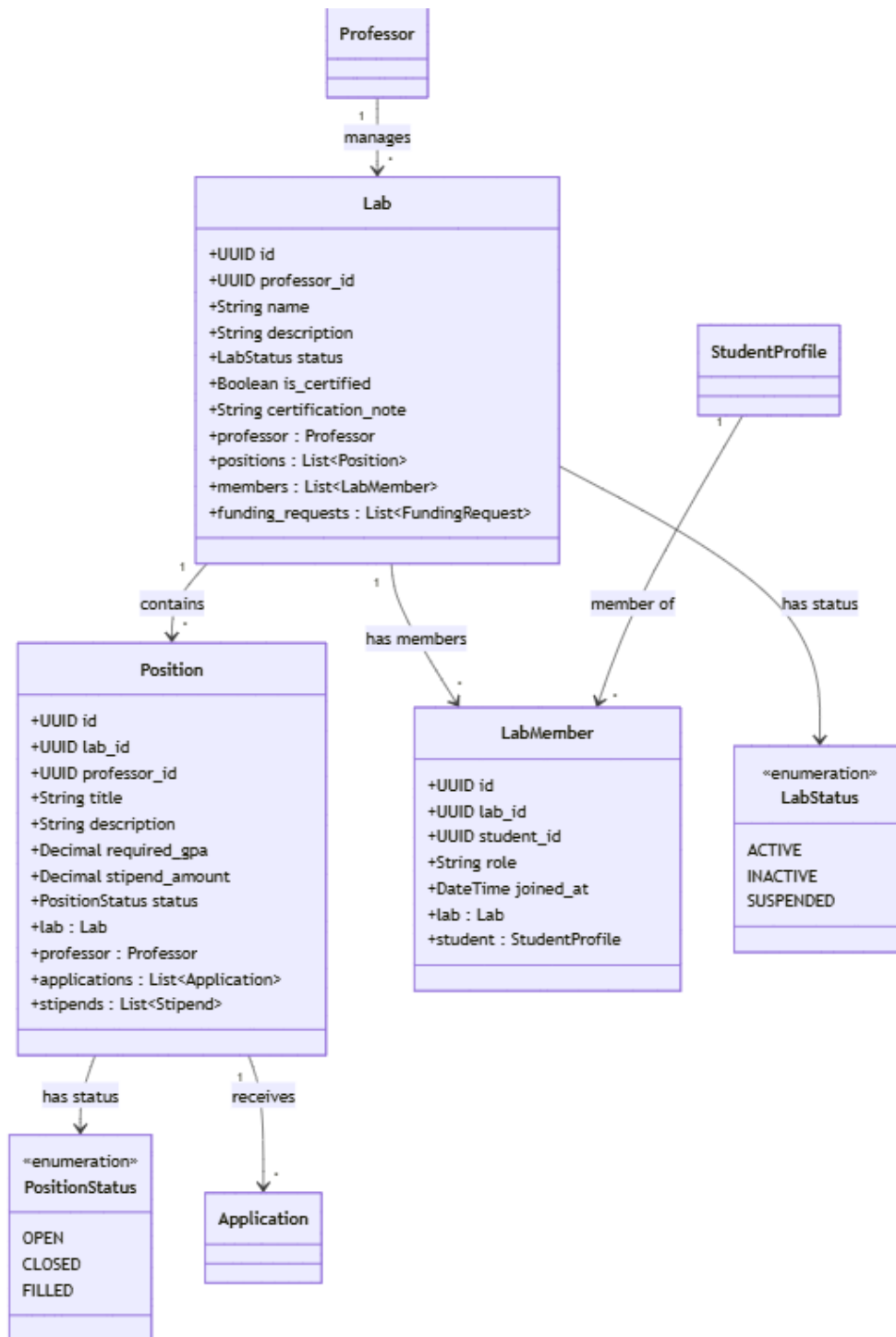


Diagram 3 — Application, Document & Stipend Domain

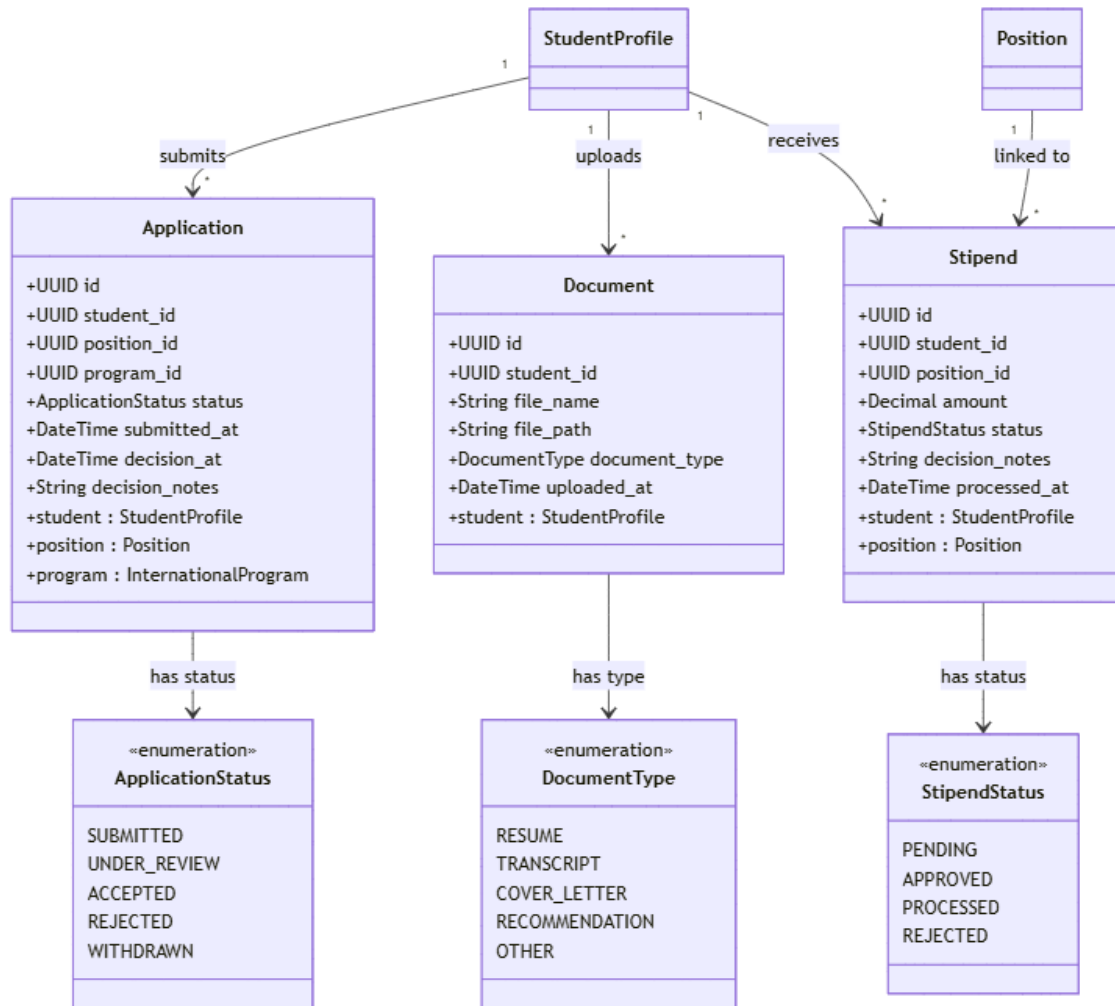


Diagram 4 — International Programs Domain

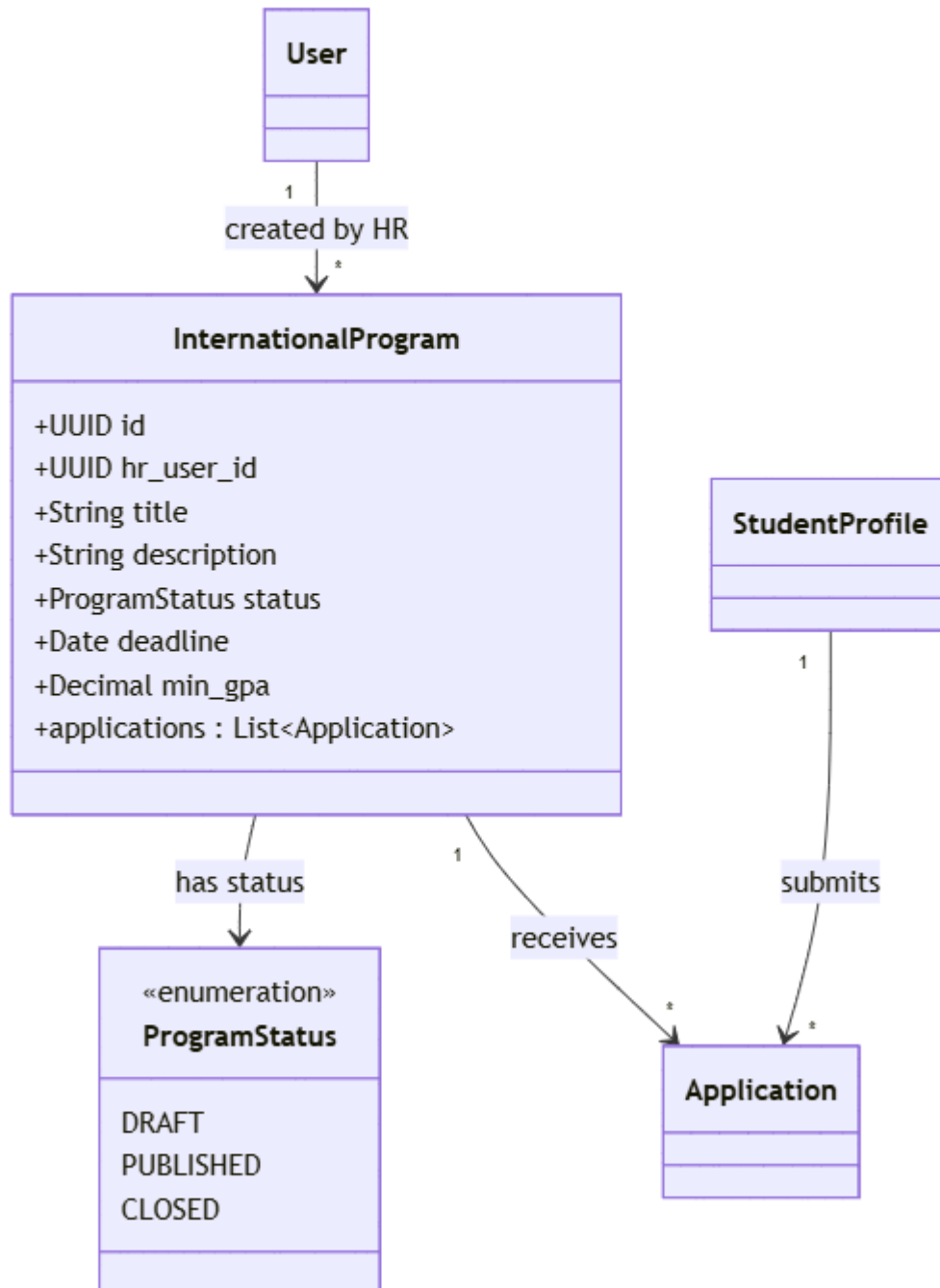
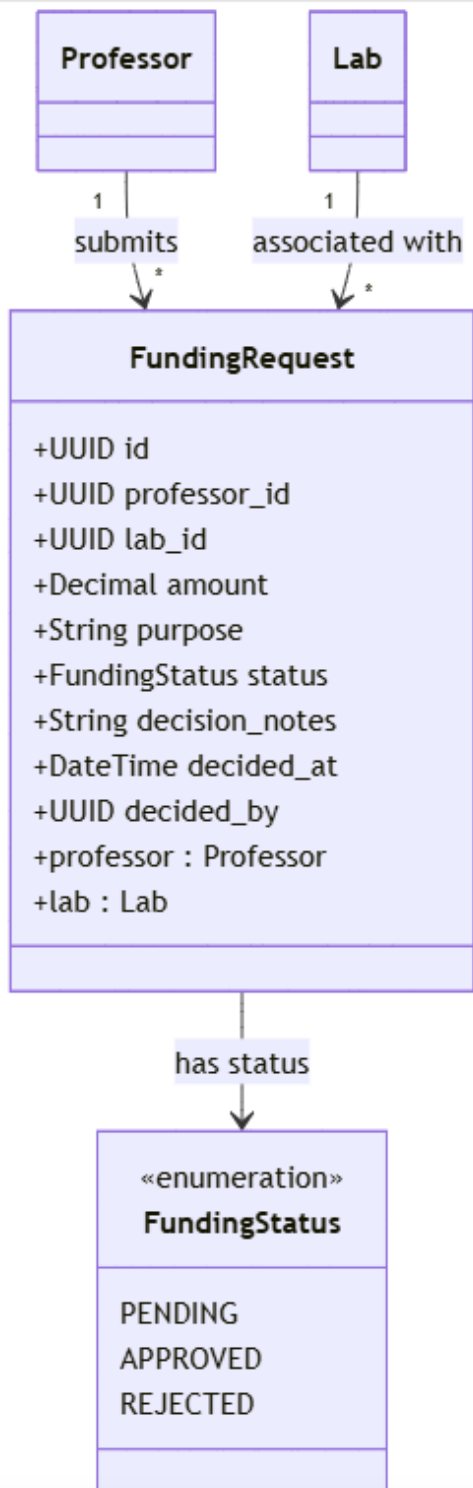
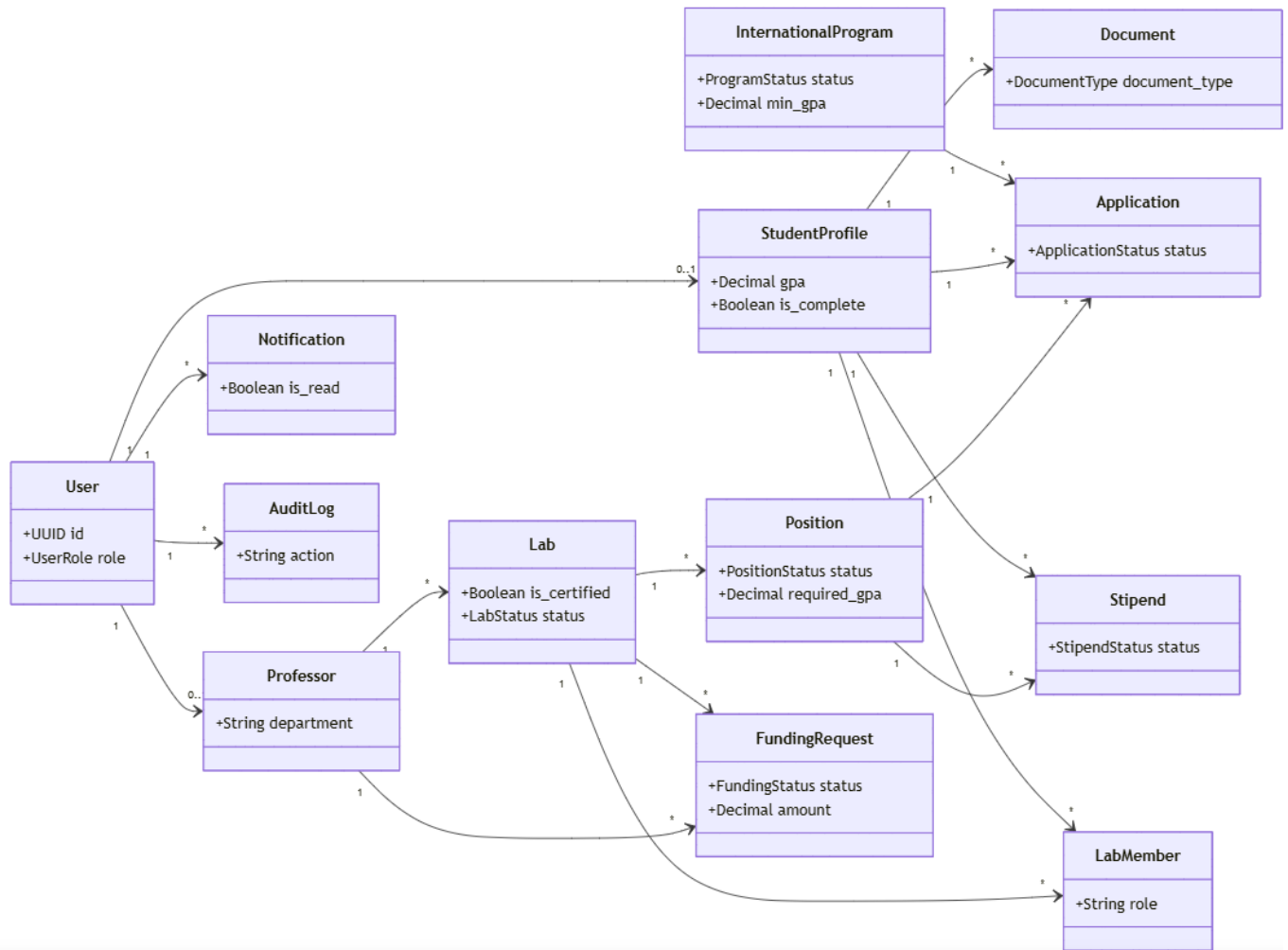


Diagram 5 — Funding Domain

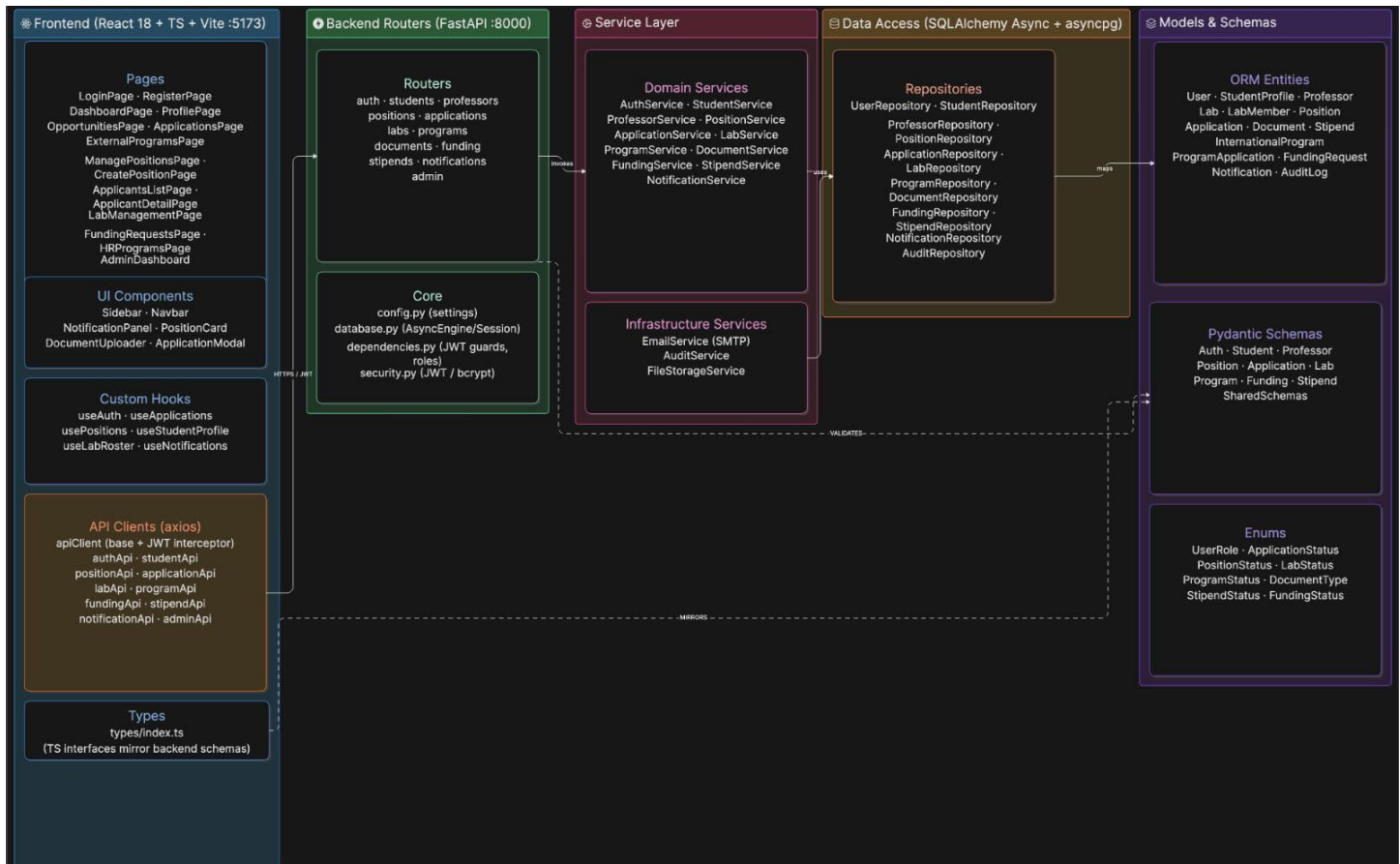


Full system overview



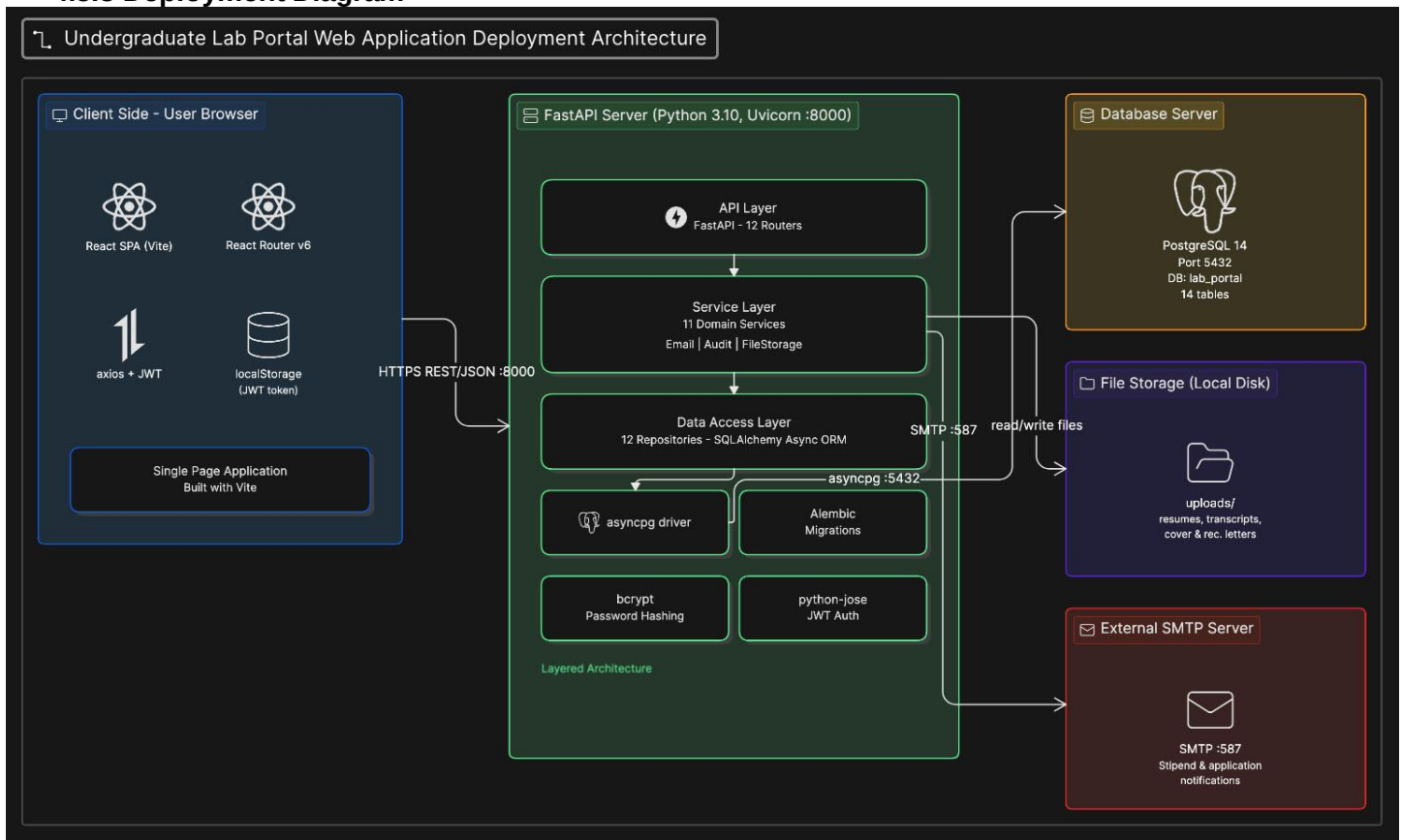
Undergraduate Lab Portal Requirements Specification

4.5.3 Component Diagrams



Undergraduate Lab Portal Requirements Specification

4.5.3 Deployment Diagram



5. Implementation Technology

Implementation Technology

The Research Platform is a dynamic Web Application. For the creation of this software, we have combined Client Side Scripting and Server Side Scripting. The communication between the client and the server takes place via the HTTPS protocol, with authentication handled through JSON Web Tokens (JWT).

For the Client Side, we have used the following technologies:

- JavaScript
- TypeScript
- React 18 (UI Component Library)
- Recharts (Data Visualization Library)
- CSS3 with custom properties for theming

For the Server Side, we have utilized FastAPI (Python), which enables seamless and high-performance interactions within the software and amongst users. Embracing an Object-Oriented Programming (OOP) approach, our architecture incorporates a robust set of features including models, migrations, routers, services, repositories, and cryptography. These components ensure a structured and efficient management of backend functionalities. The server communicates with the client via a RESTful API following the OpenAPI specification.

For the Database, we have used PostgreSQL, and the database schema is version-controlled and managed using Alembic (migration tool built on top of SQLAlchemy). Direct database inspection and administration can be performed using pgAdmin.

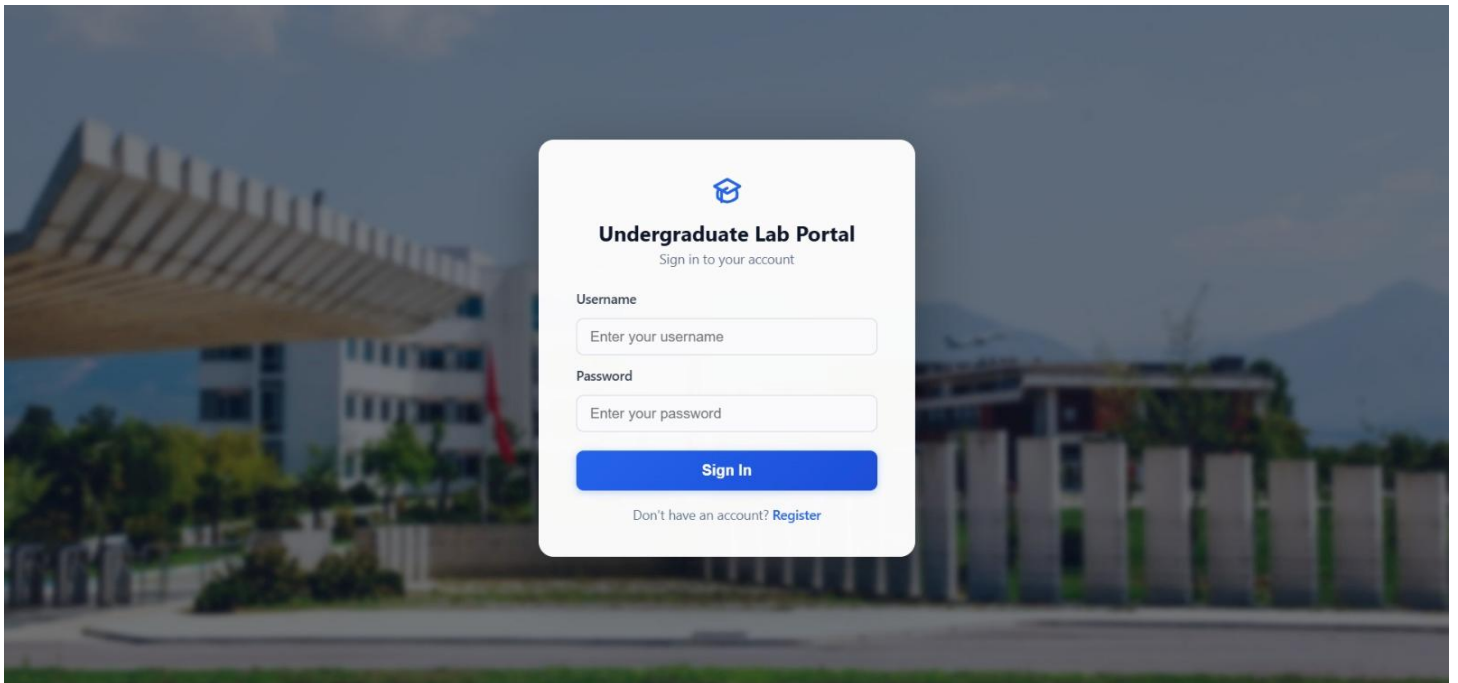
For Security, user passwords are hashed using bcrypt, and session management is handled via signed JWT access and refresh tokens transmitted over HTTPS. Role-based access control (RBAC) enforces authorization across all API endpoints, distinguishing between Student, Professor, HR Staff, and Admin roles.

Github links:

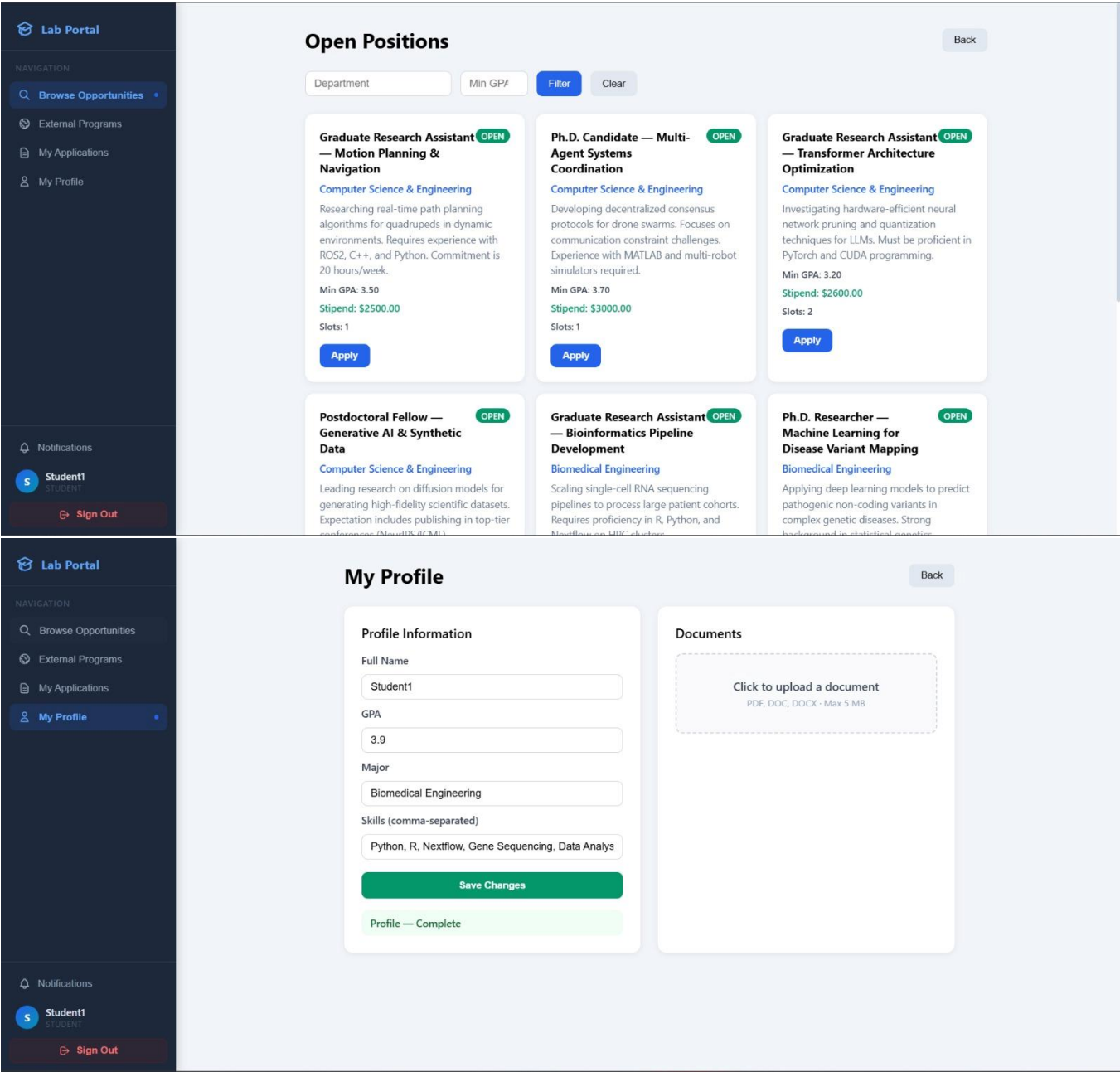
https://github.com/siborakociu/Undergraduate_Lab_Portal.git

Undergraduate Lab Portal Requirements Specification

5. Appendix - Detailed Designs



Undergraduate Lab Portal Requirements Specification



Undergraduate Lab Portal Requirements Specification

Lab Portal

NAVIGATION

Manage Positions

View Applicants

Lab Management

Request Funding

Notifications

Prof2
PROFESSOR

Sign Out

Lab Management

BackNew Lab

Computational Genomics Laboratory
Biomedical Engineering

ACTIVE

Biomaterials & Tissue Engineering Lab
Biomedical Engineering

ACTIVE

Computational Genomics Laboratory
Biomedical Engineering

ACTIVE

Cert: NOT SET

Edit

3 positions3 open0 members

Lab Status:Active

Positions in this lab

Ph.D. Candidate — Biomimetic Scaffold Degradation Modeling
Biomedical Engineering · Min GPA 3.6

Accepted students (0)
No one accepted yet.

OPEN

Ph.D. Researcher — Machine Learning for Disease Variant Mapping
Biomedical Engineering · Min GPA 3.7

OPEN

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Browse Opportunities

External Programs

My Applications

My Profile

Notifications

Student4
STUDENT

Sign Out

My Applications

Back

Position:Ph.D. Candidate — Biomimetic Scaffold Degradation Modeling

Submitted:5/27/2026

Status:SUBMITTED

Position:Graduate Research Assistant — 3D Bioprinting & Hydrogel Synthesis

Submitted:5/27/2026

Status:SUBMITTED

Position:Graduate Research Assistant — Bioinformatics Pipeline Development

Submitted:5/27/2026

Status:SUBMITTED

Position:Ph.D. Researcher — Machine Learning for Disease Variant Mapping

Submitted:5/27/2026

Status:SUBMITTED

Undergraduate Lab Portal Requirements Specification

Lab Portal

NAVIGATION

Manage Positions

View Applicants

Lab Management

Request Funding

Notifications

Prof2

PROFESSOR

Sign Out

New Position

Back

Lab

Computational Genomics Laboratory (Biomedical Engineering)

Title

Ph.D. Candidate — Biomimetic Scaffold Degradation Modeling

Description

Investigating the in vivo degradation rates of synthetic polymers using both computational modeling and wet-lab histology experiments.

Department

Biomedical Engineering

Required GPA

3.6

Stipend per student (USD)

2800

Lab Portal

NAVIGATION

Research Overview

Lab Compliance

Funding & Stipends

Reports

Notifications

Admin1

ADMIN

Sign Out

Admin Panel

Back

Research Overview

Lab Compliance

Funding & Stipends

Reports

All Labs

Expired Certs

Suspended

Autonomous Systems & Robotics Lab

ACTIVE

VALID (6/11/2027)

Computer Science & Engineering

Owner: @Prof1 - Prof1@gmail.com

Members: 0 - Positions: 4

Cert type: Robotic System Safety Clearance (RSSC)

Set Certification

Suspend (Flag)

Biomaterials & Tissue Engineering Lab

ACTIVE

VALID (8/21/2028)

Biomedical Engineering

Owner: @Prof2 - Prof2@gmail.com

Members: 0 - Positions: 1

Cert type: High-Performance Computing & Data Privacy Certification

Set Certification

Suspend (Flag)

Computational Genomics Laboratory

ACTIVE

VALID (4/10/2027)

Biomedical Engineering

Owner: @Prof2 - Prof2@gmail.com

Members: 0 - Positions: 3

Cert type: Biosafety Level 1 (BSL-1) Data Handling

Set Certification

Suspend (Flag)

Undergraduate Lab Portal Requirements Specification

